PATHOLOGY

Rapid Revision Notes · NEET PG & INI-CET

How to read these notes :

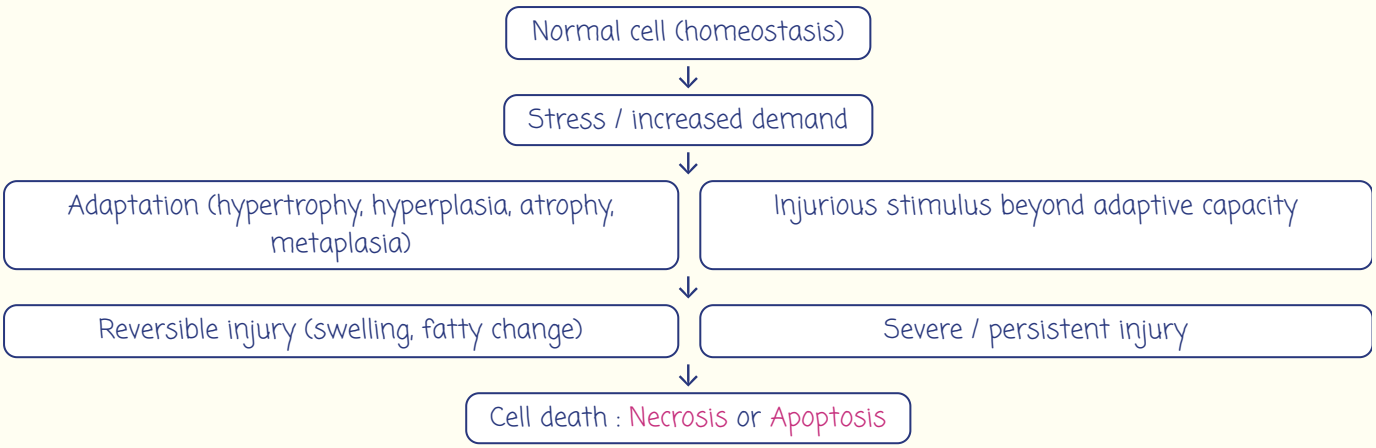
- Navy ink → core content.
- Pink ink → key terms, buzzwords, most-asked facts.
- Dark green ink / green boxes ("Filled") → filled-in high-yield points for a top rank.
- Dashed pink box → mnemonic.
- Yellow box → exam pearl / recent-pattern question.
- Numbered lists → causes, risk factors and criteria, one per line.

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CELL INJURY, ADAPTATION & CELL DEATH

Overview



Cellular Adaptations

Adaptation	Definition	Physiological example	Pathological example
Hypertrophy	↑ cell size → ↑ organ size	Uterus in pregnancy (also hyperplasia), skeletal muscle in body-builders	LVH in hypertension, aortic stenosis
Hyperplasia	↑ cell number (only in dividing cells)	Breast at puberty / lactation, liver regeneration after partial hepatectomy	Endometrial hyperplasia (unopposed estrogen), BPH, warts (HPV)
Atrophy	↓ cell size and number	Thymus after puberty, uterus after delivery	Disuse, denervation, ↓ blood supply, pressure, ↓ nutrition, ↓ endocrine
Metaplasia	One adult differentiated cell type replaced by another	—	Barrett esophagus, squamous metaplasia in smokers' bronchi
Hypoplasia	Incomplete development of an organ (fewer cells)	—	Unilateral renal hypoplasia
Agenesis / Aplasia	Complete absence of organ / its primordium; aplasia = primordium present but no development	—	Renal agenesis (Potter sequence)
Dysplasia	Disordered growth: loss of uniformity and architectural orientation	—	CIN, Barrett with dysplasia

Hypertrophy :

- Seen in permanent cells (cardiac, skeletal muscle) which cannot divide.
- Mechanism : mechanical sensors + growth factors (TGF-β, IGF-1, FGF) + vasoactive agents (α-adrenergic agonists, endothelin-1, angiotensin II).
- Signal pathways : PI3K/AKT (physiological), G-protein coupled receptor signalling (pathological).
- Switch of contractile proteins to fetal forms : α-myosin heavy chain → β-MHC (slower, energy-efficient contraction).
- ↑ ANP (atrial natriuretic peptide) gene expression in hypertrophied ventricle.

Filled

- Pure hypertrophy example : skeletal muscle, heart. Pure hyperplasia : liver regeneration. Both together : gravid uterus (smooth muscle).
- Selective hypertrophy of an organelle : ↑ smooth ER in hepatocytes with barbiturates / phenobarbital (CYP450 induction).
- Cardiac myocyte hypertrophy limit → myofibril fragmentation, loss of contractile elements → cardiac failure.

Hyperplasia :

- Occurs only in tissues capable of division (labile and stable cells).
- Driven by growth factors and stem-cell recruitment.
- Pathological hyperplasia is a fertile soil for cancer : endometrial hyperplasia → endometrial carcinoma; HPV papillomas.
- BPH is NOT pre-malignant for prostatic carcinoma.

Atrophy :

Causes :

1. Decreased workload (disuse atrophy)
2. Loss of innervation (denervation atrophy)
3. Diminished blood supply (senile atrophy of brain)
4. Inadequate nutrition (marasmus, cachexia — TNF-α)
5. Loss of endocrine stimulation (endometrium after menopause)
6. Pressure (erosion of vertebra by aortic aneurysm)

Mechanism :

- ↓ protein synthesis + ↑ protein degradation via ubiquitin-proteasome pathway.
- Autophagy (self-eating) → autophagic vacuoles.
- Residual bodies contain lipofuscin → brown atrophy (heart, liver of elderly).

Metaplasia :

- Reversible; reprogramming of stem cells, not change of differentiated cells.
- Driven by cytokines/growth factors + retinoic acid; transcription factors (SOX, CDX2).

Type	Example	Cause
Columnar → squamous (most common)	Bronchus	Cigarette smoking, vitamin A deficiency
Columnar → squamous	Endocervix (transformation zone)	Low vaginal pH
Transitional → squamous	Urinary bladder	Schistosoma haematobium, stones
Squamous → columnar (intestinal)	Barrett esophagus	GERD (acid reflux)
Gastric → intestinal	Stomach	H. pylori, autoimmune gastritis
Connective tissue	Myositis ossificans (bone in muscle)	Trauma / hemorrhage

Exam pearl

- Epithelial metaplasia is reversible; dysplasia may be reversible (low grade); carcinoma in situ is not invasive but irreversible.
- Myositis ossificans : "zonal pattern" — mature bone at periphery, immature fibroblasts at centre (opposite of osteosarcoma).
- Apocrine metaplasia in fibrocystic breast — not pre-malignant.

Cell Injury — Causes

1. Oxygen deprivation (hypoxia, ischemia) — the most common cause.
2. Physical agents (trauma, temperature, radiation, electric shock).
3. Chemical agents and drugs.
4. Infectious agents.
5. Immunologic reactions.
6. Genetic derangements.
7. Nutritional imbalances.

Filled : Hypoxia vs ischemia

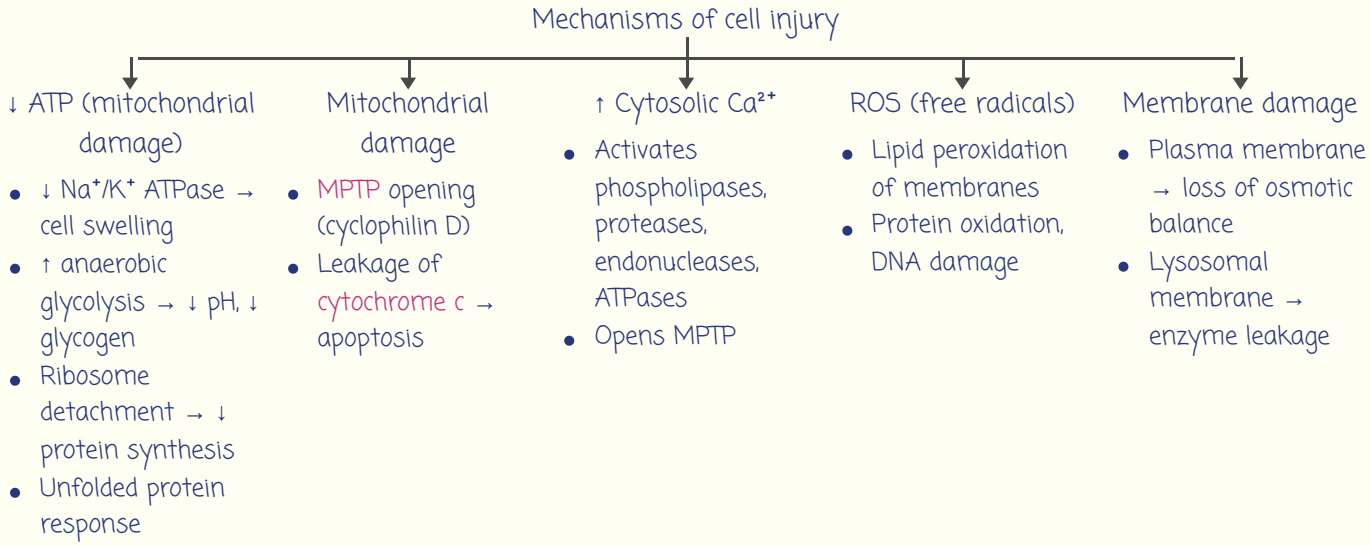
- Ischemia injures tissues faster than hypoxia — loss of both O2 and metabolic substrates (glucose) + accumulation of toxic metabolites.
- In hypoxia, anaerobic glycolysis continues; in ischemia it stops once substrates are exhausted.

Susceptibility of cells to ischemia :

Cell	Time to irreversible injury
Neurons	3-5 min
Myocardium	20-30 min (irreversible); hepatocytes 1-2 h
Fibroblasts, skeletal muscle	Many hours

- Most vulnerable neurons : CA1 of hippocampus (Sommer sector) → Purkinje cells of cerebellum → pyramidal cells of neocortex (layers 3, 5, 6).
- Most sensitive renal segment : proximal tubule (S3 segment) and thick ascending limb.
- Liver : zone 3 (centrilobular) most vulnerable to ischemia.

Mechanisms of Cell Injury



Free radicals (ROS) :

Generation :

1. Normal metabolism (reduction-oxidation reactions)
2. Absorption of radiant energy (UV, X-rays → hydrolysis of water)
3. Leukocyte respiratory burst (NADPH oxidase)
4. Enzymatic metabolism of chemicals (CCl_4 → CCl_3•)
5. Transition metals : Fenton reaction ($Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH• + OH^-$)
6. Nitric oxide (NO) → peroxynitrite ($ONOO^-$)

Removal (antioxidants) :

1. Antioxidants : vitamin E (lipid-soluble, membrane), vitamin A, ascorbic acid, glutathione.
2. Storage/transport proteins that bind free iron and copper : transferrin, ferritin, lactoferrin, ceruloplasmin.
3. Enzymes :
   - Superoxide dismutase : $O_2^- \rightarrow H_2O_2$
   - Catalase (peroxisomes) : $2H_2O_2 \rightarrow O_2 + 2H_2O$
   - Glutathione peroxidase : $H_2O_2 + 2GSH \rightarrow GSSG + 2H_2O$

> Exam pearl
> - Most reactive (damaging) free radical : hydroxyl radical (OH•).
> - Superoxide dismutase is mitochondrial (Mn-SOD) and cytosolic (Cu/Zn-SOD).
> - Free-radical injury examples : CCl_4 hepatotoxicity, reperfusion injury, radiation, oxygen toxicity (retinopathy of prematurity, BPD), acetaminophen (NAPQI), aging.

Ischemia-reperfusion injury :

1. ↑ ROS generation by damaged mitochondria and leukocytes.
2. Intracellular calcium overload (MPTP opening).
3. Inflammation : neutrophil influx with cytokines.
4. Complement activation (IgM deposits in ischemic tissue).

- Morphology in heart : contraction band necrosis (hyper-contracted myofibrils due to Ca^{2+} influx).

Chemical injury :

Direct acting	Indirect (converted to toxic metabolites by CYP450)
Mercuric chloride — binds SH groups (kidney, GI)	CCl_4 → CCl_3• — fatty liver, centrilobular necrosis
Cyanide — inhibits cytochrome oxidase	Acetaminophen → NAPQI — zone 3 necrosis, ↓ glutathione
Chemotherapeutic agents	Bromobenzene, vinyl chloride

Reversible Cell Injury

Two main features :

1. Cellular swelling (hydropic change / vacuolar degeneration) — earliest change; failure of Na^+/K^+ ATPase.
2. Fatty change (steatosis) — liver, heart, kidney (organs in lipid metabolism).

Ultrastructural changes (reversible) :

1. Plasma membrane alterations : blebbing, blunting, loss of microvilli.
2. Mitochondrial changes : swelling, small amorphous densities.
3. Dilatation of ER with detachment of polysomes.
4. Nuclear alterations : disaggregation of granular and fibrillar elements.

5. Myelin figures : phospholipid whorls from damaged membranes.

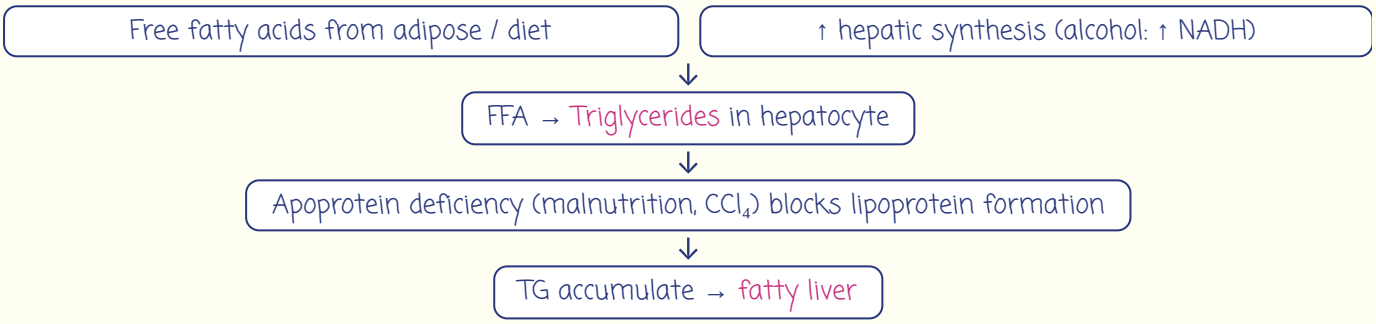
> Filled : Reversible vs irreversible — point of no return
> - Two phenomena characterize irreversibility : (1) inability to restore mitochondrial function (lack of oxidative phosphorylation) even after reperfusion, (2) profound disturbances in membrane function.
> - Large flocculent amorphous densities in mitochondria → irreversible (ischemic myocardium).
> - Leakage of enzymes (troponin, CK-MB, ALT) indicates membrane damage → irreversible injury.

Fatty change (steatosis) — liver :

Causes :

1. Alcohol (most common in developed countries)
2. Non-alcoholic fatty liver disease (obesity, diabetes, metabolic syndrome)
3. Protein malnutrition (kwashiorkor — ↓ apoprotein)
4. Toxins (CCl_4), drugs (tetracycline, valproate)
5. Starvation, hypoxia, pregnancy (acute fatty liver of pregnancy)
6. Reye syndrome (aspirin in children, microvesicular)

Pathogenesis of fatty liver



Feature	Microvesicular	Macrovesicular
Appearance	Many small droplets, central nucleus	Single large vacuole, nucleus pushed to periphery
Causes	Reye syndrome, acute fatty liver of pregnancy, valproate, tetracycline	Alcohol, obesity, diabetes, malnutrition
Significance	Mitochondrial dysfunction; can be severe	Usually benign, reversible

- Stains : fat dissolved in routine paraffin sections (clear vacuoles); demonstrated by Oil Red O and Sudan III/IV on frozen sections; osmium tetroxide (black).
- Heart : fatty change in prolonged moderate hypoxia (profound anemia) → tigered effect / "thrush-breast" heart (bands of yellow myocardium alternating with red-brown) in the papillary muscles.

Hyaline change :

- Homogeneous, glassy, pink (eosinophilic) appearance on H&E — a descriptive term, not a specific lesion.

Intracellular hyaline	Extracellular hyaline
Mallory-Denk bodies (alcoholic hepatitis — keratin)	Hyaline arteriolosclerosis (benign HTN, diabetes)
Russell bodies (immunoglobulin in plasma cells)	Hyaline in old scars, corpora albicantia
Councilman bodies (apoptotic hepatocytes)	Hyaline membranes (ARDS, HMD of newborn)
Hyaline droplets in PCT (proteinuria)	Amyloid (hyaline-like)
Zenker's hyaline degeneration of muscle (typhoid)	Hyaline cartilage-like scar

Other intracellular accumulations :

Substance	Examples
Cholesterol	Atherosclerosis (foam cells), xanthomas, cholesterolosis (strawberry gallbladder), Niemann-Pick type C
Proteins	Russell bodies, Mallory hyaline, α1-antitrypsin globules (PAS +, diastase-resistant)
Glycogen	Diabetes (Armanni-Ebstein lesion — PCT/loop of Henle), glycogen storage diseases (stain : PAS + diastase-sensitive, Best carmine)
Pigments	Lipofuscin, melanin, hemosiderin, bilirubin, carbon (see Pigments chapter)

Irreversible Injury — Necrosis

- Necrosis : death of cells in a living organism due to severe injury, always pathological, with inflammation.

Morphology of necrotic cell :

- Cytoplasm : ↑ eosinophilia (loss of RNA + denatured proteins), glassy, myelin figures, calcification.
- Nucleus :

Pyknosis (shrinkage, ↑ basophilia) → Karyorrhexis (fragmentation) → Karyolysis (fading; DNase)

Patterns of necrosis :

Type	Mechanism / features	Classic examples
Coagulative	Protein denaturation > enzymatic digestion; architecture preserved for days ("ghost cells")	Infarcts of all solid organs except brain (heart, kidney, spleen)
Liquefactive	Enzymatic digestion dominates; tissue becomes liquid, creamy (pus)	Brain infarct, abscess (bacterial/fungal)
Caseous	"Cheese-like", friable white; granuloma with central amorphous granular debris	Tuberculosis, histoplasmosis, coccidioidomycosis
Fat necrosis	Lipases release FA → combine with Ca^{2+} (saponification) → chalky white	Acute pancreatitis, traumatic fat necrosis of breast
Fibrinoid	Immune complexes + fibrin in vessel walls; bright pink, amorphous	Malignant hypertension, vasculitis (PAN), Aschoff body, preeclampsia
Gangrenous (clinical term)	Usually coagulative necrosis of limb (dry), + bacterial liquefaction (wet)	Diabetic foot, intestinal gangrene

Filled : Extra high-yield

- Gumma (tertiary syphilis) — rubbery necrosis, architecture of vessels partially preserved (ghost outlines) — vs caseation where outlines are lost.
- Caseation without TB : histoplasmosis, coccidioidomycosis, sarcoid (rarely, no), tularemia, cat-scratch (stellate microabscess).
- Fibrinoid necrosis in placenta (normal finding at term — Nitabuch layer).
- Fat necrosis in breast can mimic carcinoma clinically (hard lump + skin retraction) — history of trauma.

Apoptosis

- Programmed cell death; membrane intact, no inflammation.
- Hallmark morphology : cell shrinkage, chromatin condensation (pyknosis, most characteristic), cytoplasmic blebs, apoptotic bodies, phagocytosis by macrophages.
- DNA fragmentation : step-ladder pattern on agarose gel (internucleosomal, 180-200 bp multiples) vs smear pattern in necrosis.

Causes :

Physiological :

1. Embryogenesis (implantation, organogenesis, involution — e.g., interdigital webs)
2. Hormone-dependent involution (endometrium in menstruation, ovarian follicular atresia, breast after weaning)
3. Cell turnover in proliferating populations (intestinal crypt epithelium)
4. Elimination of self-reactive lymphocytes (negative selection in thymus)
5. Death of host cells that have served their purpose (neutrophils after acute inflammation)

Pathological :

1. DNA damage (radiation, cytotoxic drugs)
2. Accumulation of misfolded proteins (ER stress)
3. Infections — viral hepatitis (Councilman bodies)
4. Pathological atrophy in parenchymal organs after duct obstruction (pancreas, kidney, parotid)
5. Cell death by cytotoxic T cells (graft rejection)

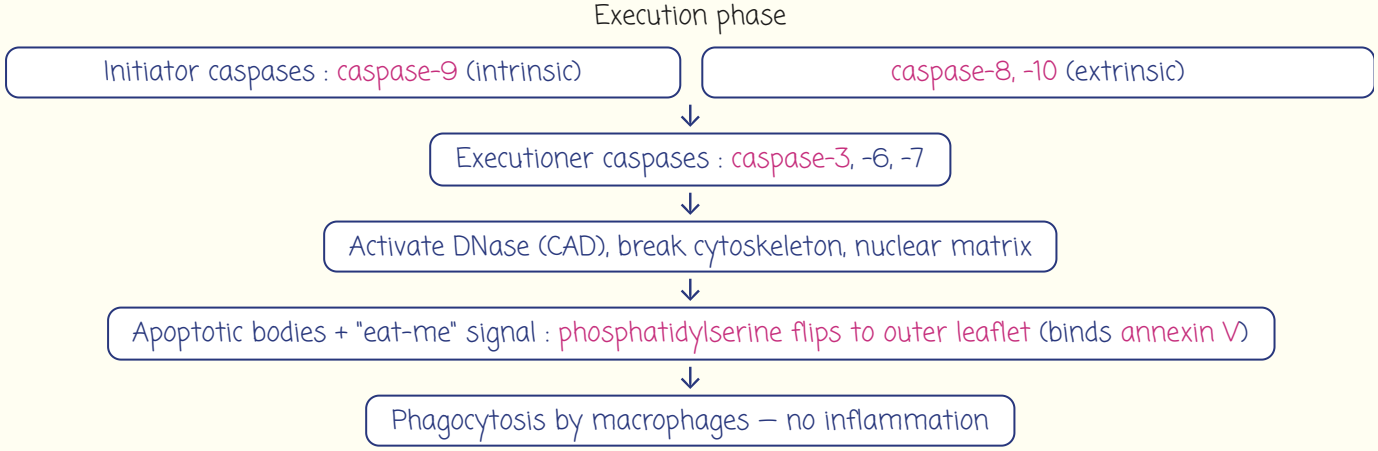
Pathways :

Intrinsic (mitochondrial) pathway :

- Most common pathway.
- Trigger : ↓ growth factors, DNA damage (p53), misfolded proteins.
- BH3-only sensors (BIM, BID, BAD, PUMA, NOXA) ↑ → activate BAX/BAK (pro-apoptotic).
- ↓ BCL-2, BCL-XL, MCL-1 (anti-apoptotic).
- Cytochrome c release → binds APAF-1 → apoptosome → caspase-9 (initiator).

Extrinsic (death receptor) pathway :

- Death receptors : TNFR1, Fas (CD95) — have death domain.
- FasL (on activated T cells) binds Fas → adaptor FADD → caspase-8 (also caspase-10).
- Inhibitor : FLIP (viral and cellular).
- Role : eliminating self-reactive lymphocytes, CTL killing.
- Defect → autoimmune lymphoproliferative syndrome (ALPS).



Filled : Rank boosters — apoptosis

- Caspase = cysteine protease that cleaves after aspartic acid.
- BCL-2 overexpression by t(14;18) → follicular lymphoma (failure of apoptosis).
- IAP (inhibitor of apoptosis proteins) block caspases; SMAC/DIABLO from mitochondria neutralize IAP.
- Markers of apoptosis : annexin V, TUNEL assay, caspase-3 cleavage, DNA ladder.
- Granzyme B from CTL activates caspases directly (and BID); perforin makes pores.
- p53 → PUMA/NOXA → BAX/BAK; also arrests cell cycle via p21.

Necrosis vs Apoptosis :

Feature	Necrosis	Apoptosis
Cell size	Enlarged (swelling)	Reduced (shrinkage)
Nucleus	Pyknosis → karyorrhexis → karyolysis	Fragmentation into nucleosome-size pieces
Plasma membrane	Disrupted	Intact; altered structure (lipid orientation)
Cellular contents	Enzymatic digestion; may leak out	Intact; released in apoptotic bodies
Inflammation	Frequent	None
Role	Always pathological	Often physiological; may be pathological
DNA on gel	Smear	Ladder
ATP	Not required	Energy-dependent

Other Forms of Cell Death

Type	Key molecules	Features / examples
Necroptosis	RIPK1, RIPK3 → MLKL	Programmed necrosis, caspase-independent; TNF; Crohn disease, pancreatitis, reperfusion; CMV inhibits
Pyroptosis	Inflammasome → caspase-1 (also 4, 5, 11) → gasdermin D pores	Fever (IL-1β, IL-18); infected cells (Salmonella, Shigella)
Ferroptosis	Iron-dependent lipid peroxidation; ↓ GPX4 / glutathione	Stroke, neurodegeneration, cancer; inhibited by ferrostatin
Autophagy	Beclin-1, LC3, ATG genes; mTOR inhibition	Starvation survival; defects in Crohn (ATG16L1), Alzheimer, Huntington
Entosis	Cell engulfed by neighbour	"Cell-in-cell" in tumours
Anoikis	Loss of integrin-matrix attachment	Prevents metastasis; resistance in cancer
NETosis	Neutrophil extracellular traps (PAD4)	Sepsis, SLE, thrombosis
Parthanatos	PARP-1 overactivation, AIF	Stroke, DNA damage

Mnemonic : Ferroptosis

"Ferro = Fe, peroxidation, GPX4 guards" — iron + lipid ROS; not apoptosis (no caspases).

Autophagy

- "Self-eating" — delivers cytoplasmic materials to lysosomes.
- Types : macroautophagy (main), microautophagy, chaperone-mediated autophagy.
- Steps : nucleation (Beclin-1 / VPS34) → elongation (ATG genes, LC3) → autophagosome → fusion with lysosome → autophagolysosome.
- Induced by nutrient deprivation (mTOR inhibition).

Cellular Aging

1. DNA damage (accumulated mutations; defective repair — Werner syndrome: DNA helicase, Bloom syndrome).
2. Replicative senescence — telomere shortening (Hayflick limit); telomerase active in germ/stem/cancer cells.
3. Defective protein homeostasis (↓ chaperones, ↓ proteasome).
4. Dysregulated nutrient sensing : caloric restriction ↑ lifespan via ↓ IGF-1 and ↓ mTOR, ↑ sirtuins (NAD-dependent deacetylases).
5. Persistent inflammation ("inflammaging"), ROS.

Filled : Progeria syndromes

Syndrome	Defect
Werner syndrome	WRN — DNA helicase (RecQ) — adult-onset premature aging
Hutchinson-Gilford progeria	LMNA (lamin A) — progerin — childhood aging
Bloom syndrome	BLM helicase
Ataxia telangiectasia	ATM kinase — radiation sensitivity
Dyskeratosis congenita	Telomerase (TERC/TERT/DKC1)

CALCIFICATION & AMYLOIDOSIS

Pathological Calcification

- Abnormal tissue deposition of calcium salts with small amounts of iron, magnesium and other minerals.
- Stain : von Kossa (black — actually stains phosphate/carbonate), Alizarin red S (red-orange, calcium). On H&E : basophilic, amorphous, granular.

Feature	Dystrophic	Metastatic
Serum calcium	Normal	Hypercalcemia
Tissue	Dead / dying (necrotic) tissue	Normal (viable) tissue
Common sites	Atheroma, damaged / aging heart valves, caseous TB, infarcts, dead parasites	Interstitial tissue of gastric mucosa, kidneys, lungs, systemic arteries, pulmonary veins
Mechanism	Initiation in membrane-bound vesicles (phospholipids + phosphatase) or extracellular	Loss of acid (H^+) → alkaline compartment favours deposition
Clinical	Usually a marker of prior injury	Nephrocalcinosis, respiratory deficits

Mnemonic : Sites of metastatic calcification

"Calcium loves alkaline tissues" — tissues that excrete acid have an internal alkaline compartment : gastric mucosa (HCl), kidney (H^+), lung (CO_2).

Causes of metastatic calcification (hypercalcemia) :

1. ↑ Parathyroid hormone — primary hyperparathyroidism, PTHrP from tumours (squamous cell lung carcinoma)
2. Bone destruction — multiple myeloma, leukemia, skeletal metastases, Paget disease (accelerated turnover), immobilization
3. Vitamin D-related — vitamin D intoxication, sarcoidosis (macrophages activate vitamin D), idiopathic hypercalcemia of infancy (Williams syndrome)
4. Renal failure — phosphate retention → secondary hyperparathyroidism
5. Milk-alkali syndrome (excess calcium + absorbable antacids)
6. Aluminium intoxication (chronic dialysis)

Dystrophic calcification — examples :

1. Atherosclerotic plaques (advanced atheroma)
2. Aging or damaged heart valves (calcific aortic stenosis, bicuspid aortic valve)
3. Caseous necrosis in TB (Ghon complex, calcified lymph nodes)
4. Fat necrosis (pancreatitis — "chalky white")
5. Mönckeberg medial calcific sclerosis (medium muscular arteries, > 50 y)
6. Dead parasites — hydatid cyst wall, cysticercus, Schistosoma eggs
7. Thrombi (phleboliths in veins)
8. Tumours — psammoma bodies

Psammoma bodies :

- Laminated, concentric calcified structures (dystrophic) formed by progressive deposition on single necrotic cells.

Mnemonic : Psammoma bodies — "PSaMMOMa"

Papillary carcinoma of thyroid · Serous cystadenocarcinoma of ovary · Meningioma · Malignant Mesothelioma · also prolactinoma, papillary RCC, somatostatinoma (duodenum), endometrial serous carcinoma.

Filled

- Calciphylaxis (calcific uremic arteriolopathy) — end-stage renal disease, ↑ Ca × P product, medial calcification of small arteries → painful skin necrosis.
- Calcinosis cutis in CREST / limited scleroderma, dermatomyositis (children).
- Heterotopic ossification = bone formation (not just calcium) in soft tissue — myositis ossificans.
- Michaelis-Gutmann bodies (target-like, calcified, PAS+, von Kossa +) — malakoplakia of bladder (defective macrophage lysosomal killing of E. coli).
- Schaumann bodies (laminated calcium + protein) and asteroid bodies — sarcoidosis.

Amyloidosis

- Extracellular deposition of misfolded proteins which aggregate into insoluble fibrils with a β-pleated sheet configuration.
- All amyloid : ~95% fibril protein + 5% P component (serum amyloid P, pentraxin) + glycosaminoglycans (heparan sulfate).

Physical nature :

- EM : non-branching fibrils, 7.5-10 nm diameter, indefinite length.
- X-ray crystallography / infrared spectroscopy : cross-β-pleated sheet conformation (responsible for Congo red birefringence).

Staining :

Stain	Result
H&E	Pale eosinophilic, amorphous, hyaline extracellular material
Congo red (light microscopy)	Pink / red (salmon-pink)
Congo red under polarized light	Apple-green birefringence (most specific)
Gross (Lugol's iodine)	Mahogany brown → blue-violet with dilute H_2SO_4 (starch-like, hence "amyloid" by Virchow)
Thioflavin T / S	Fluorescence (UV)
Crystal violet / methyl violet	Metachromasia (rose-pink on violet)
Immunohistochemistry / mass spectrometry	Typing of amyloid (gold standard for type)

Classification of amyloid proteins :

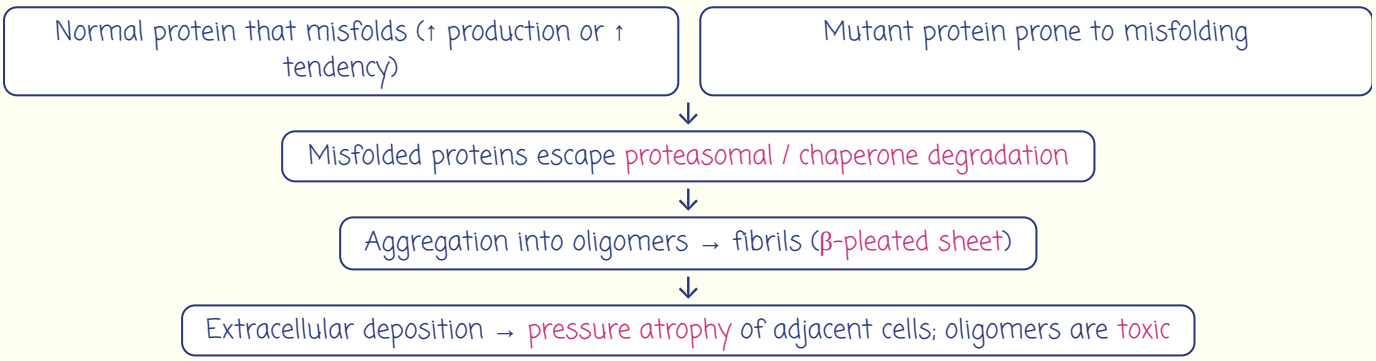
Protein	Precursor	Clinical setting
AL (amyloid light chain)	Ig light chains (λ > κ)	Primary / plasma cell dyscrasia — multiple myeloma, B-cell neoplasms
AA (amyloid-associated)	SAA (acute phase, liver; IL-1/IL-6)	Reactive systemic (secondary) — chronic inflammation
Aβ2m (β2-microglobulin)	β2-microglobulin (MHC-I)	Long-term hemodialysis — carpal tunnel, joints
ATTR	Transthyretin (normal or mutant)	Familial amyloid polyneuropathy; senile systemic amyloidosis (heart, wild-type TTR)
Aβ (amyloid β)	APP (amyloid precursor protein)	Alzheimer disease, cerebral amyloid angiopathy
ACal	(Pro)calcitonin	Medullary carcinoma thyroid

Protein	Precursor	Clinical setting
AIAPP	Amylin (islet amyloid polypeptide)	Type 2 diabetes (islets)
AANF	Atrial natriuretic factor	Isolated atrial amyloidosis (elderly)
APrP	Prion protein	Spongiform encephalopathies
AA (mutant pyrin)	SAA	Familial Mediterranean fever (MEFV — pyrin)
ALys / AFib / AApoA1	Lysozyme, fibrinogen, apo A1	Hereditary renal amyloidosis

Causes of reactive systemic (AA) amyloidosis :

1. Tuberculosis (most common in India), bronchiectasis, chronic osteomyelitis, lepromatous leprosy
2. Rheumatoid arthritis (most common in West), ankylosing spondylitis
3. Inflammatory bowel disease (Crohn > UC)
4. Skin popping (s.c. heroin abuse)
5. Non-immunocyte tumours : renal cell carcinoma, Hodgkin lymphoma
6. Familial Mediterranean fever, other auto-inflammatory syndromes

Pathogenesis :



Exam pearl

- Primary AL amyloid : most common form in developed countries; occurs in 5-15% of multiple myeloma.
- Bence Jones protein (free light chains) in urine/serum in AL amyloidosis — even without overt myeloma ("monoclonal gammopathy").
- AL amyloidosis = Congo red + ; AA = Congo red loses affinity after potassium permanganate ($KMnO_4$) treatment; AL is permanganate-resistant.

Morphology (organ involvement) :

Organ	Features
Kidney (most common, most serious)	Enlarged, pale, waxy; deposits in glomeruli (mesangium, subendothelial) → nephrotic syndrome; later shrunken. Peritubular and vessel walls.
Spleen	Sago spleen (deposits in splenic follicles — tapioca-like granules); Lardaceous spleen (deposits in red pulp / sinusoid walls — map-like).
Liver	Hepatomegaly; deposits in space of Disse → pressure atrophy of hepatocytes; liver function usually preserved.
Heart	Restrictive cardiomyopathy, conduction disturbances, low voltage ECG with thick walls; "sparkling / granular" echo; most common cause of death in AL amyloid. Subendocardial deposits.

Organ	Features
Others	Macroglossia (AL), carpal tunnel syndrome, adrenal insufficiency, GI malabsorption, periorbital purpura ("raccoon eyes", AL), pinch purpura

Diagnosis — biopsy sites :

1. Abdominal fat pad aspiration — preferred screening (simple, safe)
2. Rectal biopsy (submucosal vessels) — 75% positive
3. Gingival / labial salivary gland biopsy
4. Renal biopsy — most sensitive in renal disease
5. Bone marrow biopsy (AL)
6. Serum / urine immunofixation and free light chain assay; scintigraphy (SAP scan, Tc-PYP for ATTR cardiac)

Filled : Rank boosters

- Localized amyloidosis : nodular lung, larynx, skin, bladder, tongue; endocrine amyloid (medullary Ca thyroid, islet in T2DM, pituitary prolactinoma).
- Senile cardiac amyloidosis (wild-type ATTR) — men > 70 y; ATTR V122I mutation in 4% African Americans.
- Drug for ATTR : Tafamidis (TTR stabilizer), Patisiran (siRNA), Inotersen.
- β2-microglobulin is not removed by cuprophane dialysis membranes.
- Amyloid in Alzheimer : Aβ42 — neuritic plaques; APP gene on chromosome 21 → early Alzheimer in Down syndrome.
- Hereditary : Familial Mediterranean fever — autosomal recessive, MEFV (pyrin), AA amyloid; Muckle-Wells (cryopyrin) — AA.
- Most common cause of death in systemic amyloidosis overall : renal failure (AA) / cardiac (AL).

INFLAMMATION

Overview

- Protective response involving host cells, blood vessels, proteins and mediators, intended to eliminate the initial cause and the necrotic cells, and initiate repair.
- Cardinal signs : Rubor, Calor, Tumor, Dolor (Celsus) + Functio laesa (loss of function — Virchow).

Feature	Acute	Chronic
Onset	Fast : minutes-hours	Slow : days
Cellular infiltrate	Mainly neutrophils	Monocytes / macrophages, lymphocytes, plasma cells
Tissue injury, fibrosis	Usually mild, self-limited	Often severe and progressive
Local & systemic signs	Prominent	Less
Hallmark	Exudation (edema)	Tissue destruction + repair (angiogenesis, fibrosis) simultaneously

Stimuli for acute inflammation :

1. Infections (bacterial, viral, fungal, parasitic) and microbial toxins
2. Tissue necrosis (ischemia, trauma, burns)
3. Foreign bodies (splinters, sutures, urate / cholesterol crystals)
4. Immune reactions (hypersensitivity, autoimmunity)

Recognition of microbes and damaged cells :

Receptor	Ligand / function
Toll-like receptors (TLRs)	PAMPs; TLR4 → LPS (with CD14), TLR2 → peptidoglycan/lipoteichoic acid, TLR5 → flagellin, TLR3 → dsRNA, TLR7/8 → ssRNA, TLR9 → CpG DNA
NOD-like receptors / inflammasome (NLRP3)	DAMPs — urate crystals (gout), cholesterol, ATP, K^+ efflux → caspase-1 → IL-1β
C-type lectin receptors	Fungal glycans (Dectin-1)
RIG-like receptors	Viral RNA (cytosol)

Filled

- Gain-of-function NLRP3 (cryopyrin) mutations → cryopyrin-associated periodic syndromes (Muckle-Wells, FCAS) → treated with anakinra (IL-1R antagonist).
- TLRs : all are on cell surface except TLR3, 7, 8, 9 (endosomal — nucleic acids).

Vascular Events

1. Transient vasoconstriction (seconds) — first, neurogenic.
2. Vasodilation (histamine, NO) — erythema, warmth (earliest consistent change).
3. Increased vascular permeability → exudate → hallmark of acute inflammation.
4. Stasis (slowing of circulation, ↑ viscosity) → margination of leukocytes.

Mechanisms of increased vascular permeability :

Mechanism	Mediators	Vessel	Timing
Endothelial cell contraction (most common)	Histamine, bradykinin, leukotrienes, substance P	Post-capillary venules	Immediate transient (15-30 min)

Mechanism	Mediators	Vessel	Timing
Endothelial retraction (cytoskeletal reorganization)	IL-1, TNF, IFN-γ	Venules	Delayed (4–6 h), prolonged (24 h+)
Direct endothelial injury	Burns, toxins	All (venules, capillaries, arterioles)	Immediate sustained
Delayed prolonged leakage	Mild thermal injury, X-rays, UV (sunburn), some bacterial toxins	Venules + capillaries	Begins 2–12 h, lasts hours–days
Leukocyte-mediated injury	ROS, proteases	Venules, pulmonary capillaries	Late
Transcytosis (VVO)	VEGF	Venules	—
Leakage from new vessels	VEGF (angiogenesis)	Regenerating vessels	Until mature

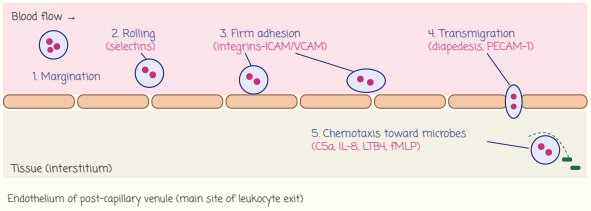
Exudate vs transudate :

Feature	Exudate	Transudate
Mechanism	↑ vascular permeability (inflammation)	↑ hydrostatic / ↓ oncotic pressure
Protein	> 3 g/dL	< 3 g/dL
Specific gravity	> 1.020	< 1.012
Cells	Many (inflammatory)	Few
Light's criteria (pleural)	Fluid/serum protein > 0.5; fluid/serum LDH > 0.6; fluid LDH > 2/3 upper limit of normal serum LDH	Not met

Exam pearl

- Lymphatics proliferate in inflammation; lymphangitis (red streaks) + lymphadenitis — e.g., streptococcal cellulitis.
- Edema due to inflammation = exudate; pus = purulent exudate rich in neutrophils, dead cells, microbes.

Cellular Events — Leukocyte Recruitment



Steps of leukocyte extravasation (labelled)

Adhesion molecules :

Endothelial molecule	Leukocyte ligand	Role
P-selectin (CD62P) — from Weibel-Palade bodies (histamine, thrombin)	Sialyl-Lewis X (PSGL-1)	Rolling
E-selectin (CD62E) — induced by IL-1, TNF	Sialyl-Lewis X	Rolling, adhesion
GlyCam-1, CD34	L-selectin (CD62L)	Rolling (homing of lymphocytes to HEV)
ICAM-1 (Ig family)	LFA-1 (CD11a/CD18), Mac-1 (CD11b/CD18) — β2 integrins	Firm adhesion, arrest, transmigration

Endothelial molecule	Leukocyte ligand	Role
VCAM-1	VLA-4 (β1 integrin)	Adhesion (eosinophils, monocytes, lymphocytes)
PECAM-1 (CD31)	PECAM-1	Transmigration (diapedesis)

- Integrin affinity ↑ by chemokines (IL-8) displayed on endothelium ("inside-out" signalling).
- Leukocytes pierce basement membrane by collagenases.
- Neutrophils dominate first 6-24 h, replaced by monocytes in 24-48 h (exception : Pseudomonas — neutrophils persist; viral — lymphocytes first; allergy — eosinophils).

Leukocyte adhesion deficiency (LAD) :

Type	Defect	Features
LAD-1	β2 integrin (CD18) — ITGB2	Delayed separation of umbilical cord (> 30 days), recurrent bacterial infections without pus, neutrophilia, periodontitis
LAD-2	Sialyl-Lewis X (fucosyl transferase — fucose metabolism)	Milder infections, Bombay blood group, mental retardation, short stature
LAD-3	Kindlin-3 (integrin activation)	LAD-1 like + bleeding (Glanzmann-like)

Chemotaxis :

- Exogenous : N-formyl methionine peptides (bacterial), lipids.
- Endogenous : C5a, LTB4, IL-8 (CXCL8), kallikrein, platelet activating factor.
- Receptors : 7-transmembrane G-protein coupled; movement via actin polymerization at leading edge (filopodia).

Phagocytosis & killing :

Recognition & attachment (opsonins : IgG Fc, C3b, collectins/MBL) → Engulfment (phagosome) → Fusion with lysosome (phagolysosome) → Killing & degradation

Killing mechanism	Details
Oxygen-dependent (respiratory burst)	NADPH oxidase (phagocyte oxidase) : $O_2 \rightarrow O_2^- \rightarrow H_2O_2$; Myeloperoxidase (MPO) : $H_2O_2 + Cl^- \rightarrow HOCl$ (hypochlorous acid / bleach) — most potent bactericidal system
Nitric oxide	iNOS (macrophages) → NO + O_2^- → peroxynitrite
Oxygen-independent	Lysozyme, lactoferrin, major basic protein (eosinophils), defensins, BPI, elastase, cathelicidins
NETs	Extracellular chromatin + granule proteins (PAD4 → citrullinated histones); anti-NET antibodies in SLE

Defects in leukocyte function :

Disease	Defect	Key features
Chronic granulomatous disease (CGD)	NADPH oxidase (gp91phox — X-linked, CYBB, 70%; autosomal recessive p47phox)	Infections with catalase-positive organisms (S. aureus, Aspergillus, Nocardia, Serratia, Burkholderia); granulomas. Test : NBT test (no blue), dihydrorhodamine (DHR) flow cytometry (preferred)
Chédiak-Higashi syndrome	LYST — lysosomal trafficking (AR)	Giant granules in neutrophils, partial albinism, peripheral neuropathy, bleeding, NK defect

Disease	Defect	Key features
Myeloperoxidase deficiency	MPO	Most common inherited phagocyte defect; mostly asymptomatic; Candida infections in diabetics; NBT normal
LAD 1/2	Integrins / selectin ligands	See above
Hyper-IgE (Job) syndrome	STAT3 (AD)	Cold staphylococcal abscesses, eczema, ↑ IgE, retained primary teeth, coarse facies, ↓ Th17

Mnemonic : Catalase-positive organisms in CGD — "Cats Need PLACESS"

Nocardia, Pseudomonas, Listeria, Aspergillus, Candida, E. coli, Staph aureus, Serratia (+ Burkholderia cepacia).

Chemical Mediators

Mediator	Source	Principal actions
Histamine	Mast cells, basophils, platelets	Vasodilation, ↑ permeability (immediate), endothelial activation
Serotonin (5-HT)	Platelets, enterochromaffin cells	Vasoconstriction, ↑ permeability
PGI_2 (prostacyclin)	Endothelium	Vasodilation, inhibits platelet aggregation
TXA_2	Platelets	Vasoconstriction, platelet aggregation
PGE_2	Mast cells, leukocytes	Pain, fever, vasodilation
PGD_2	Mast cells	Vasodilation, ↑ permeability, chemotaxis
LTB_4	Leukocytes	Chemotaxis, leukocyte activation
LTC_4, D_4, E_4 (SRS-A)	Leukocytes, mast cells	Bronchospasm, ↑ permeability, vasoconstriction
Lipoxins (LXA_4, B_4)	Leukocytes + platelets (transcellular)	Anti-inflammatory — inhibit neutrophil recruitment
PAF	Leukocytes, mast cells	Platelet aggregation, bronchospasm, ↑ permeability, vasodilation (low dose)
TNF, IL-1	Macrophages, endothelium, mast cells	Endothelial activation, fever, acute phase response, cachexia (TNF)
IL-6	Macrophages	Acute phase proteins (liver)
Chemokines (IL-8)	Leukocytes, macrophages	Chemotaxis
Bradykinin	Plasma (kininogen)	↑ Permeability, pain, vasodilation, bronchoconstriction
C3a, C5a	Plasma (complement)	Anaphylatoxins → mast cell histamine release; C5a : chemotaxis
C3b	Complement	Opsonin
NO	Endothelium (eNOS), macrophages (iNOS)	Vasodilation, microbial killing, ↓ platelet adhesion

Arachidonic acid pathway :

Membrane phospholipids

↓

↓ Phospholipase A2 (inhibited by corticosteroids via lipocortin / annexin A1)

↓

Arachidonic acid

↓

Cyclooxygenase (COX-1, COX-2) — blocked by NSAIDs, aspirin (irreversible)

5-Lipoxygenase — blocked by zileuton

↓

PGG_2 → PGH_2 → PGI_2, TXA_2, PGD_2, PGE_2, $PGF_2\alpha$

5-HPETE → LTA_4 → LTB_4 ; LTC_4 → D_4 → E_4 (receptor blocked by montelukast) ; lipoxins (12-LO)

Filled

- Aspirin acetylates COX → also triggers aspirin-triggered lipoxins (15-epi-LX) — anti-inflammatory.
- Resolvins, protectins (from omega-3 fatty acids) → resolution of inflammation.
- Aspirin-exacerbated respiratory disease (Samter triad) : asthma + nasal polyps + aspirin sensitivity — shunting to LT pathway.
- Most important mediators of fever : IL-1, TNF, IL-6 → COX in hypothalamus → PGE_2 (reset thermostat).

Plasma protease systems :

Hageman factor (factor XII) activated by collagen, basement membrane, activated platelets

↓

Kinin system : prekallikrein → kallikrein → bradykinin (degraded by ACE / kininase)

Clotting system : thrombin → fibrinopeptides

Fibrinolytic : plasmin → fibrin split products, C3 cleavage

↓

Complement activation (plasmin, kallikrein)

- ACE inhibitors → ↑ bradykinin → cough, angioedema.
- C1-inhibitor deficiency → hereditary angioedema (↑ bradykinin; ↓ C4).

Complement system :

Pathway	Trigger	Key steps
Classical	Antigen-antibody (IgM > IgG3 > IgG1), C1q	C1 → C4, C2 → C3 convertase (C4b2a)
Alternative	Microbial surfaces (LPS), spontaneous C3 hydrolysis	C3b + factor B, factor D → C3bBb (stabilized by properdin)
Lectin	Mannose-binding lectin on microbes	MASP → C4, C2

- Common : C3 convertase → C5 convertase → C5b-C9 (MAC) → lysis.
- Regulators : C1 inhibitor, DAF (CD55) and CD59 (MIRL) (↓ in PNH), factor H, factor I.

Deficiency	Result
C1q, C2, C4 (early classical)	SLE-like immune complex disease (C2 = most common deficiency)
C3	Severe pyogenic infections, immune complex GN
C5-C9 (MAC)	Recurrent Neisseria infections
C1 inhibitor	Hereditary angioedema
DAF / CD59	PNH

Deficiency	Result
Factor H / I	Atypical HUS, C3 glomerulopathy

Morphologic Patterns of Acute Inflammation

Pattern	Features	Examples
Serous	Thin, cell-poor fluid	Skin blister (burn), serous effusions, viral
Fibrinous	Fibrin deposition in body cavities	Bread-and-butter pericarditis (rheumatic, uremic), lobar pneumonia, fibrinous pleuritis
Suppurative / purulent	Pus — neutrophils, necrotic debris	Abscess (S. aureus), acute appendicitis
Ulcer	Local defect / excavation of surface	Peptic ulcer, aphthous ulcer
Pseudomembranous	Necrotic epithelium + fibrin + exudate membrane	C. difficile colitis, diphtheria
Catarrhal	Mucus hypersecretion	Common cold, allergic rhinitis
Hemorrhagic	Exudate with RBCs	Plague, anthrax, meningococcal septicemia, influenza pneumonia
Membranous (croupous)	Fibrin + necrotic cells on surface	Diphtheria (also pseudomembranous)
Cellulitis	Diffuse spread through tissue planes	Streptococcus pyogenes (hyaluronidase, streptokinase)

Exam pearl

- Abscess : central necrosis + neutrophils, surrounded by zone of preserved neutrophils→ vascular dilatation→ fibroblast proliferation (pyogenic membrane).
- Outcome of acute inflammation : resolution, chronic inflammation, fibrosis (scarring), abscess.

Chronic Inflammation

Causes :

1. Persistent infections — TB, Treponema, fungi, viruses, parasites (delayed-type hypersensitivity)
2. Hypersensitivity diseases — autoimmune (RA, IBD), allergic (asthma)
3. Prolonged exposure to toxic agents — exogenous (silica), endogenous (cholesterol in atherosclerosis)
4. Some diseases not conventionally inflammatory — Alzheimer, metabolic syndrome, cancer

Cells of chronic inflammation :

Cell	Key points
Macrophage (dominant cell)	Monocyte → tissue; half-life of monocyte ~1 day, macrophage months-years. M1 (classical) : IFN-γ, LPS → microbicidal (NO, ROS). M2 (alternative) : IL-4, IL-13 → repair, fibrosis (TGF-β), anti-inflammatory (IL-10)
Lymphocytes	Th1 (IFN-γ)→ macrophage activation; Th2 (IL-4, 5, 13) → eosinophils; Th17 (IL-17) → neutrophils
Plasma cells	Antibody production; Russell bodies; "cartwheel / clock-face" chromatin
Eosinophils	Eotaxin (CCL11) chemotaxis; major basic protein — toxic to parasites, epithelium
Mast cells	FcεRI (IgE); histamine — allergy, anaphylaxis
Neutrophils	Chronic osteomyelitis, chronic lung damage (smoking) — "acute on chronic"

Tissue macrophage names :

Tissue	Name
Liver	Kupffer cells
Spleen, lymph nodes	Sinus histiocytes
Lung	Alveolar macrophages (dust cells)
CNS	Microglia
Bone	Osteoclasts
Skin	Langerhans cells (dendritic; Birbeck granules, CD1a, S100, langerin CD207)
Connective tissue	Histiocytes
Kidney	Mesangial cells
Placenta	Hofbauer cells

Granulomatous Inflammation

- Granuloma : focus of chronic inflammation consisting of aggregates of activated macrophages (epithelioid cells) ± giant cells, rim of lymphocytes ± plasma cells, ± central necrosis.
- Epithelioid cells : activated macrophages with abundant pink cytoplasm, slipper/shoe-sole shaped nuclei.
- Key mediators : IFN-γ (Th1) converts macrophages to epithelioid cells; TNF-α maintains granuloma (anti-TNF therapy → TB reactivation).

Types of giant cells :

Giant cell	Features	Seen in
Langhans giant cell	Nuclei in horseshoe / peripheral arrangement	TB, sarcoidosis
Foreign body giant cell	Nuclei scattered haphazardly	Sutures, talc, foreign bodies
Touton giant cell	Ring of nuclei + foamy (lipid) cytoplasm outside ring	Xanthoma, xanthogranuloma, fat necrosis
Reed-Sternberg cell	Owl-eye nucleoli, bilobed nucleus	Hodgkin lymphoma
Aschoff giant cell	Multinucleate Anitschkow cells	Rheumatic heart disease
Warthin-Finkeldey cell	Grape-like nuclei	Measles, lymphoid tissue
Tumour giant cells	Pleomorphic, hyperchromatic	Malignant tumours
Osteoclast-like giant cells	Many nuclei, uniform	Giant cell tumour of bone

Types of granuloma & examples :

Category	Examples
Caseating	Tuberculosis, histoplasmosis, coccidioidomycosis
Non-caseating	Sarcoidosis, Crohn disease, berylliosis, leprosy (tuberculoid), foreign body
Suppurative (stellate)	Cat-scratch disease (Bartonella henselae), LGV (Chlamydia L1-L3), tularemia, yersinia, melioidosis
Foreign body	Sutures, talc (IV drug use), silica
Immune granuloma	TB, leprosy, syphilis, sarcoid, schistosomiasis
Necrotizing granulomatous vasculitis	Granulomatosis with polyangiitis (Wegener)

Category	Examples
Palisading granuloma	Rheumatoid nodule, granuloma annulare, necrobiosis lipoidica

Mnemonic : Causes of granulomas — "TB SCHLEP"

Tuberculosis · Berylliosis, Brucella · Sarcoidosis, Syphilis, Schistosoma · Crohn, Cat-scratch · Histoplasma & other fungi · Leprosy, LGV · EGPA & GPA (vasculitis) · Parasites, foreign Particles (talc, sutures, silica).

Filled : Rank boosters — granuloma

- Granulomatous mastitis, orchitis — idiopathic types.
- Hepatic granulomas : TB, sarcoidosis, drugs (allopurinol, quinidine), Q fever (fibrin-ring / doughnut granuloma), PBC.
- Fibrin-ring (doughnut) granuloma : Q fever (Coxiella), also Hodgkin, CMV, EBV, allopurinol.
- Chronic granulomatous disease — granulomas due to failure of killing, not a "granuloma disease" in the classical sense.
- Sarcoid granulomas : asteroid bodies, Schaumann bodies, Hamazaki-Wesenberg bodies; ACE ↑, hypercalcemia (1α-hydroxylase in macrophages).

Systemic Effects of Inflammation (Acute Phase Response)

1. Fever — pyrogens (LPS → IL-1, TNF → PGE_2 in hypothalamus)
2. Acute phase proteins (liver, IL-6) :
   - C-reactive protein (CRP) — opsonin, activates complement (pentraxin)
   - Fibrinogen → ↑ ESR (rouleaux formation)
   - Serum amyloid A (SAA) — AA amyloid
   - Hepcidin — ↓ iron absorption/release → anemia of chronic disease
   - Ferritin, haptoglobin, ceruloplasmin, α1-antitrypsin, complement
   - Negative acute phase reactants : albumin, transferrin, transthyretin
3. Leukocytosis (↑ release from marrow — IL-1, TNF; later CSFs) :
   - Neutrophilia — bacterial infection
   - Lymphocytosis — viral (mumps, rubella), pertussis, TB
   - Eosinophilia — asthma, allergy, parasites
   - Leukopenia — typhoid, some viruses, rickettsiae, protozoa
   - Leukemoid reaction : > 40,000-100,000/μL
4. ↑ heart rate, BP; ↓ sweating; rigors; anorexia; somnolence
5. Septic shock — TNF, IL-1 → DIC, hypotension, metabolic disturbances

Exam pearl

- ESR is ↑ by fibrinogen (most), globulins; ↓ by sickle cells, polycythemia, spherocytes, hypofibrinogenemia, CHF.
- Procalcitonin ↑ specifically in bacterial sepsis (not viral).
- hs-CRP : cardiovascular risk marker.

TISSUE REPAIR & HEALING

Overview

- Repair = restoration of tissue architecture and function after injury.
- Two processes : Regeneration (replacement by same cells) and Scar formation (fibrosis) (connective tissue deposition).

Cell cycle & proliferative capacity :

Tissue type	Cell cycle	Examples
Labile (continuously dividing)	Continuously cycling	Surface epithelia (skin, GIT, urinary), bone marrow / hematopoietic cells, endometrium
Stable (quiescent)	G0, enter G1 when stimulated	Liver, kidney, pancreas, endothelium, fibroblasts, smooth muscle
Permanent	Terminally differentiated, non-dividing	Neurons, cardiac muscle, (skeletal muscle – limited via satellite cells)

Cell cycle checkpoints

G1 – restriction point (Rb) : most important, cyclin D-CDK4/6 phosphorylates Rb → releases E2F

↓

G1/S checkpoint – p53 → p21 halts damaged cells (DNA repair or apoptosis)

↓

S (DNA synthesis) → G2/M checkpoint (DNA damage)

↓

M (mitosis) – spindle assembly checkpoint

Stem cells :

- Embryonic stem cells – pluripotent; induced pluripotent stem cells (iPSC) created by Oct3/4, Sox2, KLF4, c-MYC (Yamanaka factors).
- Adult (tissue) stem cell niches :

Tissue	Niche
Skin	Bulge of hair follicle, basal layer
Intestine	Crypt base (above Paneth cells; Lgr5+)
Liver	Canals of Hering (oval cells)
Cornea	Limbus
Brain	Subventricular zone, dentate gyrus
Muscle	Satellite cells
Bone marrow	Hematopoietic (CD34+) and mesenchymal stem cells

Growth factors :

Factor	Main actions
EGF / TGF-α	Epithelial and fibroblast mitogen; re-epithelialization, granulation tissue
HGF (scatter factor)	Hepatocyte, epithelial proliferation; cell motility (c-MET receptor)

Factor	Main actions
VEGF	Angiogenesis (most important), ↑ permeability; induced by hypoxia (HIF-1α)
PDGF	Chemotactic for neutrophils, macrophages, fibroblasts, SMC; fibroblast proliferation
FGF (basic, FGF-2)	Angiogenesis, wound contraction, ECM synthesis, migration
TGF-β	Most important fibrogenic agent; ↑ collagen, fibronectin; ↓ MMP; anti-inflammatory
KGF (FGF-7)	Keratinocyte migration, proliferation
Angiopoietin 1/2	Vessel maturation and stabilization (Tie2)

Extracellular Matrix (ECM)

Component	Key facts
Collagen	Most abundant protein. Type I — skin, bone, tendon (most common); type II — cartilage; type III — granulation tissue, blood vessels (reticulin); type IV — basement membrane; type VII — anchoring fibrils
Elastin	Fibrillin-1 scaffold (Marfan)
Fibronectin	Adhesion of cells to ECM; granulation tissue, provisional matrix
Laminin	Most abundant glycoprotein in basement membrane
Proteoglycans	Hyaluronan (cushioning), heparan sulfate
Integrins	Cell-ECM attachment, signalling

Mnemonic : Collagen types — "SCAB"

Type I : Skin, bone, tendon (bone) · Type II : Cartilage (cartwolage) · Type III : Arteries, granulation ("three", reticulin) · Type IV : Basement membrane ("under the floor").

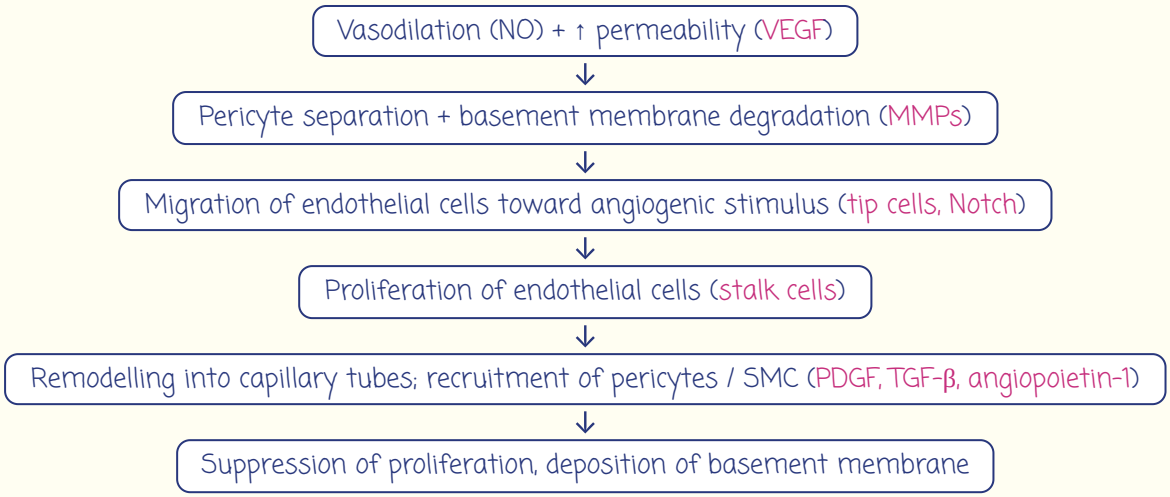
Filled : Collagen synthesis — disorders

Step	Location	Defect
Hydroxylation of proline/lysine	RER	Scurvy (vitamin C)
Glycosylation, triple helix (procollagen)	RER	Osteogenesis imperfecta (COL1A1/A2)
Cleavage of terminal propeptides	Outside cell	Ehlers-Danlos (dermatosparaxis type)
Cross-linking by lysyl oxidase (Cu^{2+})	Outside cell	Menkes disease (ATP7A), EDS
- Alport syndrome — type IV collagen (α5, X-linked). Vascular EDS — type III collagen (COL3A1) — arterial / uterine rupture.		

Liver Regeneration

- After partial hepatectomy (up to 60%) liver regenerates by compensatory hyperplasia of remaining hepatocytes (priming : TNF, IL-6 → G0→G1; growth factors : HGF, TGF-α → progression).
- If hepatocyte proliferation impaired (chronic injury) → regeneration from progenitor (oval) cells in canals of Hering.

Angiogenesis



Scar Formation

Granulation tissue :

- Hallmark of healing: pink, soft, granular gross appearance.
- Microscopy : proliferating fibroblasts + new thin-walled capillaries (angiogenesis) in loose ECM, with macrophages; type III collagen.
- Appears by 3-5 days.

Steps :

1. Angiogenesis
2. Formation of granulation tissue (fibroblast migration and proliferation)
3. Remodelling of connective tissue (type III → type I collagen by MMPs (zinc-dependent); inhibited by TIMPs)

> Exam pearl
> - Myofibroblasts (α-SMA +) → wound contraction (most important in secondary intention).
> - Tensile strength : 10% at 1 week; 70-80% at 3 months (maximum, never reaches 100%).
> - Fibronectin forms provisional scaffold; collagen type III first, then type I.

Cutaneous Wound Healing

Feature	Primary union (first intention)	Secondary union (second intention)
Wound	Clean, surgical, edges approximated	Large, gaping, tissue loss, infection
Inflammation	Less	More intense
Granulation tissue	Scanty	Abundant
Wound contraction	Absent / minimal	Prominent (myofibroblasts)
Scar	Thin, linear	Large, may be deforming
Infection risk	Low	High
Healing time	Short	Long

Timeline of healing by first intention :

Time	Event
24 h	Neutrophils at margins; blood clot (fibrin) + scab; basal cells begin mitosis

Time	Event
24–48 h	Epithelial cells migrate from edges, fuse in midline under scab → thin continuous layer
Day 3	Neutrophils replaced by macrophages; granulation tissue invades incision; collagen fibres vertical, do not bridge
Day 5	Neovascularization maximal; collagen fibrils bridge the incision; epidermis normal thickness
Week 2	Continued collagen accumulation, fibroblast proliferation; "blanching" (↓ vessels, ↓ edema)
End of 1 month	Cellular connective tissue without inflammatory cells, covered by intact epidermis; dermal appendages lost permanently

Factors Affecting Healing

Local :

1. Infection — most important cause of delayed healing
2. Mechanical factors — early motion, tension
3. Foreign bodies — sutures, fragments
4. Size, location and type of wound (face heals faster; poor blood supply delays)
5. Ischemia / poor perfusion (atherosclerosis, venous stasis)

Systemic :

1. Nutrition — protein deficiency, vitamin C deficiency (scurvy), zinc deficiency
2. Glucocorticoids — ↓ TGF-β, ↓ collagen, anti-inflammatory (but useful to ↓ fibrosis in corneal infections)
3. Diabetes mellitus — microangiopathy, ↓ neutrophil function, infection
4. Age, anemia, uremia, malignancy, chemo- and radiotherapy

Complications of wound healing :

Complication	Key features
Deficient scar	Wound dehiscence (abdominal surgery — vomiting, coughing, ileus), ulceration
Hypertrophic scar	Excess collagen, confined to wound margins, regresses over time
Keloid	Collagen extends beyond original wound margins, does not regress, recurs after excision; thick, glassy, hyalinized type I and III collagen bundles; African descent, earlobes, sternum
Exuberant granulation (proud flesh)	Protrudes above skin, blocks re-epithelialization — cauterize / excise
Desmoid (aggressive fibromatosis)	Locally aggressive fibroblast proliferation; β-catenin / APC (Gardner syndrome)
Contracture	Exaggerated contraction — burns, palms, soles, anterior thorax
Marjolin ulcer	Squamous cell carcinoma in chronic scar/burn

Filled

- Keloid — ↑ TGF-β; treatment : intralesional triamcinolone, silicone sheeting, radiotherapy after excision.
- Mast cells are numerous in keloids (pruritus).
- Fibrosis in parenchymal organs (cirrhosis, IPF) driven by TGF-β — "persistent repair".
- Hepatic stellate (Ito) cells → myofibroblasts → collagen in space of Disse (cirrhosis).

Healing at Specific Sites

Fracture healing :

Hematoma formation (fibrin clot) — immediate, platelets release PDGF, TGF-β, FGF

↓

Soft (procallus) callus — granulation tissue; osteoprogenitor cells; end of 1st week

↓

Hard (bony) callus / provisional callus — woven bone + hyaline cartilage; maximum at end of 2-3 weeks

↓

Remodelling — woven → lamellar bone along lines of weight-bearing (osteoclasts + osteoblasts); months

- Complications : malunion, delayed union, non-union → pseudarthrosis (cleft lined by synovial-like cells), infection (open fracture), fat embolism.
- Factors impairing : movement at fracture site, poor blood supply (scaphoid, femoral neck, talus), infection, interposition of soft tissue, nutrition.

Other sites :

Site	Healing features
Nervous system (CNS)	No regeneration of neurons; gliosis by astrocytes (gemistocytes) — not fibroblasts
Peripheral nerve	Wallerian degeneration distal; Schwann cell guidance; regrowth 1-3 mm/day; traumatic (amputation) neuroma if misdirected
Cardiac muscle	Fibrous scar (no regeneration)
Skeletal muscle	Limited regeneration from satellite cells
Liver	Regeneration (if framework intact) or fibrosis
Kidney	Tubular epithelium regenerates (ATN — if basement membrane intact); glomeruli do not
Lung	Type II pneumocytes regenerate alveolar epithelium (stem cells for type I)
Cartilage	Poor healing (avascular) — fibrocartilage
Mucosa (GIT)	Rapid regeneration from crypt stem cells
Cornea	Limbal stem cells

Exam pearl

- Healing of nerve : Bands of Büngner (Schwann cell columns guiding axons).
- Chromatolysis — swelling of neuron cell body, dispersion of Nissl substance, nucleus pushed to periphery — axonal reaction.

HEMODYNAMIC DISORDERS, THROMBOSIS & SHOCK

Edema

- Accumulation of excess fluid in interstitial tissue or body cavities.
- Generalized severe edema = anasarca. In cavities : hydrothorax, hydropericardium, hydroperitoneum (ascites).

Pathophysiologic categories :

Mechanism	Causes
↑ Hydrostatic pressure	Impaired venous return : CHF, constrictive pericarditis, ascites in cirrhosis, venous obstruction (DVT), external pressure; arteriolar dilation (heat)
↓ Plasma oncotic pressure (hypoproteinemia)	Nephrotic syndrome (most common), cirrhosis, protein malnutrition (kwashiorkor), protein-losing gastroenteropathy
Lymphatic obstruction	Inflammatory (filariasis → elephantiasis), neoplastic, post-surgical (axillary dissection → arm edema), post-irradiation; peau d'orange in breast cancer
Sodium retention	Excessive salt intake with renal insufficiency, ↑ renin-angiotensin-aldosterone (CHF), acute GN
Inflammation	Acute & chronic inflammation, angiogenesis (exudate)

Exam pearl

- Renal edema (nephrotic) : generalized, first appears in loose connective tissue — periorbital (eyelids), morning.
- Cardiac edema : dependent (legs in standing, sacrum in recumbent).
- Pulmonary edema : left ventricular failure (most common) — heavy wet lungs (2-3× normal), frothy, blood-tinged fluid on cut section.
- Brain edema : flattened gyri, narrowed sulci, compressed ventricles → herniation.
- Pitting edema — interstitial fluid; non-pitting — lymphedema, myxedema.

Hyperemia & Congestion

Feature	Hyperemia (active)	Congestion (passive)
Mechanism	Arteriolar dilation → ↑ inflow	Impaired venous outflow
Colour	Red (erythema) — oxygenated blood	Blue-red (cyanosis) — deoxygenated
Examples	Exercise, inflammation, blushing	CHF (systemic), venous obstruction (local)

Chronic venous congestion (CVC) — organs :

Organ	Cause	Morphology
Lung	Left heart failure, mitral stenosis	Acute : engorged alveolar capillaries, septal edema. Chronic : thick fibrotic septa, hemosiderin-laden macrophages ("heart failure cells") in alveoli → brown induration of lung
Liver	Right heart failure, (Budd-Chiari)	Nutmeg liver : congested red-brown centrilobular areas (zone 3) alternating with pale/fatty periportal areas. Centrilobular hemorrhagic necrosis; cardiac sclerosis / cardiac cirrhosis (long-standing)
Spleen	Right heart failure, portal hypertension	Congestive splenomegaly (up to 500-1000 g), firm; Gamna-Gandy bodies (siderofibrotic nodules — hemosiderin + calcium on collagen)
Kidney	Right heart failure	Enlarged, cyanotic; medulla congested

Filled

- Heart failure cells — demonstrated with Perls Prussian blue (hemosiderin, blue).
- Zone 3 necrosis : congestion (RHF), shock, acetaminophen, CCl_4, alcohol, halothane. Zone 1 : phosphorus poisoning, eclampsia (periportal fibrin). Zone 2 : yellow fever.

Hemorrhage

Term	Size / site
Petechiae	1-2 mm — ↑ capillary pressure, thrombocytopenia, platelet dysfunction
Purpura	≥ 3 mm (up to 1 cm) — trauma, vasculitis, vascular fragility
Ecchymoses (bruise)	> 1-2 cm subcutaneous hematomas; colour changes : red-blue (Hb) → blue-green (bilirubin) → golden-brown (hemosiderin)
Hematoma	Blood accumulation within tissue
Hemothorax, hemopericardium, hemoperitoneum, hemarthrosis	Blood in cavities

- Clinical significance : volume (> 20% rapid loss → hemorrhagic shock), site (intracranial), chronic → iron deficiency anemia.

Hemostasis — Normal

Arteriolar vasoconstriction (reflex + endothelin)

↓

Primary hemostasis : platelet adhesion (vWF-GpIb) → shape change, granule release (ADP, TXA_2) → aggregation (GpIIb/IIIa - fibrinogen) → platelet plug

↓

Secondary hemostasis : tissue factor (factor III) → coagulation cascade → thrombin → fibrin

↓

Clot stabilization & resorption : fibrin polymerization (factor XIIIa), platelet contraction; t-PA → plasmin limits clot

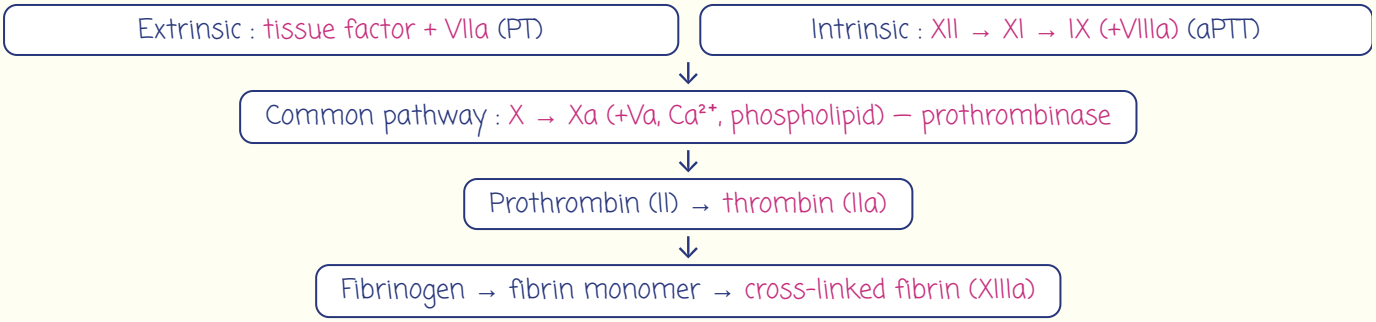
Endothelium — anti- vs pro-thrombotic :

Antithrombotic (normal)	Prothrombotic (injured / activated)
PGI_2, NO (inhibit platelets); ADPase (CD39)	vWF (platelet adhesion)
Thrombomodulin → thrombin activates protein C	Tissue factor (induced by TNF, IL-1, LPS)
Heparin-like molecules → ↑ antithrombin III	PAI-1 (↓ fibrinolysis)
TFPI (tissue factor pathway inhibitor)	
t-PA (fibrinolysis)	

Platelets :

- α-granules : fibrinogen, fibronectin, factor V, vWF, PF4, PDGF, TGF-β, P-selectin.
- Dense (δ) granules : ADP, ATP, Ca^{2+}, serotonin, histamine, epinephrine.
- Disorders : Bernard-Soulier (GpIb) — giant platelets, no ristocetin aggregation; Glanzmann thrombasthenia (GpIIb/IIIa) — no aggregation with ADP, normal ristocetin.

Coagulation cascade :



- Vitamin K-dependent factors : II, VII, IX, X, protein C, protein S (γ-carboxylation of glutamate). Shortest half-life : factor VII (~6 h) → PT prolongs first with warfarin.
- Thrombin : most important — converts fibrinogen, activates V, VIII, XI, XIII, platelets (PAR receptors); with thrombomodulin activates protein C.
- Natural anticoagulants : antithrombin III (inhibits IIa, IXa, Xa, XIa, XIIa; ↑ by heparin), protein C + S (inactivate Va and VIIIa), TFPI.

Thrombosis

Virchow's triad :

1. Endothelial injury — most important; dominant in heart and arterial circulation (MI, ulcerated plaque, vasculitis, hypertension, toxins of smoking)
2. Abnormal blood flow (stasis or turbulence) — stasis : venous thrombi; turbulence : arterial and cardiac thrombi
3. Hypercoagulability (thrombophilia)

Hypercoagulable states :

Primary (genetic) :

1. Factor V Leiden (Arg506Gln; resistance to activated protein C) — most common inherited cause
2. Prothrombin G20210A mutation (↑ prothrombin) — second most common
3. Deficiency of antithrombin III (heparin resistance), protein C, protein S
4. ↑ factor VIII, IX, XI, fibrinogen
5. Hyperhomocysteinemia (MTHFR — also acquired : folate / B12 / B6 deficiency)

Secondary (acquired) :

1. Prolonged bed rest / immobilization, surgery, fractures, burns
2. Malignancy (Trousseau syndrome — migratory thrombophlebitis; mucin-secreting adenocarcinomas — pancreas)
3. Antiphospholipid antibody syndrome
4. Heparin-induced thrombocytopenia (HIT) — antibodies to heparin-PF4 complex, day 5–14
5. Pregnancy & postpartum, oral contraceptives, estrogen therapy
6. Nephrotic syndrome (loss of antithrombin III)
7. Myeloproliferative neoplasms (polycythemia vera, JAK2), PNH, sickle cell
8. Smoking, obesity, cardiomyopathy, atrial fibrillation, prosthetic valves, DIC

Filled : Antiphospholipid syndrome

- Antibodies : lupus anticoagulant, anticardiolipin, anti-β2-glycoprotein I.
- In vitro : prolonged aPTT (paradoxically) ; in vivo : thrombosis.
- Features : recurrent arterial/venous thrombosis, recurrent miscarriages, thrombocytopenia, false-positive VDRL, livedo reticularis.
- Catastrophic APS — multiorgan microvascular thrombosis.
- Treatment : lifelong warfarin (not DOACs in triple-positive).

Morphology of thrombi :

- Lines of Zahn — alternating pale layers of platelets + fibrin and dark layers of RBCs — indicate thrombosis in flowing blood (antemortem).

Feature	Antemortem thrombus	Postmortem clot
Lines of Zahn	Present	Absent
Attachment	Firmly attached to vessel wall	Not attached; gelatinous
Appearance	Dry, granular, friable, grey-red	Rubbery, dark red "currant jelly" dependent portion + yellow "chicken fat" supernatant
Shape	Shape of vessel ± tail	Cast of vessel

Type	Features
Arterial / cardiac (mural)	Endothelial injury/turbulence; begin at injury; grow retrograde; white / pale (platelet-rich); occlusive in coronary, cerebral, femoral arteries
Venous (phlebothrombosis)	Stasis; grow in direction of flow (toward heart); red / stasis thrombi (RBC-rich); deep veins of leg (90%)
Vegetations	Thrombi on heart valves — infective endocarditis, NBTE (marantic), Libman-Sacks (SLE)

Fate of thrombus :

1. Propagation — accumulates more platelets, fibrin → obstruction
2. Embolization — dislodges, travels
3. Dissolution — fibrinolysis (recent thrombi)
4. Organization and recanalization — ingrowth of endothelial cells, SMC, fibroblasts; new capillary channels
5. Septic infection → mycotic aneurysm

> Exam pearl
> - Most common site of DVT : deep veins of calf; most significant (embolism) : popliteal, femoral, iliac veins (above knee).
> - Superficial thrombophlebitis (saphenous varicosities) rarely embolize.
> - Phlegmasia alba dolens (milk leg) — iliofemoral thrombosis in pregnancy.

Disseminated Intravascular Coagulation (DIC)

- Consumption coagulopathy : widespread microvascular thrombi → consumption of platelets and factors → bleeding.

Causes :

1. Obstetric complications — abruptio placentae, retained dead fetus, amniotic fluid embolism, septic abortion, eclampsia
2. Infections — Gram-negative sepsis (most common), meningococcemia, malaria, Rocky Mountain spotted fever
3. Neoplasms — acute promyelocytic leukemia (M3), adenocarcinomas (pancreas, prostate, lung)
4. Massive tissue injury — trauma, burns, extensive surgery
5. Others — snake bite, acute intravascular hemolysis, shock, giant hemangioma (Kasabach-Merritt), heat stroke

Release of tissue factor (placenta, tumour, APL granules) | Endothelial injury (sepsis, LPS, TNF)

↓

Widespread thrombin generation → microthrombi (fibrin) in small vessels

↓

Consumption of platelets, fibrinogen, V, VIII → bleeding | Microangiopathic hemolytic anemia (schistocytes), ischemic organ damage

↓

Secondary fibrinolysis → ↑ D-dimer, FDPs (most specific : D-dimer)

- Lab : ↓ platelets, ↑ PT and aPTT, ↓ fibrinogen, ↑ D-dimer, schistocytes.
- Organs : kidney (cortical necrosis), adrenal (Waterhouse-Friderichsen in meningococcemia), pituitary (Sheehan), brain, lungs.

Embolism

- Detached intravascular solid, liquid or gaseous mass carried by blood to a distant site. 99% from thrombi (thromboembolism).

Pulmonary thromboembolism :

1. Source : 95% from deep leg veins (above knee)
2. 60-80% clinically silent (small)
3. Occluding > 60% of pulmonary circulation → sudden death, acute right heart failure (cor pulmonale), cardiovascular collapse
4. Saddle embolus — at bifurcation of main pulmonary artery
5. Small-medium artery obstruction → hemorrhage (bronchial circulation intact); infarction only when circulation compromised (LVF)
6. Paradoxical embolism — venous embolus via ASD / PFO into systemic circulation
7. Multiple emboli over time → pulmonary hypertension

Systemic thromboembolism :

- 80% from intracardiac mural thrombi (2/3 due to LV wall infarcts, 1/4 left atrial dilation + atrial fibrillation).
- Sites : lower extremities (75%), brain (10%), intestines, kidneys, spleen.

Other emboli :

Type	Key features
Fat & marrow embolism	Long bone fractures (tibia, femur), burns, CPR; syndrome after 1-3 days : pulmonary insufficiency, neurologic symptoms, anemia, thrombocytopenia, petechial rash (upper body, axilla); Oil Red O / Sudan stain on frozen section; fatal in ~10%
Air embolism	> 100 cc needed for clinical effect; obstetric procedures, chest wall injury, IV lines. Decompression sickness (caisson disease) — nitrogen bubbles : bends (joints), chokes (lungs); chronic : aseptic necrosis of femoral head, tibia, humerus
Amniotic fluid embolism	Mortality ~80%; sudden dyspnea, cyanosis, shock, seizures, coma, DIC; fetal squamous cells, lanugo hair, fat (vernix), mucin in maternal pulmonary microcirculation
Tumour, bone marrow, atheromatous (cholesterol), foreign body (talc), septic	Atheroembolism : cholesterol clefts, after angiography; livedo reticularis, renal failure, eosinophilia

Infarction

- Area of ischemic necrosis caused by occlusion of either arterial supply or venous drainage. 99% due to thrombotic or embolic events.

Feature	Red (hemorrhagic) infarct	White (anemic / pale) infarct
Circulation	Venous occlusion; dual blood supply; loose tissues; previously congested tissue; reperfusion	Arterial occlusion in solid organs with end-arterial circulation
Examples	Lung, small intestine, ovarian torsion, testicular torsion, liver (dual), brain (embolic, reperfusion)	Heart, spleen, kidney
Shape	Wedge-shaped with occluded vessel at apex, periphery as base	
Histology	Coagulative necrosis (except brain : liquefactive)	Coagulative necrosis

Factors affecting development of infarct :

1. Anatomy of vascular supply — dual (lung, liver) or parallel (radial + ulnar artery) supply protects; end-arteries (kidney, spleen) do not
2. Rate of occlusion — slow occlusion allows collaterals (coronary)
3. Tissue vulnerability to hypoxia — neurons (3-4 min) > myocardium (20-30 min) > fibroblasts
4. Oxygen content of blood — anemia, cyanosis, CHF worsen

- Septic infarction → abscess (infected emboli from IE).
- Earliest histologic change in infarct : none in 4-12 h; acute inflammation at margins within hours to 1-2 days.

Gangrene

Feature	Dry gangrene	Wet gangrene	Gas gangrene
Mechanism	Arterial occlusion, slow	Venous + arterial occlusion, infection (putrefaction)	Clostridium perfringens (α-toxin, lecithinase)
Site	Limbs (toes)	Bowel (strangulated hernia, volvulus), diabetic foot, lung	Contaminated deep wounds, muscle (myonecrosis)
Line of demarcation	Clear	Poorly defined	Absent
Appearance	Dry, shrunken, black (iron sulfide from Hb)	Moist, swollen, foul-smelling	Crepitus, gas bubbles
Spread	Slow	Rapid	Very rapid
Prognosis	Good	Poor (septicemia)	Poor

- Examples : Fournier gangrene (scrotum/perineum — synergistic), noma (cancrum oris) (malnourished children), Meleney's synergistic gangrene.

Shock

- State of cellular hypoperfusion due to ↓ cardiac output or ↓ effective circulating blood volume → impaired tissue perfusion and hypoxia.

Type	Mechanism	Examples
Cardiogenic	Myocardial pump failure	MI, arrhythmias, cardiac tamponade, pulmonary embolism (obstructive)
Hypovolemic	Inadequate blood or plasma volume	Hemorrhage, burns, vomiting, diarrhea
Septic (distributive)	Peripheral vasodilation + pooling, endothelial activation/injury, DIC	Gram-negative > Gram-positive sepsis, fungi

Type	Mechanism	Examples
Neurogenic	Loss of vascular tone	Spinal cord injury, anesthetic accident
Anaphylactic	IgE-mediated systemic vasodilation, ↑ permeability	Drugs, insect stings, foods

Pathogenesis of septic shock :

1. Inflammatory mediators — PAMPs (LPS) → TLRs → TNF, IL-1, IL-6, IFN-γ, HMGB1; complement C3a, C5a
2. Endothelial activation and injury — capillary leak, edema, ↑ tissue factor → DIC
3. Immune suppression — CARS (compensatory anti-inflammatory response) — IL-10, lymphocyte apoptosis
4. Metabolic abnormalities — insulin resistance, hyperglycemia; relative adrenal insufficiency
5. Organ dysfunction — ARDS, AKI, liver, heart (↓ contractility)

Stages of shock :

Nonprogressive (compensated) : ↑ sympathetic (tachycardia, peripheral vasoconstriction, cool skin), RAAS, ADH — vital organ perfusion maintained

↓

Progressive : tissue hypoperfusion → anaerobic glycolysis → lactic acidosis → arteriolar dilation, blood pooling; DIC, ARDS

↓

Irreversible : lysosomal enzyme leakage, myocardial depression, ischemic bowel → bacteremia; ATN — death

Morphology of shock :

Organ	Change
Brain	Ischemic encephalopathy — watershed infarcts
Heart	Subendocardial hemorrhage, necrosis, contraction band necrosis
Kidney	Acute tubular necrosis (most common cause of oliguria)
Lung	Relatively resistant in hypovolemic shock; DAD (shock lung, ARDS) in septic shock — hyaline membranes
Adrenal	Lipid depletion of cortical cells (stress); Waterhouse-Friderichsen hemorrhage
GIT	Patchy mucosal hemorrhages, necrosis (hemorrhagic enteropathy)
Liver	Fatty change, central (zone 3) hemorrhagic necrosis

Filled : Warm vs cold shock

- Septic shock — early "warm" shock: vasodilation, warm flushed skin, ↑ cardiac output.
- Cardiogenic / hypovolemic — "cold" shock : cool clammy skin, ↓ CO, ↑ SVR.
- qSOFA : RR ≥ 22, altered mentation, SBP ≤ 100. Septic shock = vasopressors to keep MAP ≥ 65 + lactate > 2 mmol/L despite fluids.

Fluid & Electrolyte Derangements

Disorder	Key causes	Key features
Hyponatremia (< 135)	SIADH (small cell lung Ca, CNS, drugs), CHF, cirrhosis, diuretics, Addison	Cerebral edema, seizures; too-rapid correction → osmotic demyelination (central pontine myelinolysis)
Hypernatremia (> 145)	Water loss, diabetes insipidus	Brain shrinkage, hemorrhage

Disorder	Key causes	Key features
Hypokalemia	Diuretics, vomiting, hyperaldosteronism, insulin, β-agonists	U waves, flat T, muscle weakness, ileus
Hyperkalemia	Renal failure, Addison, ACEi, tissue lysis, acidosis	Peaked T waves, wide QRS, arrhythmia
Hypocalcemia	Hypoparathyroidism, CKD, vitamin D deficiency, pancreatitis	Tetany (Chvostek, Trousseau), prolonged QT
Hypercalcemia	Hyperparathyroidism, malignancy	"Stones, bones, groans, moans", short QT
Metabolic acidosis (↑ AG)	MUDPILES : methanol, uremia, DKA, paraldehyde, isoniazid/iron, lactic, ethylene glycol, salicylates	Kussmaul breathing
Metabolic alkalosis	Vomiting, diuretics, hyperaldosteronism	Hypokalemia

Mnemonic : Normal anion gap acidosis — "HARDASS"

Hyperalimentation · Acetazolamide · RTA · Diarrhea · Addison · Spironolactone · Saline infusion.

NEOPLASIA

Definitions & Nomenclature

- Neoplasia : "new growth" — disorder of cell growth triggered by acquired (or inherited) mutations affecting a single cell and its clonal progeny; growth is autonomous and persists after the stimulus is removed (Willis).
- Components : parenchyma (neoplastic cells — determine behaviour and name) + stroma (connective tissue, vessels — support). Abundant collagenous stroma = desmoplasia → scirrhous (stony hard) tumours, e.g. breast carcinoma.

Tissue of origin	Benign	Malignant
Fibrous tissue	Fibroma	Fibrosarcoma
Fat	Lipoma (most common soft tissue tumour)	Liposarcoma (most common adult soft-tissue sarcoma along with UPS)
Cartilage	Chondroma	Chondrosarcoma
Bone	Osteoma	Osteosarcoma
Smooth muscle	Leiomyoma	Leiomyosarcoma
Skeletal muscle	Rhabdomyoma	Rhabdomyosarcoma (most common sarcoma of childhood)
Blood vessels	Hemangioma	Angiosarcoma
Squamous epithelium	Squamous papilloma	Squamous cell carcinoma
Glandular epithelium	Adenoma, cystadenoma	Adenocarcinoma
Melanocytes	Nevus	Melanoma
Hematopoietic	—	Leukemia, lymphoma
Germ cells	Mature teratoma	Immature teratoma, seminoma

Misnomers (malignant despite benign-sounding names) :

1. Lymphoma 2. Melanoma 3. Mesothelioma 4. Seminoma 5. Hepatoma (HCC) 6. Astrocytoma, glioma 7. Myeloma

Tumour-like lesions :

Term	Definition	Example
Hamartoma	Disorganized mass of mature tissue indigenous to the site	Pulmonary hamartoma (cartilage — "popcorn calcification"), Peutz-Jeghers polyp
Choristoma (heterotopia)	Normal tissue at an abnormal site	Pancreatic rest in stomach wall, Meckel's gastric mucosa
Teratoma	From totipotent germ cells — tissues of more than one germ layer	Ovarian mature cystic teratoma (dermoid)
Mixed tumour	Divergent differentiation of one germ layer	Pleomorphic adenoma of salivary gland, fibroadenoma
Blastoma	Tumour of embryonic tissue	Retinoblastoma, nephroblastoma

Benign vs Malignant

Feature	Benign	Malignant
Differentiation	Well differentiated, resembles tissue of origin	Varies; anaplasia in poorly differentiated
Rate of growth	Slow, may stop	Erratic, rapid; ↑ mitoses
Local invasion	Encapsulated, expansile, cohesive	Locally invasive, infiltrative, no capsule
Metastasis	Absent	Frequently present — the single most definitive feature

Anaplasia (lack of differentiation) — features :

1. Pleomorphism — variation in size and shape of cells and nuclei
2. Abnormal nuclear morphology — hyperchromasia, ↑ N:C ratio (up to 1:1, normal 1:4–1:6), coarse chromatin, prominent nucleoli
3. Mitoses — numerous, atypical (tripolar, quadripolar, multipolar spindles)
4. Loss of polarity
5. Tumour giant cells, ischemic necrosis

Dysplasia :

- Disordered growth : pleomorphism, hyperchromasia, ↑ mitoses, loss of architectural orientation.
- Carcinoma in situ : dysplastic changes involve the full thickness of epithelium, but basement membrane intact.
- Dysplasia ≠ cancer; low-grade dysplasia may regress.

Filled : Exceptions to "benign never metastasizes / malignant always metastasizes"

- Malignant tumours that rarely metastasize : basal cell carcinoma of skin, gliomas (primary CNS tumours), verrucous carcinoma, dermatofibrosarcoma protuberans.
- "Benign" tumours that may metastasize : pleomorphic adenoma (metastasizing PA), giant cell tumour of bone (lung), leiomyoma (benign metastasizing), ameloblastoma.
- Most common malignant tumour of liver, lung, brain, bone, adrenal = metastasis.

Invasion & Metastasis

Steps of invasion of ECM :

1. Loosening of tumour cells — ↓ E-cadherin (cell-cell adhesion; CDH1 mutation in diffuse gastric cancer, lobular breast cancer), epithelial-mesenchymal transition (EMT) — SNAIL, TWIST
2. Local degradation of basement membrane / ECM — MMP-2 & MMP-9 (type IV collagenases), cathepsin D, uPA
3. Changes in attachment to ECM — integrins, laminin receptors
4. Migration (locomotion) — autocrine motility factors (HGF/c-MET), cleavage products of ECM

Routes of spread :

Route	Typical tumours
Seeding of body cavities	Ovarian carcinoma (peritoneum), CNS tumours (CSF — medulloblastoma), pseudomyxoma peritonei (appendiceal mucinous), Krukenberg tumour (gastric → ovary, transcoelomic/lymphatic)
Lymphatic spread	Carcinomas (most common route). Sentinel lymph node biopsy — breast, melanoma
Hematogenous spread	Sarcomas; carcinomas : RCC (renal vein), HCC (portal / hepatic vein), follicular thyroid, choriocarcinoma

Exam pearl

- Sarcomas spread via blood; carcinomas via lymphatics — exceptions : carcinomas via blood : RCC, HCC, follicular Ca thyroid, choriocarcinoma; sarcomas via lymph : rhabdomyosarcoma, synovial sarcoma, epithelioid sarcoma, clear cell sarcoma, angiosarcoma.
- Most common sites of hematogenous metastasis : liver and lung. Bone metastasis : prostate (osteoblastic), breast (mixed), lung, kidney (lytic), thyroid (lytic).
- Virchow's node (left supraclavicular) — gastric carcinoma; Sister Mary Joseph nodule — umbilical metastasis.
- Seed and soil hypothesis — Paget; CXCR4-CXCL12 guides breast cancer to bone / lung.

Molecular Basis of Cancer — Hallmarks

Mnemonic : Hallmarks of cancer (Hanahan & Weinberg)

1. Self-sufficiency in growth signals (oncogenes)
2. Insensitivity to growth-inhibitory signals (tumour suppressors)
3. Altered cellular metabolism (Warburg effect)
4. Evasion of apoptosis
5. Limitless replicative potential (telomerase)
6. Sustained angiogenesis
7. Invasion and metastasis
8. Evasion of immune surveillance — Enabling : genomic instability, tumour-promoting inflammation.

Oncogenes (gain of function — one allele sufficient) :

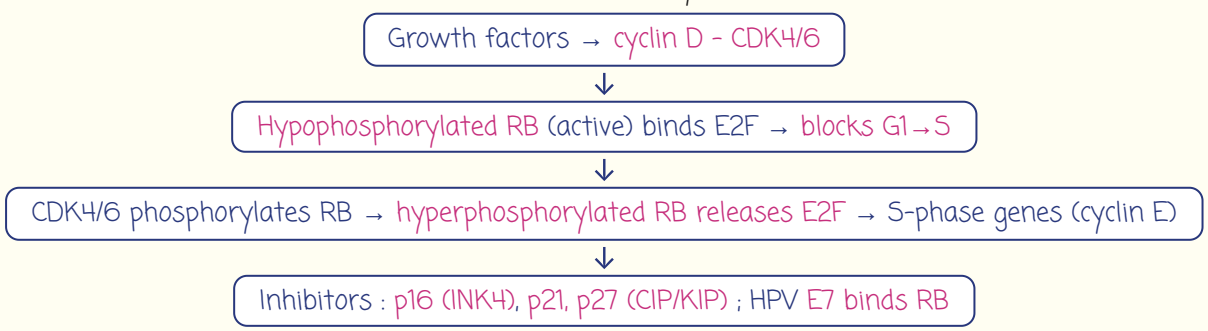
Category	Oncogene	Mechanism	Associated tumours
Growth factor	SIS (PDGF-β)	Overexpression	Astrocytoma, osteosarcoma
Growth factor receptor	ERBB2 (HER2/neu)	Amplification	Breast, gastric, ovarian
	EGFR (ERBB1)	Point mutation / amplification	Lung adenocarcinoma (non-smoking women), GBM, SCC head/neck
	RET	Point mutation / rearrangement	MEN 2A, 2B, familial medullary thyroid; papillary thyroid (RET/PTC)
	KIT	Point mutation	GIST, mastocytosis, seminoma
	ALK	Translocation	Lung adenocarcinoma (EML4-ALK), anaplastic large cell lymphoma (NPM-ALK), neuroblastoma
Signal transduction (GTP binding)	KRAS	Point mutation	Pancreas (>90%), colon, lung
	NRAS / HRAS	Point mutation	Melanoma, hematologic / bladder, kidney
Signal transduction (kinase)	BRAF (V600E)	Point mutation	Melanoma, hairy cell leukemia, papillary thyroid, colon (serrated)
Non-receptor tyrosine kinase	ABL	t(9;22) BCR-ABL	CML, some ALL
	JAK2 (V617F)	Point mutation	Polycythemia vera, ET, myelofibrosis
Nuclear transcription factor	c-MYC	t(8;14)	Burkitt lymphoma
	N-MYC	Amplification	Neuroblastoma
	L-MYC	Amplification	Small cell carcinoma lung

Category	Oncogene	Mechanism	Associated tumours
Cell cycle	Cyclin D1	t(11;14) / amplification	Mantle cell lymphoma, breast, esophagus
	CDK4	Amplification	Melanoma, sarcoma (liposarcoma)
Anti-apoptotic	BCL-2	t(14;18)	Follicular lymphoma

Tumour suppressor genes (loss of function — "two-hit hypothesis", Knudson) :

Gene	Function	Familial syndrome	Sporadic tumours
RB (13q14)	Inhibits G1→S (binds E2F)	Familial retinoblastoma, osteosarcoma	Retinoblastoma, osteosarcoma, breast, colon, lung
TP53 (17p13)	"Guardian of the genome" — cell cycle arrest (p21), apoptosis, senescence	Li-Fraumeni syndrome (sarcomas, breast, brain, adrenocortical, leukemia)	Most common mutated gene in human cancer (>50%)
APC (5q21)	Inhibits WNT/β-catenin	Familial adenomatous polyposis	Colon (70-80%), stomach, pancreas
NF1 (17q11)	Neurofibromin — GAP for RAS	Neurofibromatosis type 1	Neuroblastoma, JMML
NF2 (22q12)	Merlin (cytoskeleton, contact inhibition)	NF2 (bilateral acoustic schwannoma)	Schwannoma, meningioma
WT1 (11p13)	Transcription factor	WAGR, Denys-Drash	Wilms tumour
VHL (3p)	Ubiquitin ligase — degrades HIF-1α	von Hippel-Lindau (hemangioblastoma, clear cell RCC, pheochromocytoma)	Clear cell RCC
BRCA1 (17q21), BRCA2 (13q12)	DNA repair (homologous recombination)	Familial breast and ovarian carcinoma; BRCA2 — male breast, pancreas, prostate	Rare
PTEN (10q23)	Inhibits PI3K/AKT	Cowden syndrome	Endometrial, prostate, GBM
CDKN2A (9p21)	p16/INK4a (CDK inhibitor), ARF	Familial melanoma	Pancreas, esophagus, melanoma, GBM
SMAD4 (DPC4, 18q)	TGF-β signalling	Juvenile polyposis	Pancreas (55%), colon
STK11 (LKB1)	Kinase (AMPK)	Peutz-Jeghers syndrome	—
PTCH (9q22)	Inhibits Hedgehog	Gorlin syndrome (basal cell nevus)	Basal cell carcinoma, medulloblastoma
CDH1	E-cadherin	Familial gastric (diffuse) carcinoma	Gastric, lobular breast
TSC1/TSC2	Hamartin / tuberin — inhibit mTOR	Tuberous sclerosis	—
MEN1	Menin	MEN 1	Parathyroid, pituitary, pancreatic NET

RB and the cell cycle



p53 response to DNA damage

DNA damage (radiation, UV, chemicals) → ATM / ATR kinases activate p53 (releases from MDM2)

↓

p21 → cell cycle arrest (G1) + DNA repair (GADD45) | PUMA, NOXA, BAX → apoptosis | Senescence (miR-34)

↓

Repair successful → cell resumes | Repair fails → apoptosis/senescence

↓

HPV E6 degrades p53 ; MDM2 amplification in sarcomas (liposarcoma)

DNA repair defects :

Defect	Syndrome / tumour
Mismatch repair (MLH1, MSH2, MSH6, PMS2)	Lynch syndrome (HNPCC) — colon (right-sided), endometrium, ovary; microsatellite instability
Nucleotide excision repair	Xeroderma pigmentosum (UV — pyrimidine dimers) — skin SCC, BCC, melanoma
Homologous recombination	BRCA1/2, Bloom (BLM), ataxia telangiectasia (ATM), Fanconi anemia
Base excision repair	MUTYH-associated polyposis

Warburg effect, telomerase, angiogenesis :

- Warburg effect (aerobic glycolysis) — cancer cells use glycolysis even with adequate oxygen; basis of FDG-PET; driven by PI3K/AKT, MYC; IDH1/2 mutation → oncometabolite 2-hydroxyglutarate (gliomas, AML, cholangiocarcinoma).
- Telomerase reactivation in 85-95% cancers (TERT promoter mutations — melanoma, GBM).
- Angiogenic switch — VEGF, bFGF; inhibitors : angiostatin, endostatin, thrombospondin-1 (p53 induces). Drug : bevacizumab (anti-VEGF).

Carcinogenesis

Multistep carcinogenesis (chemical)

Initiation — exposure to carcinogen → permanent, irreversible DNA damage (mutation); initiator alone is not sufficient

↓

Promotion — promoters (phorbol esters (TPA), hormones, phenols, drugs) induce proliferation of initiated cells; reversible, non-mutagenic

↓

Progression → benign → malignant tumour; additional mutations (tumour heterogeneity)

Chemical carcinogens :

Class	Agent	Cancer
Direct-acting	Alkylating agents (cyclophosphamide, busulfan, nitrosoureas)	AML / MDS, lymphoma
Indirect (procarcinogens)	Polycyclic aromatic hydrocarbons (benzo[a]pyrene — tobacco, smoked food)	Lung, oral, larynx, bladder
	Aromatic amines, azo dyes (β-naphthylamine — aniline dye, rubber)	Bladder (urothelial)

Class	Agent	Cancer
	Aflatoxin B1 (Aspergillus flavus – ground nuts, grain)	HCC (TP53 codon 249 G:T mutation)
	Nitrosamines (smoked fish, preserved foods)	Gastric, esophageal
Others	Vinyl chloride	Angiosarcoma of liver
	Asbestos	Mesothelioma, lung carcinoma (synergistic with smoking)
	Arsenic	Skin SCC, lung, angiosarcoma liver
	Benzene	AML
	Chromium, nickel	Lung, nasal sinus
	Betel nut (areca)	Oral SCC (submucous fibrosis)
	Diethylstilbestrol (DES) in utero	Clear cell adenocarcinoma of vagina
	Naphthylamine, cyclophosphamide (acrolein)	Bladder carcinoma

Radiation carcinogenesis :

- UV-B → pyrimidine (thymine) dimers → skin cancer (SCC, BCC, melanoma); xeroderma pigmentosum.
- Ionizing radiation → leukemias (most common; AML, CML – except CLL), papillary thyroid carcinoma (children), breast, lung; latent period leukemia 7-12 years; solid tumours decades.

Oncogenic viruses & microbes :

Agent	Mechanism	Cancers
HPV (16, 18 – high risk)	E6 → degrades p53; E7 → binds and inactivates RB; integration into host genome	Cervical carcinoma, anal, oropharyngeal (tonsil), penile, vulvar
EBV	LMP-1 (mimics CD40) → NF-κB; EBNA-2	Burkitt lymphoma (endemic), nasopharyngeal carcinoma, Hodgkin (mixed cellularity), post-transplant lymphoproliferative disorder, gastric carcinoma
HBV, HCV	Chronic inflammation, regeneration; HBx protein	Hepatocellular carcinoma (HCV also → B-cell lymphoma)
HTLV-1	Tax protein (activates transcription, inactivates p16)	Adult T-cell leukemia/lymphoma (Japan, Caribbean)
HHV-8 (KSHV)	vFLIP, vGPCR	Kaposi sarcoma, primary effusion lymphoma, multicentric Castleman
Merkel cell polyomavirus	Large T antigen	Merkel cell carcinoma
H. pylori	CagA; chronic inflammation	Gastric adenocarcinoma, MALT lymphoma
Schistosoma haematobium	Chronic inflammation	Squamous cell carcinoma of bladder
Clonorchis sinensis / Opisthorchis	Chronic inflammation	Cholangiocarcinoma

Hereditary predisposition :

Category	Examples
Autosomal dominant cancer syndromes	Retinoblastoma, Li-Fraumeni, FAP, Lynch, MEN, NF1, NF2, VHL, BRCA
Defective DNA repair (AR)	Xeroderma pigmentosum, ataxia telangiectasia, Bloom, Fanconi
Familial cancers (uncertain inheritance)	Breast, ovary, colon (non-syndromic)

Grading & Staging

Feature	Grading	Staging
Based on	Degree of differentiation (histology) + number of mitoses	Size of primary, lymph node spread, distant metastasis (TNM)
Clinical value	Less	Greater prognostic value
Examples	Gleason (prostate), Nottingham / Bloom-Richardson (breast), Fuhrman / WHO-ISUP (RCC)	TNM (AJCC/UICC), Dukes (colon), FIGO (gynaecological), Ann Arbor (lymphoma), Clark & Breslow (melanoma)

Clinical Aspects — Host Effects

Cancer cachexia :

- Loss of body fat, lean mass, weakness, anorexia, anemia; TNF-α (cachectin), IL-1, IL-6, proteolysis-inducing factor, lipid-mobilizing factor.

Paraneoplastic syndromes :

Syndrome	Mechanism	Tumour
Cushing syndrome	ACTH / ACTH-like	Small cell carcinoma lung, pancreatic, neural tumours, carcinoid
SIADH (hyponatremia)	ADH / ANP	Small cell carcinoma lung, intracranial neoplasms
Hypercalcemia	PTHrP, TGF-α, TNF, IL-1, calcitriol (lymphoma)	Squamous cell carcinoma lung, breast, renal, ovarian; Hodgkin / T-cell lymphoma (1,25-OH vitamin D)
Hypoglycemia	Insulin / IGF-2	Fibrosarcoma, mesothelioma, HCC, other mesenchymal tumours
Carcinoid syndrome	Serotonin, bradykinin	Carcinoid tumour (with liver metastases), pancreatic, gastric, small cell
Polycythemia	Erythropoietin	Renal cell carcinoma, cerebellar hemangioblastoma, HCC, leiomyoma uterus, pheochromocytoma
Lambert-Eaton myasthenic syndrome	Antibodies against presynaptic voltage-gated Ca^{2+} channels	Small cell carcinoma lung
Myasthenia gravis	Anti-AChR antibodies	Thymoma
Paraneoplastic cerebellar degeneration	Anti-Yo (Purkinje cells), anti-Hu, anti-Tr	Ovarian, breast; SCLC (Hu); Hodgkin (Tr)
Anti-NMDA receptor encephalitis	Anti-NMDAR	Ovarian teratoma
Acanthosis nigricans	? EGF	Gastric carcinoma, lung, uterus
Leser-Trélat sign	Sudden eruption of multiple seborrheic keratoses	GI adenocarcinoma (gastric)
Dermatomyositis	Immunologic	Lung, breast, ovary
Hypertrophic pulmonary osteoarthropathy (clubbing)	Unknown	Lung carcinoma (adenocarcinoma)
Trousseau phenomenon (migratory thrombophlebitis)	Hypercoagulability (mucin, TF)	Pancreatic carcinoma, lung
NBTE (marantic endocarditis)	Hypercoagulability	Advanced mucin-secreting adenocarcinomas

Syndrome	Mechanism	Tumour
Pure red cell aplasia	Immunologic	Thymoma
Opsoclonus-myoclonus	Anti-Ri / Hu	Neuroblastoma (children), breast
Hyperthyroidism (TSH-like)	hCG	Choriocarcinoma, hydatidiform mole

Mnemonic : Lung carcinoma paraneoplastics

- Small cell → "ACTH, ADH, Antibodies" : Cushing, SIADH, Lambert-Eaton (VGCC), anti-Hu encephalomyelitis / sensory neuropathy.
- Squamous cell → Stones : PTHrP → hypercalcemia.
- Adenocarcinoma → Arthropathy : clubbing / hypertrophic pulmonary osteoarthropathy, Trousseau.

Laboratory Diagnosis of Cancer

1. Histologic examination — excision / incision / core biopsy; frozen section (intra-operative, quick — margins, nature of lesion)
2. Cytology — FNAC (breast, thyroid, lymph nodes, salivary); exfoliative (Pap smear, sputum, urine, effusions)
3. Immunohistochemistry (IHC) — identify cell of origin, prognostic/predictive markers
4. Flow cytometry — leukemias, lymphomas (CD markers), DNA ploidy
5. Molecular diagnosis — PCR (BCR-ABL monitoring), FISH (HER2, N-MYC, translocations), NGS, liquid biopsy (ctDNA)
6. Tumour markers (serum)

Immunohistochemistry markers :

Marker	Tumour / cell type
Cytokeratin	Carcinomas, mesothelioma (epithelial)
Vimentin	Sarcomas (mesenchymal), RCC, endometrial, melanoma
Desmin	Muscle tumours (rhabdomyosarcoma, leiomyosarcoma)
Myogenin, MyoD1	Rhabdomyosarcoma (most specific)
S-100	Melanoma, schwannoma, Langerhans cell histiocytosis, neural, chondroid
HMB-45, Melan-A, SOX10	Melanoma
LCA (CD45)	Lymphomas / leukocytes
CD20 (B), CD3 (T), CD15 & CD30 (Reed-Sternberg)	Lymphomas
CD31, CD34, ERG, Factor VIII	Vascular tumours (angiosarcoma)
Chromogranin, synaptophysin, CD56, INSM1	Neuroendocrine tumours (carcinoid, SCLC)
GFAP	Glial tumours (astrocytoma)
CD117 (c-KIT), DOG-1	GIST
TTF-1	Lung adenocarcinoma, small cell carcinoma, thyroid
Thyroglobulin, PAX8	Thyroid (PAX8 also kidney, Müllerian)
CK7+/CK20−	Lung, breast, ovary (non-mucinous), thyroid, endometrium
CK7−/CK20+	Colorectal carcinoma (CDX2 +), Merkel cell (dot-like)
ER, PR, HER2, GATA3	Breast carcinoma
PSA, NKX3.1	Prostatic carcinoma
Calretinin, WT1, D2-40	Mesothelioma

Marker	Tumour / cell type
CD99 (MIC2) + FLI-1	Ewing sarcoma / PNET
AFP, Glypican-3, HepPar-1, Arginase-1	HCC
p63, p40, CK5/6	Squamous cell carcinoma
CD1a, langerin (CD207)	Langerhans cell histiocytosis
Inhibin	Sex-cord stromal tumours (granulosa cell), adrenocortical
PLAP, OCT3/4, SALL4	Seminoma / dysgerminoma (germ cell)
Ki-67 (MIB-1)	Proliferation index

Tumour markers (serum) :

Marker	Tumour
α-fetoprotein (AFP)	HCC, yolk sac (endodermal sinus) tumour, (NTDs in pregnancy)
β-hCG	Choriocarcinoma, hydatidiform mole, germ cell tumours (seminoma – low)
CEA	Colorectal, pancreas, stomach, breast, lung (monitoring recurrence)
CA-125	Ovarian (serous) carcinoma
CA 19-9	Pancreatic, biliary, colon
CA 15-3 / CA 27-29	Breast carcinoma
PSA	Prostate carcinoma (also BPH, prostatitis)
Calcitonin	Medullary carcinoma of thyroid
Thyroglobulin	Follow-up of differentiated (papillary, follicular) thyroid Ca after thyroidectomy
Catecholamines, VMA, metanephrines	Pheochromocytoma, neuroblastoma (HVA, VMA)
5-HIAA (urine)	Carcinoid syndrome
LDH	Lymphoma, dysgerminoma, Ewing, germ cell tumours
Immunoglobulins (M-band), β2-microglobulin	Multiple myeloma
S-100, LDH	Melanoma
Neuron-specific enolase (NSE)	Small cell carcinoma, neuroblastoma
Placental ALP	Seminoma
HE4	Ovarian cancer (with CA-125 – ROMA)
Bombesin	Small cell lung Ca, neuroblastoma, gastric

Filled : Rank boosters — neoplasia

- Tumour of highest incidence in India : males — lip/oral cavity & lung; females — breast > cervix (GLOBOCAN).
- Most common cancer worldwide (incidence) : lung (2022 now breast close second); most common cause of cancer death : lung.
- Cancer stem cells — CD133 (brain), CD44+/CD24− (breast).
- Driver vs passenger mutations ; kataegis, chromothripsis (catastrophic shattering).
- Oncogene addiction — basis of targeted therapy (imatinib — BCR-ABL, KIT; trastuzumab — HER2; vemurafenib — BRAF; crizotinib — ALK).
- Immune checkpoints : PD-1/PD-L1, CTLA-4 — tumours evade T cells; drugs : pembrolizumab, nivolumab, ipilimumab; MSI-high tumours respond best.
- Tumour antigens : products of mutated genes (neoantigens), overexpressed (HER2, tyrosinase), viral (HPV E6/E7), oncofetal (AFP, CEA), cancer-testis antigens (MAGE), differentiation antigens (CD20, PSA).
- Effector cells against tumours : CD8+ CTL (most important), NK cells, macrophages (M1).

IMMUNOPATHOLOGY

Organisation of the Immune System

Feature	Innate immunity	Adaptive immunity
Specificity	Pathogen-associated molecular patterns (PAMPs)	Highly specific antigens
Memory	None (trained immunity limited)	Memory
Cells	Neutrophils, macrophages, NK cells, dendritic cells, mast cells, ILCs	T and B lymphocytes
Molecules	Complement, TLRs, antimicrobial peptides, cytokines (IFN-α/β)	Antibodies, TCR, BCR
Speed	Minutes-hours	Days

Lymphocytes :

Cell	Markers	Key functions
T cells	CD3, TCR (αβ — 95%; γδ — mucosal)	Cell-mediated immunity; mature in thymus
CD4+ helper T	CD4, recognize antigen on MHC class II	Help B cells, macrophages; CD4:CD8 = 2:1
CD8+ cytotoxic T	CD8, recognize MHC class I	Kill virus-infected / tumour cells (perforin, granzyme, FasL)
Treg	CD4, CD25 (IL-2Rα), FOXP3	Suppress immunity (IL-10, TGF-β, CTLA-4); FOXP3 mutation → IPEX
B cells	CD19, CD20, CD21 (EBV receptor, CR2), surface IgM & IgD	Humoral immunity → plasma cells (CD138)
NK cells	CD16 (FcγRIII), CD56; no TCR	Kill cells lacking MHC-I ("missing self"); ADCC; IL-12 activates; produce IFN-γ
Dendritic cells	CD11c, MHC II	Most potent antigen-presenting cells; follicular DC (antigen display for B cells, CD21, CD35 — not APC proper)

CD4+ T helper subsets :

Subset	Induced by	Signature cytokines	Functions / disease
Th1	IL-12, IFN-γ (T-bet)	IFN-γ, IL-2	Activate macrophages (M1), granulomas, DTH; IgG opsonizing
Th2	IL-4 (GATA-3)	IL-4, IL-5, IL-13	IgE switching, eosinophils, helminths, allergy, asthma, M2
Th17	IL-1, IL-6, IL-23, TGF-β (RORγt)	IL-17, IL-22	Neutrophil recruitment, extracellular bacteria & fungi; psoriasis, IBD, RA; ↓ in hyper-IgE (STAT3)
Tfh	IL-6, IL-21 (Bcl-6)	IL-21, IL-4	Germinal centre B-cell help
Treg	TGF-β, IL-2 (FOXP3)	IL-10, TGF-β	Tolerance

MHC (HLA) molecules — chromosome 6 :

Feature	Class I	Class II
Loci	HLA-A, B, C	HLA-DP, DQ, DR
Distribution	All nucleated cells + platelets (not RBCs)	APCs (dendritic cells, macrophages, B cells), activated T cells, thymic epithelium
Structure	α chain + β2-microglobulin	α + β chains

Feature	Class I	Class II
Antigen	Endogenous (cytosolic) — proteasome → TAP	Exogenous — endocytosed, invariant chain
Presents to	CD8+ T cells	CD4+ T cells

- Class III region : complement (C2, C4, factor B), TNF, HSP70.

HLA associations :

HLA	Disease
B27	Ankylosing spondylitis (90%), reactive arthritis, psoriatic arthritis, IBD arthritis ("PAIR")
DR3	Type 1 DM, Graves, SLE, celiac, myasthenia, Addison, Sjögren, dermatitis herpetiformis
DR4	Rheumatoid arthritis, type 1 DM, pemphigus vulgaris
DR3/DR4 heterozygote	Highest risk of type 1 DM
DQ2 / DQ8	Celiac disease (DQ2 in 95%)
DR2 (DR15)	Multiple sclerosis, Goodpasture, SLE, narcolepsy, hay fever
DR5	Pernicious anemia, Hashimoto, JRA
B51	Behçet disease
Cw6	Psoriasis
B*57:01	Abacavir hypersensitivity
B*15:02	Carbamazepine SJS/TEN (Asians)
B*58:01	Allopurinol SJS/TEN
A3	Hemochromatosis (HFE linked)

Immunoglobulins :

Ig	Key features
IgG	Most abundant in serum (75%); crosses placenta; opsonization, complement (IgG3 > IgG1), secondary response; 4 subclasses (IgG4-related disease)
IgA	Most produced overall; secretory IgA (dimer + J chain + secretory component) in mucosa, colostrum; most common primary Ig deficiency
IgM	Pentamer; first antibody in primary response; best complement activator; largest; B-cell receptor (monomer); isohemagglutinins
IgE	Binds FcεRI on mast cells, basophils; type I hypersensitivity, helminths; lowest serum concentration; heat labile
IgD	B-cell receptor (naive B cells)

Hypersensitivity Reactions

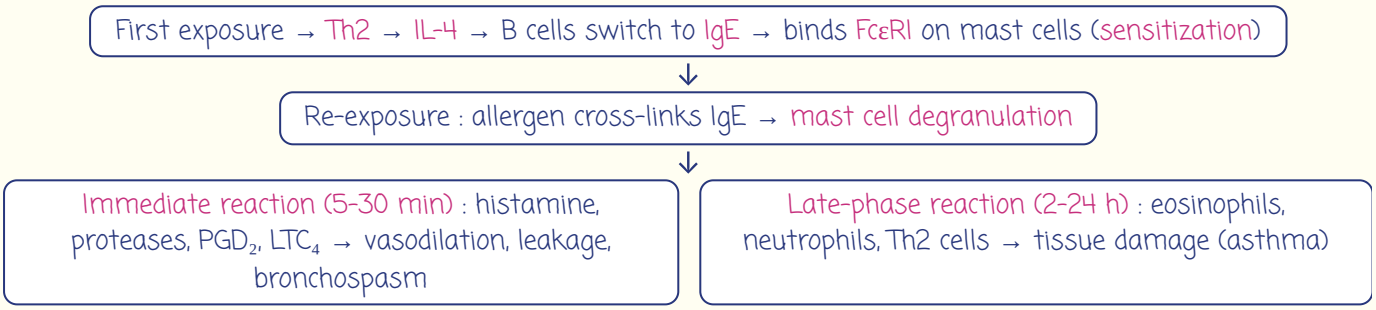
Type	Mechanism	Mediators	Examples
Type I (immediate / anaphylactic)	IgE on mast cells cross-linked by allergen; Th2	Histamine, leukotrienes, PGD_2, cytokines	Anaphylaxis, bronchial asthma (atopic), allergic rhinitis, urticaria, food allergy, Casoni test
Type II (antibody-mediated / cytotoxic)	IgG / IgM against fixed antigens on cells or ECM	Complement (MAC), opsonization, ADCC, receptor dysfunction	AIHA, transfusion reactions, HDN (Rh), ITP, Goodpasture, Graves (stimulating), myasthenia gravis (blocking), pemphigus vulgaris, bullous pemphigoid, acute rheumatic fever, pernicious anemia, hyperacute rejection

Type	Mechanism	Mediators	Examples
Type III (immune complex)	Antigen-antibody complexes deposit in vessels → complement activation → neutrophils	Complement (C3a, C5a), neutrophil enzymes	SLE, post-streptococcal GN, PAN, serum sickness, Arthus reaction, reactive arthritis, HSP (IgA), cryoglobulinemia, hypersensitivity pneumonitis (III + IV)
Type IV (T-cell / delayed)	CD4+ (DTH — Th1/Th17) and CD8+ (cytotoxicity); no antibodies	IFN-γ, IL-17, TNF; perforin/granzyme	Tuberculin (Mantoux) test, contact dermatitis (poison ivy, nickel), type 1 DM, MS, RA, IBD, Hashimoto, granulomas (TB), acute cellular graft rejection, Lepromin test, patch test

Mnemonic : Hypersensitivity — "ACID"

Type I : Anaphylactic / Allergic · Type II : Cytotoxic (antibody) · Type III : Immune complex · Type IV : Delayed (cell-mediated). Antibody in I, II, III; no antibody in IV.

Type I — phases :



Type III — systemic vs local :

- Serum sickness (systemic) : 5-10 days after foreign protein (horse anti-thymocyte globulin) — fever, urticaria, arthralgia, proteinuria, lymphadenopathy, ↓ C3.
- Arthus reaction (local) : localized necrotizing vasculitis at site of injection (hours) in previously immunized host — fibrinoid necrosis; e.g. after tetanus booster.
- Lesions : fibrinoid necrosis of vessels; immunofluorescence : granular deposits (lumpy-bumpy) vs linear in anti-GBM (type II).

Filled

- Type V hypersensitivity (some texts) : stimulatory antibodies — Graves disease (TSH receptor) (classified under type II in Robbins).
- Type II drugs : penicillin-induced hemolysis (hapten), methyldopa (autoimmune), quinidine (immune complex/innocent bystander).
- Contact dermatitis is type IV but is mediated by both CD4 and CD8 T cells; Langerhans cells are the APCs.
- Jones-Mote reaction (cutaneous basophil hypersensitivity) — variant of type IV.

Autoimmunity

Mechanisms :

1. Failure of central tolerance — AIRE mutation (thymic expression of tissue antigens) → APS-1 (APECED)
2. Failure of peripheral tolerance — anergy, Tregs (FOXP3 → IPEX), CTLA-4, PD-1, deletion by apoptosis (Fas/FasL → ALPS)
3. Genetic susceptibility — HLA, PTPN22 (most common non-HLA; RA, T1DM), NOD2 (Crohn), IL-23R

4. Infections — molecular mimicry (rheumatic fever — streptococcal M protein); polyclonal B-cell activation (EBV); epitope spreading
5. Tissue injury releasing sequestered antigens (sympathetic ophthalmia, post-vasectomy anti-sperm)
6. Female predominance (hormones, X-chromosome genes)

Autoantibodies — high yield :

Antibody	Disease
ANA	SLE (95-100%, sensitive), other CTDs
Anti-dsDNA, anti-Smith (Sm)	SLE (specific); anti-dsDNA correlates with nephritis / activity
Anti-histone	Drug-induced lupus (hydralazine, procainamide, isoniazid)
Anti-Ro (SS-A), anti-La (SS-B)	Sjögren; neonatal lupus (congenital heart block)
Anti-centromere	Limited scleroderma (CREST)
Anti-Scl-70 (topoisomerase I)	Diffuse systemic sclerosis (lung fibrosis)
Anti-RNA polymerase III	Diffuse SSc — scleroderma renal crisis
Anti-U1 RNP	Mixed connective tissue disease
Anti-Jo-1 (histidyl-tRNA synthetase)	Polymyositis / dermatomyositis (antisynthetase — ILD)
Anti-Mi-2	Dermatomyositis
Anti-CCP (ACPA)	Rheumatoid arthritis (most specific); RF (IgM anti-Fc of IgG) — sensitive
c-ANCA (PR3)	Granulomatosis with polyangiitis (Wegener)
p-ANCA (MPO)	Microscopic polyangiitis, EGPA (Churg-Strauss), UC, PSC, crescentic GN
Anti-GBM (α3 chain of type IV collagen)	Goodpasture syndrome
Anti-mitochondrial (AMA, M2)	Primary biliary cholangitis
Anti-smooth muscle, anti-LKM1	Autoimmune hepatitis type 1, type 2
Anti-TPO, anti-thyroglobulin	Hashimoto thyroiditis
TSH receptor (TRAb / TSI)	Graves disease
Anti-AChR (85%), anti-MuSK	Myasthenia gravis
Anti-VGCC	Lambert-Eaton
Anti-GAD65, anti-islet, anti-insulin, ZnT8	Type 1 DM
Anti-tTG, anti-endomysial (IgA), anti-deamidated gliadin	Celiac disease
Anti-intrinsic factor, anti-parietal cell	Pernicious anemia
Anti-desmoglein 3 (& 1)	Pemphigus vulgaris
Anti-BP180, BP230 (hemidesmosome)	Bullous pemphigoid
Anti-phospholipase A2 receptor (PLA2R)	Primary membranous nephropathy
Anti-aquaporin-4	Neuromyelitis optica

ANA immunofluorescence patterns :

Pattern	Antigen	Disease
Homogeneous / diffuse	dsDNA, histones	SLE, drug-induced lupus
Rim / peripheral	dsDNA	SLE (most specific pattern)
Speckled (most common, least specific)	Sm, RNP, Ro, La	SLE, MCTD, Sjögren

Pattern	Antigen	Disease
Nucleolar	RNA	Systemic sclerosis
Centromeric	Centromere proteins	CREST

Systemic Lupus Erythematosus (SLE)

- Prototype multisystem autoimmune disease (type III > type II); F : M = 9 : 1; reproductive age.
- Defective clearance of apoptotic bodies (nuclear antigens) + type I interferon signature + UV light, drugs.
- Libman-Sacks endocarditis : small, sterile vegetations on both surfaces of valve (mitral).
- LE cell : neutrophil engulfing denatured nucleus (hematoxylin body).
- Onion-skin lesion of spleen (periarterial fibrosis).

Lupus nephritis (ISN/RPS classes) :

Class	Features
I	Minimal mesangial
II	Mesangial proliferative
III	Focal (< 50% glomeruli)
IV	Diffuse proliferative (≥ 50%) — most common and most severe; wire-loop lesions (subendothelial deposits), full-house immunofluorescence (IgG, IgA, IgM, C3, C1q)
V	Membranous — nephrotic syndrome
VI	Advanced sclerosing (≥ 90%)

Mnemonic : SLE criteria (classic 11) — "SOAP BRAIN MD"

Serositis · Oral ulcers · Arthritis · Photosensitivity · Blood disorders · Renal disorder · ANA · Immunologic (dsDNA, Sm, aPL) · Neurologic (seizure, psychosis) · Malar rash · Discoid rash. (EULAR/ACR 2019 : ANA ≥ 1:80 entry criterion.)

Other Autoimmune Diseases

Disease	Key pathology
Sjögren syndrome	Lymphocytic sialadenitis (focus score ≥ 1), dry eyes / mouth; anti-Ro/La; ↑ risk of marginal zone (MALT) B-cell lymphoma (40×)
Systemic sclerosis	Fibrosis (TGF-β), microvascular damage; skin, GIT (esophageal dysmotility — lower 2/3), lung (ILD — leading cause of death), kidney (renal crisis — onion-skinning); CREST : calcinosis, Raynaud, esophageal dysmotility, sclerodactyly, telangiectasia
Inflammatory myopathies	Dermatomyositis — perifascicular atrophy, CD4 + B cells, complement on capillaries, heliotrope rash, Gottron papules, malignancy association. Polymyositis — CD8 T cells endomysial. Inclusion body myositis — rimmed vacuoles, >50 y
Polyarteritis nodosa	Necrotizing vasculitis of medium arteries, spares lungs, HBV (30%), segmental, transmural fibrinoid necrosis — lesions of different stages, microaneurysms ("rosary beads")
IgG4-related disease	Storiform fibrosis, obliterative phlebitis, IgG4+ plasma cells (>30-50/HPF); autoimmune pancreatitis type 1, Riedel thyroiditis, retroperitoneal fibrosis, Mikulicz

Transplant Rejection

Type	Timing	Mechanism	Morphology
Hyperacute	Minutes-hours	Pre-formed antibodies (ABO, anti-HLA from transfusions, pregnancy, previous transplant) — type II	Thrombosis, fibrinoid necrosis, infarction; cyanotic, mottled graft; prevented by cross-match
Acute cellular	Weeks-months (usually < 6 months)	CD8+ CTL and CD4+ (type IV)	Interstitial mononuclear infiltrate, tubulitis, endotheliitis (endarteritis)
Acute antibody-mediated (humoral)	Days-months	Donor-specific antibodies (DSA)	Vasculitis, glomerulitis, peritubular capillaritis, C4d deposition in peritubular capillaries
Chronic	Months-years	T cells (IFN-γ) + antibodies	Arteriosclerosis (intimal thickening — graft vasculopathy), interstitial fibrosis, tubular atrophy; transplant glomerulopathy; bronchiolitis obliterans (lung)

Graft-versus-host disease (GVHD) :

- Requirements (Billingham criteria) : 1. immunocompetent graft (T cells) 2. host antigens recognized as foreign 3. immunocompromised host.
- Settings : allogeneic hematopoietic stem cell transplant (most common), liver/small bowel transplant, transfusion of non-irradiated blood to immunodeficient recipients.
- Acute GVHD (< 100 days) : skin (rash), liver (jaundice — bile duct damage), GIT (bloody diarrhea) — epithelial cell apoptosis.
- Chronic GVHD : scleroderma-like skin, sicca, obliterative bronchiolitis.
- Prevention : irradiation / leukoreduction of blood products, T-cell depletion.

Filled

- Most immunogenic graft : bone marrow > skin > ...; cornea — immunologically privileged (avascular).
- Order of importance in HLA matching (kidney) : HLA-DR > B > A.
- C4d — marker of antibody-mediated rejection.
- CMV is the most common opportunistic infection after solid organ transplant; EBV → PTLD.
- Graft-versus-leukemia effect — beneficial T-cell action in allogeneic HSCT.

Primary Immunodeficiencies

Disease	Defect	Key features
X-linked agammaglobulinemia (Bruton)	BTK (tyrosine kinase) — B-cell maturation arrest (pre-B)	Boys after 6 months (maternal IgG wanes); absent B cells (CD19), all Ig ↓, absent germinal centres, tonsils; recurrent bacterial (sinopulmonary), enteroviral (polio vaccine-paralysis, echovirus encephalitis), Giardia
Common variable immunodeficiency (CVID)	Heterogeneous (TACI, ICOS); B cells present, defective differentiation	Onset after 2nd decade; ↓ IgG ± IgA, IgM; ↑ autoimmunity, lymphoma, gastric Ca; hyperplastic B-cell follicles
Selective IgA deficiency	IgA	Most common PID (1:700); mostly asymptomatic; sinopulmonary, Giardia; anaphylaxis on blood transfusion (anti-IgA antibodies); ↑ celiac, autoimmunity

Disease	Defect	Key features
Hyper-IgM syndrome	CD40L (X-linked, T cell) or CD40, AID	↑ IgM, ↓ IgG/A/E; Pneumocystis pneumonia, Cryptosporidium
DiGeorge syndrome (22q11.2 deletion)	Thymic hypoplasia (3rd & 4th pharyngeal pouches)	CATCH-22 : cardiac (truncus, TOF, interrupted aortic arch), abnormal facies, thymic aplasia (↓ T cells), cleft palate, hypocalcemia (absent parathyroids → tetany); TBX1
SCID	X-linked (most common, 50-60%) : common γ chain (IL-2Rγ); AR : ADA deficiency (adenosine deaminase) → toxic dATP	Thrush, diaper rash, failure to thrive, opportunistic (Candida, Pneumocystis, CMV); absent thymic shadow, ↓ T cells; BCG-osis; treatment : HSCT, gene therapy (ADA)
Wiskott-Aldrich syndrome	WAS gene (X-linked) → cytoskeleton	Thrombocytopenia (small platelets), eczema, recurrent infections; ↓ IgM, ↑ IgA & IgE; ↑ lymphoma
Ataxia telangiectasia	ATM (DNA repair) AR	Cerebellar ataxia, oculocutaneous telangiectasia, IgA deficiency, ↑ AFP, radiation sensitivity, lymphoma
Chronic mucocutaneous candidiasis	T-cell defect to Candida (AIRE, STAT1 GOF, IL-17)	Candida skin, nails, mucosa
IPEX	FOXP3 (Treg)	Immune dysregulation, polyendocrinopathy, enteropathy, X-linked
ALPS	Fas / FasL	Lymphadenopathy, splenomegaly, autoimmune cytopenias; ↑ double-negative (CD4–CD8–) T cells
Bare lymphocyte syndrome	MHC class II (CIITA, RFX)	↓ CD4 T cells, SCID-like
Complement / phagocyte defects	See Inflammation chapter	C5-C9 → Neisseria; CGD; LAD

Mnemonic : B vs T-cell deficiency infections

- B cell (antibody) : encapsulated bacteria → "SHiN SKiS" (S. pneumoniae, H. influenzae b, N. meningitidis, Salmonella, Klebsiella, GBS) + enteroviruses + Giardia.
- T cell : viruses, fungi (Candida, Pneumocystis), intracellular bacteria (Mycobacteria, Listeria), protozoa.

Secondary Immunodeficiency & HIV / AIDS

Causes of secondary immunodeficiency :

1. HIV infection (most important)
2. Malnutrition (most common worldwide), aging
3. Malignancy → leukemia, lymphoma (Hodgkin → T-cell defect), myeloma
4. Drugs → corticosteroids, chemotherapy, immunosuppressants, biologics (anti-TNF)
5. Diabetes mellitus, chronic renal failure, cirrhosis
6. Splenectomy → encapsulated organisms (OPSI)
7. Protein loss → nephrotic syndrome, protein-losing enteropathy, burns

HIV → virology :

- Retrovirus (Lentivirus); HIV-1 (worldwide; group M → subtype C most common in India), HIV-2 (West Africa).
- Genes : gag (p24 capsid, p17 matrix), pol (reverse transcriptase, integrase, protease), env (gp120 → binds CD4; gp41 → fusion). Regulatory : tat, rev, nef (↓ CD4 and MHC-I), vif, vpr, vpu.
- Receptor : CD4 + co-receptor CCR5 (macrophage-tropic, R5 → early, transmission) or CXCR4 (T-tropic, X4 → late). CCR5 Δ32 homozygous → resistance.

Transmission :

1. Sexual contact (most common worldwide) — heterosexual in India
2. Parenteral — IV drug users (needle sharing), blood transfusion (risk ~90-95% per unit — highest per-exposure), hemophiliacs (factor concentrates)
3. Mother-to-child (vertical) — intrapartum (most common), transplacental, breast milk
4. Occupational — needle-stick injury (~0.3%); mucous membrane (0.09%)

Natural history :

Acute (primary) HIV syndrome : 3-6 weeks — fever, sore throat, lymphadenopathy, rash; high viremia, ↓ CD4; seroconversion

↓

Chronic (clinical latency) phase : 7-10 years; persistent generalized lymphadenopathy; virus replicates in lymphoid tissue (follicular dendritic cells); slow CD4 decline

↓

AIDS : CD4 < 200/μL or AIDS-defining illness — opportunistic infections, tumours, wasting, encephalopathy

Lab diagnosis :

1. ELISA (4th gen : p24 antigen + antibody) — screening; window shortened to ~2-3 weeks
2. Rapid tests (3 different antigen principles — NACO strategy III)
3. Western blot (confirmatory, historical) — p24 + gp41 + gp120/160
4. HIV RNA PCR (viral load) — most sensitive; diagnosis in infants < 18 months (DNA PCR), monitoring therapy
5. p24 antigen — earliest marker (before antibodies)
6. CD4 count — staging, prophylaxis (Pneumocystis prophylaxis at CD4 < 200)

Opportunistic infections by CD4 count :

CD4/ μL	Infections
< 500	Tuberculosis (any CD4 — most common OI in India), oral candidiasis, herpes zoster, Kaposi sarcoma, oral hairy leukoplakia (EBV)
< 200	Pneumocystis jirovecii pneumonia, HIV encephalopathy, histoplasmosis, coccidioidomycosis
< 100	Toxoplasma encephalitis (ring-enhancing lesions), Cryptococcal meningitis, esophageal candidiasis, cryptosporidiosis, PML (JC virus), primary CNS lymphoma (EBV)
< 50	CMV retinitis, Mycobacterium avium complex (MAC), bacillary angiomatosis

HIV-associated neoplasms :

1. Kaposi sarcoma (HHV-8) — most common in untreated
2. B-cell lymphomas — DLBCL, Burkitt, primary CNS lymphoma (EBV), primary effusion lymphoma (HHV-8), plasmablastic lymphoma
3. Invasive cervical carcinoma (HPV) — AIDS-defining
4. Anal carcinoma (HPV)

Morphology :

- Lymph nodes : follicular hyperplasia (early) → follicular involution / "burnt-out" lymphocyte depletion (late).
- CNS : microglial nodules, multinucleated giant cells (HIV encephalitis); vacuolar myelopathy; HIV-associated neurocognitive disorder.

Filled : Handling of infected material & universal precautions

- Treat all specimens as potentially infectious (standard precautions); gloves, gown, mask, eye shield.
- Sharps in puncture-proof containers; never recap needles.
- Disinfection : 1% sodium hypochlorite (spills, 10,000 ppm with organic matter), 2% glutaraldehyde, 70% alcohol; autoclaving for reusable instruments; HIV is heat-labile (56 °C for 30 min).
- Biomedical waste (India, 2016 rules) : yellow — anatomical, soiled; red — contaminated recyclable plastics; white (translucent) — sharps; blue — glassware.
- Post-exposure prophylaxis (PEP) : start within 2 h (ideally), up to 72 h; TDF + 3TC + DTG for 28 days; test at baseline, 6 weeks, 3 months, 6 months.
- Health education : condoms, safe blood (screening all units for HIV, HBV, HCV, syphilis, malaria), harm reduction for IDUs, PPTCT — lifelong ART for pregnant women + infant nevirapine prophylaxis.

INFECTIOUS DISEASES

Patterns of Tissue Response to Infection

Pattern	Features	Examples
Suppurative (purulent)	Neutrophils, pus, abscess	Pyogenic bacteria — S. aureus, Streptococci, Gram-negative rods
Mononuclear / granulomatous	Lymphocytes, plasma cells, macrophages, granulomas	Viruses, intracellular bacteria (TB), spirochetes, fungi, helminths
Cytopathic-cytoproliferative	Cell necrosis / proliferation, inclusion bodies, multinucleated cells	Viruses — herpes, CMV, HPV (koilocytes), adenovirus
Tissue necrosis	Rapid, severe necrosis	Clostridium perfringens, Entamoeba histolytica, HSV encephalitis (temporal lobes)
Chronic inflammation & scarring	Fibrosis	HBV (cirrhosis), Schistosoma (pipe-stem fibrosis)
No reaction	Immunosuppression	PCP, MAC in AIDS

Special stains for organisms :

Stain	Organisms
Gram stain	Bacteria
Ziehl-Neelsen (acid-fast)	Mycobacteria, Nocardia (modified — weakly AF), Cryptosporidium, Isospora, Cyclospora (modified)
Fite-Faraco	M. leprae (weakly acid fast)
Auramine-rhodamine	Mycobacteria (fluorescent — sensitive screening)
GMS (Grocott methenamine silver)	Fungi, Pneumocystis (black)
PAS	Fungi, Whipple disease (Tropheryma — PAS+ macrophages), glycogen, Entamoeba trophozoites
Mucicarmine / India ink / Alcian blue	Cryptococcus capsule
Warthin-Starry / Dieterle silver	Spirochetes (Treponema), Bartonella, H. pylori, Legionella
Giemsa	Malaria, Leishmania (LD bodies), Toxoplasma, Chlamydia, Borrelia, Rickettsia, H. pylori
Toluidine blue	Pneumocystis cysts

Tuberculosis

- Mycobacterium tuberculosis — slender, acid-fast (mycolic acid), aerobic, cord factor (trehalose dimycolate) — virulence, serpentine cords; sulfatides prevent phagolysosome fusion; lipoarabinomannan (LAM) — inhibits macrophage activation.
- India : highest burden worldwide (~27% global cases).

Pathogenesis (immunity & hypersensitivity) :

Inhaled bacilli → alveolar macrophages → bacilli multiply (block phagolysosome maturation — ESX-1 secretion system)

↓

Week 3 : macrophages present antigen (MHC II) + IL-12 → Th1 cells → IFN-γ

↓

IFN-γ → macrophage activation : NO (iNOS), ROS, autophagy, defensins (cathelicidin — vitamin D dependent)

↓

TNF → granuloma formation (epithelioid cells, Langhans giant cells) + caseous necrosis (hypersensitivity)

↓

Containment (latent TB) — positive tuberculin test

Failure (HIV, ↓ IFN-γ / IL-12 pathway, anti-TNF) → progressive disease

Filled : Genetic susceptibility

- Mendelian susceptibility to mycobacterial disease (MSMD) : IFN-γ receptor, IL-12/IL-12R, STAT1 defects → disseminated BCG / NTM.
- NRAMP1 (SLC11A1) polymorphisms; vitamin D receptor polymorphisms.

Primary vs secondary TB :

Feature	Primary TB	Secondary (post-primary / reactivation) TB
Host	Unsensitized (first exposure), children	Previously sensitized
Site	Ghon focus : lower part of upper lobe / upper part of lower lobe (subpleural)	Apex of upper lobes (Simon / Assmann focus) — high O_2 tension
Lymph nodes	Hilar lymphadenopathy prominent	Minimal
Lesion	Ghon complex = Ghon focus + hilar node; calcified = Ranke complex	Cavitation (caseous necrosis erodes into bronchus)
Course	95% heal (fibrocalcification); progressive primary TB in immunocompromised	Cavitary, fibrocaseous; spread

Complications / patterns of progression :

1. Progressive pulmonary TB — apical cavities, fibrocaseous TB
2. Tuberculous bronchopneumonia (galloping consumption)
3. Miliary TB — hematogenous (pulmonary vein → systemic : liver, bone marrow, spleen, adrenals, meninges, kidneys, fallopian tubes, epididymis); millet-seed 2 mm lesions; choroid tubercles on fundoscopy
4. Isolated organ TB — meninges, kidneys, adrenals (Addison disease), bones (Pott disease — spine, thoracolumbar), fallopian tubes (tubal infertility)
5. Lymphadenitis (scrofula) — most common extrapulmonary TB; cervical nodes; collar-stud abscess
6. Intestinal TB — ileocecal; transverse (girdle) ulcers, strictures
7. Pleural — effusion (exudate, lymphocytic, ↑ ADA > 40 U/L), empyema, tuberculous pleuritis
8. Laryngitis, Rasmussen aneurysm (pulmonary artery in cavity wall → massive hemoptysis), aspergilloma in old cavity, amyloidosis (AA)

Diagnosis :

1. Sputum smear (ZN) — needs 10^4 bacilli/mL
2. Culture — Löwenstein-Jensen (gold standard, 4-8 weeks), MGIT (liquid, faster)

3. CBNAAT / GeneXpert MTB/RIF — rpoB mutation (rifampicin resistance); 2 h; Truenat (India)
4. Tuberculin skin test (Mantoux, 5 TU PPD) — type IV; read at 48–72 h; ≥ 10 mm positive (≥ 5 mm in HIV)
5. IGRA (QuantiFERON-TB Gold) — ESAT-6, CFP-10 (not in BCG — not affected by BCG vaccination)
6. Histopathology — caseating granulomas; LAM urine test (HIV)

> Exam pearl
> - False negative Mantoux : HIV / immunosuppression, sarcoidosis, measles, Hodgkin lymphoma, malnutrition, miliary TB (overwhelming).
> - Lupus vulgaris — cutaneous TB (apple-jelly nodules on diascopy); scrofuloderma, erythema induratum (Bazin) — tuberculid.
> - Atypical (NTM) mycobacteria : M. avium-intracellulare (AIDS), M. kansasii (lung), M. marinum (fish tank granuloma), M. ulcerans (Buruli ulcer), M. fortuitum/chelonae (post-surgical).

Leprosy (Hansen Disease)

- Mycobacterium leprae — weakly acid-fast (5% H_2SO_4 decolorization); grows at low temperature (30–33 °C) → skin, peripheral nerves, nose, testes, anterior eye; cannot be cultured in vitro (armadillo, mouse footpad); globi = clumps of bacilli in macrophages.
- Spread : respiratory droplets (nasal secretions) — most important.

Feature	Tuberculoid (TT)	Lepromatous (LL)
Immune response	Strong Th1 (IFN-γ, IL-2)	Th2 (IL-4, IL-10) — weak CMI, ↑ antibodies
Lesions	Few, well-defined, hypopigmented, anaesthetic macules	Multiple, symmetric nodules, leonine facies, madarosis (eyebrow loss), saddle nose
Nerves	Early asymmetric nerve involvement (thickened)	Late, symmetric glove-and-stocking
Histology	Well-formed non-caseating granulomas, nerve destruction; few/no bacilli	Foamy macrophages (lepra / Virchow cells) packed with bacilli (globi); Grenz zone (clear subepidermal zone)
Lepromin test	Positive	Negative
Bacillary index	0	5–6+

- Ridley-Jopling classification : TT - BT - BB - BL - LL (+ indeterminate).
- Lepra reactions : Type 1 (reversal reaction — type IV) — in borderline; Type 2 — erythema nodosum leprosum (ENL) — immune complex (type III) in LL/BL; treat with thalidomide.
- Lucio phenomenon — diffuse LL, necrotizing vasculitis.

Bacterial Infections

Pyogenic infections :

Organism	Lesions
Staphylococcus aureus	Abscesses, furuncle, carbuncle, impetigo, osteomyelitis, acute IE (normal valves, IV drug users — tricuspid), TSS (TSST-1), scalded skin syndrome (exfoliatin — desmoglein 1), food poisoning (enterotoxin B, 1–6 h), necrotizing pneumonia (PVL)
Streptococcus pyogenes	Pharyngitis, cellulitis, erysipelas, necrotizing fasciitis, scarlet fever (erythrogenic toxin); sequelae : rheumatic fever, PSGN
Streptococcus pneumoniae	Lobar pneumonia, otitis, meningitis (adults)

Organism	Lesions
Viridans streptococci	Subacute IE (damaged valves), dental caries (S. mutans)

Typhoid (enteric) fever :

- Salmonella Typhi (and Paratyphi A, B) — fecal-oral; enters via M cells of Peyer patches → mononuclear phagocytes → bacteremia → gallbladder (carrier state).
- Pathology : hyperplasia of Peyer patches → oval ulcers along the long axis of ileum (vs transverse in TB) → perforation (3rd week), hemorrhage; typhoid nodules (aggregates of macrophages — liver, spleen, marrow) ; Zenker's degeneration of rectus abdominis; rose spots; relative bradycardia; leukopenia.
- Diagnosis : blood culture (1st week — best), bone marrow culture (most sensitive), Widal (2nd week onward), stool/urine culture (3rd-4th week).

Mnemonic : Typhoid diagnosis by week — "BASU"
Week 1 Blood culture · Week 2 Antibodies (Widal) · Week 3 Stool culture · Week 4 Urine culture.

Diphtheria :

- Corynebacterium diphtheriae — exotoxin (phage-encoded, tox gene) — ADP-ribosylation of EF-2 → inhibits protein synthesis.
- Grey-white pseudomembrane on tonsils / pharynx / larynx — bleeds on removal; bull neck; toxin → myocarditis (most common cause of death), polyneuropathy (palatal palsy).
- Lab : Albert stain (metachromatic / volutin granules), Loeffler serum slope, potassium tellurite medium; toxigenicity — Elek test.

Gram-negative infections & bacillary dysentery :

Disease	Organism	Pathology
Bacillary dysentery	Shigella (S. dysenteriae type 1 — Shiga toxin)	Colon (rectosigmoid); M cell entry; superficial irregular ulcers, pseudomembranes; HUS; reactive arthritis
Cholera	Vibrio cholerae	Toxin → ↑ cAMP (Gs); no mucosal damage, "rice-water stools"
Plague	Yersinia pestis	Hemorrhagic lymphadenitis (bubo), pneumonia
Pseudomonas	P. aeruginosa	Burns, CF, necrotizing vasculitis (ecthyma gangrenosum), otitis externa (diabetics)
Klebsiella	K. pneumoniae	Lobar pneumonia in alcoholics — "red currant jelly" sputum, cavitation
Whooping cough	Bordetella pertussis	Lymphocytosis, laryngotracheobronchitis
Melioidosis	Burkholderia pseudomallei	Pneumonia, abscesses — diabetics (India — rising)

Syphilis :

- Treponema pallidum — spirochete; dark-field microscopy; silver stain; cannot be cultured.

Stage	Features	Pathology
Primary (~3 weeks)	Hard (indurated), painless chancre, regional lymphadenopathy	Plasma cell-rich infiltrate, obliterative endarteritis
Secondary (2-10 weeks)	Generalized maculopapular rash (palms & soles), condyloma lata, mucous patches, lymphadenopathy	Plasma cells, endarteritis; most infectious
Latent	Asymptomatic	—

Stage	Features	Pathology
Tertiary (5-20 y)	Cardiovascular (aortitis → ascending aortic aneurysm, aortic regurgitation), neurosyphilis (tabes dorsalis, general paresis, Argyll Robertson pupil), gumma	Endarteritis obliterans of vasa vasorum → "tree-bark" aorta; gumma — rubbery necrosis
Congenital	Early : snuffles, rash, hepatosplenomegaly; late : Hutchinson triad (notched incisors, interstitial keratitis, 8th nerve deafness), saber shins, saddle nose, mulberry molars	Diffuse fibrosis (pneumonia alba), osteochondritis, periostitis

- Serology : non-treponemal (VDRL, RPR — screening, follow-up; biological false positive : SLE, APS, pregnancy, malaria, leprosy, IV drug use), treponemal (FTA-ABS, TPHA — confirmation; remain positive for life).

Viral Infections

Virus	Inclusion / histology	Key pathology
Poliovirus	—	Anterior horn cells → asymmetric flaccid paralysis; no sensory loss; enters via oropharynx/intestine (Peyer patches)
Herpes simplex (HSV-1, 2)	Cowdry type A (intranuclear) inclusions; multinucleated giant cells (Tzanck smear) — 3 Ms : multinucleation, molding, margination of chromatin	Vesicles, temporal lobe hemorrhagic necrotizing encephalitis (HSV-1), keratitis, neonatal herpes (HSV-2); latency in sensory ganglia (trigeminal, sacral)
VZV	Cowdry A, Tzanck +	Chickenpox; shingles (dermatomal), latency in dorsal root ganglia
CMV	"Owl-eye" intranuclear inclusion + cytoplasmic inclusions; cytomegaly	Most common congenital infection (periventricular calcification, deafness); retinitis, colitis, pneumonitis in AIDS / transplant
EBV	—	Infectious mononucleosis : atypical lymphocytes (Downey cells — reactive CD8 T cells), heterophile antibody (Paul-Bunnell, Monospot); splenic rupture; ampicillin rash
Rabies	Negri bodies — eosinophilic cytoplasmic inclusions in Purkinje cells (cerebellum) and pyramidal cells of hippocampus (Ammon horn)	Travels retrograde along nerves; encephalitis (furious — hydrophobia; paralytic) ; Babes nodules
Measles (rubeola)	Warthin-Finkeldey giant cells (lymphoid tissue), Koplik spots (buccal mucosa)	Pneumonia (giant cell pneumonia), SSPE (years later), immune amnesia; vitamin A ↓ mortality
HPV	Koilocytes (perinuclear halo, raisinoid nucleus)	Warts, CIN, cervical carcinoma
Adenovirus	Smudge cells (intranuclear)	Pharyngoconjunctival fever, hemorrhagic cystitis
Molluscum contagiosum	Molluscum (Henderson-Patterson) bodies — large eosinophilic cytoplasmic	Umbilicated papules
Parvovirus B19	Giant pronormoblasts with glassy intranuclear inclusions	Aplastic crisis (sickle cell, HS), hydrops fetalis, erythema infectiosum ("slapped cheek")
HBV	Ground-glass hepatocytes (HBsAg)	Hepatitis, cirrhosis, HCC
Smallpox / vaccinia	Guarnieri bodies (cytoplasmic)	—

Exam pearl

- Intranuclear inclusions : herpes (HSV, VZV, CMV), adenovirus, parvovirus, polyoma (JC — PML, oligodendrocytes), measles (both).
- Cytoplasmic inclusions : rabies (Negri), poxviruses (Guarnieri, molluscum), CMV (also), measles, RSV.
- Both nuclear + cytoplasmic : CMV, measles.

Rickettsial & Chlamydial infections :

Organism	Disease	Pathology
Rickettsia prowazekii	Epidemic typhus (louse)	Endothelial infection → vasculitis; typhus nodules (brain — microglial) ; rash starts on trunk
R. rickettsii	Rocky Mountain spotted fever (tick)	Rash begins on wrists/ankles → spreads centrally, palms & soles
Orientia tsutsugamushi	Scrub typhus (India — mite, chigger)	Eschar, lymphadenopathy, ARDS, myocarditis; Weil-Felix OX-K
Coxiella burnetii	Q fever (inhalation; no rash)	Fibrin-ring granulomas, culture-negative endocarditis
Chlamydia trachomatis A-C	Trachoma	Follicular conjunctivitis, scarring — leading infectious blindness
C. trachomatis D-K	Urethritis, PID, inclusion conjunctivitis, neonatal pneumonia	Elementary body (infectious) / reticulate body (replicative); inclusions (Halberstaedter-Prowazek)
C. trachomatis L1-L3	Lymphogranuloma venereum	Stellate (suppurative) granulomas, inguinal buboes, groove sign, rectal strictures
Chlamydophila psittaci	Psittacosis	Atypical pneumonia (birds)

Fungal & Opportunistic Infections

Fungus	Morphology in tissue	Disease / notes
Candida albicans	Pseudohyphae + budding yeast; germ tube test +	Thrush, esophagitis, vaginitis, disseminated (neutropenia), endocarditis (IV drug use)
Aspergillus fumigatus	Septate hyphae branching at acute angles (45°)	Aspergilloma (fungal ball in old TB cavity — "air crescent"), ABPA (asthma, ↑ IgE, central bronchiectasis), invasive (angioinvasive, neutropenia — target lesions); aflatoxin (A. flavus)
Mucor / Rhizopus (Mucormycosis)	Broad, ribbon-like, aseptate (pauciseptate) hyphae, right-angle (90°) branching	Rhino-orbito-cerebral — diabetic ketoacidosis (COVID-associated outbreak in India), angioinvasion → infarction, black necrotic eschar
Cryptococcus neoformans	Yeast with thick mucoid capsule (India ink, mucicarmine); narrow-based budding	Meningitis in AIDS ("soap-bubble" lesions in brain), pigeon droppings; latex agglutination (capsular antigen)
Histoplasma capsulatum	Small intracellular yeasts in macrophages (2-5 μm)	Caseating granulomas, calcified, mimics TB; Ohio/Mississippi valley; bird/bat droppings
Blastomyces dermatitidis	Broad-based budding, thick double-contoured wall	Lung, skin (pseudoepitheliomatous hyperplasia)
Coccidioides immitis	Spherules with endospores (large)	Desert (San Joaquin valley fever), granulomas
Paracoccidioides	"Mariner's / pilot wheel" (multiple budding)	South America
Pneumocystis jirovecii	Foamy, frothy (cotton-candy) intra-alveolar exudate; cysts on GMS (crushed ping-pong ball / cup-shaped)	AIDS (CD4 < 200) — interstitial pneumonia; ↑ LDH; BAL; treat cotrimoxazole
Rhinosporidium seeberi	Large sporangia with endospores (strawberry-like polyps)	Nasal / ocular polyps — India (Tamil Nadu), pond bathing
Sporothrix schenckii	Cigar-shaped yeast, asteroid bodies	Rose gardener's disease — lymphocutaneous

Fungus	Morphology in tissue	Disease / notes
Madurella / Nocardia (Mycetoma)	Grains (granules) in sinus tracts — black (eumycetoma), white/red (actinomycetoma)	Madura foot — swelling + sinuses + grains
Actinomyces israelii	Sulfur granules (Gram + filaments, not acid fast)	Cervicofacial (lumpy jaw), IUCD-associated pelvic

Mnemonic : Aspergillus vs Mucor

"Aspergillus = Acute angle, septate" · "Mucor = Massive ribbons, 90°, no septa, Metabolic (DKA)".

Parasitic Diseases

Parasite	Key pathology
Plasmodium falciparum	Cerebral malaria — cytoadherence (PfEMP1 binds ICAM-1, CD36) → sequestration in brain capillaries, ring hemorrhages, Dürck granulomas; black-water fever (massive hemolysis, hemoglobinuria); malarial pigment (hemozoin) in macrophages → slate-grey spleen, liver; multiple rings, appliqué (accolé) forms, banana-shaped gametocytes; all stages of RBC infected
P. vivax / ovale	Hypnozoites (liver) → relapse; Schüffner dots; Duffy antigen receptor (vivax)
P. malariae	Quartan, nephrotic syndrome (immune complex)
Wuchereria bancrofti / Brugia malayi (Filaria)	Lymphatic obstruction → lymphedema, elephantiasis, hydrocele, chyluria; nocturnal periodicity; tropical pulmonary eosinophilia (↑ IgE, eosinophils)
Entamoeba histolytica (amoebiasis)	Flask-shaped ulcers (narrow neck, broad base) in cecum and ascending colon; trophozoites with ingested RBCs (erythrophagocytosis); amoebic liver abscess (right lobe, "anchovy sauce" pus); amoeboma
Leishmania donovani (Kala-azar)	LD bodies (amastigotes) in macrophages — spleen, liver, marrow; massive splenomegaly, hyperpigmentation, pancytopenia, hypergammaglobulinemia (aldehyde test); post-kala-azar dermal leishmaniasis; vector : sandfly (Phlebotomus argentipes); rK39 test
Taenia solium (Cysticercosis)	Neurocysticercosis — most common cause of acquired epilepsy in India; cysts with scolex (hooklets); "starry sky" on CT/MRI; calcified dead cysts
Echinococcus granulosus (Hydatid)	Cyst — outer laminated membrane (acellular, PAS +), inner germinal layer with brood capsules and protoscolices (hydatid sand), host fibrous pericyst; liver (right lobe) > lung; rupture → anaphylaxis; Casoni test (type I)
E. multilocularis	Alveolar hydatid — malignant-like infiltration
Schistosoma haematobium	Bladder — granulomas around terminal-spined eggs → SCC of bladder
S. mansoni / japonicum	Pipe-stem (Symmers) periportal fibrosis → presinusoidal portal HTN
Trichinella spiralis	Encysted larvae in skeletal muscle (diaphragm, extraocular), eosinophilia, periorbital edema
Toxoplasma gondii	Ring-enhancing brain lesions (AIDS); congenital : chorioretinitis, hydrocephalus, diffuse intracranial calcifications; lymphadenitis (Piringer-Kuchinka triad)
Trypanosoma cruzi	Chagas disease — myocarditis, megaesophagus, megacolon (destruction of myenteric plexus)
Strongyloides	Hyperinfection in steroid therapy / HTLV-1

Filled : Rank boosters — infections

- Most common opportunistic infection in AIDS in India : tuberculosis; worldwide (untreated) : PCP historically.
- Bacillary angiomatosis (Bartonella henselae) — vascular proliferation in AIDS; mimics Kaposi; neutrophils + clumps of bacteria (Warthin-Starry).
- Whipple disease — Tropheryma whipplei; PAS-positive foamy macrophages in lamina propria, arthritis, CNS, oculomasticatory myorhythmia.
- Leptospirosis (Weil disease) — jaundice, renal failure, conjunctival suffusion, pulmonary hemorrhage.
- Toxic shock syndrome — superantigen binds MHC-II + TCR Vβ outside groove → massive cytokine release.
- Cat-scratch disease — stellate necrotizing granulomas (B. henselae).
- Nipah virus (Kerala) — encephalitis, endothelial syncytia; bats (Pteropus).

GENETIC, NUTRITIONAL, ENVIRONMENTAL & PIGMENT DISORDERS

Modes of Inheritance

Mode	Key features	Examples
Autosomal dominant	Manifest in heterozygotes; 50% risk to offspring; variable expressivity, incomplete penetrance; usually structural proteins / receptors	Marfan, familial hypercholesterolemia, ADPKD, NF1, NF2, Huntington, hereditary spherocytosis, achondroplasia, VHL, FAP, tuberous sclerosis, myotonic dystrophy, osteogenesis imperfecta, acute intermittent porphyria, MEN, Li-Fraumeni
Autosomal recessive	Both alleles; 25% risk; consanguinity; complete penetrance, early onset; mostly enzyme defects	Cystic fibrosis, sickle cell, thalassemia, PKU, Wilson, hemochromatosis, α1-AT deficiency, lysosomal & glycogen storage diseases (except Hunter, Fabry), albinism, ARPKD
X-linked recessive	Males affected; no male-to-male transmission; carrier females	Duchenne/Becker, hemophilia A & B, G6PD deficiency, Fabry, Hunter, Lesch-Nyhan, Bruton, WAS, CGD (most), Ocular albinism, red-green colour blindness
X-linked dominant	Affected father → all daughters	Vitamin D-resistant (hypophosphatemic) rickets, Rett syndrome, fragile X (X-linked with anticipation), Alport (most)
Mitochondrial	Maternal inheritance; heteroplasmy	Leber hereditary optic neuropathy, MELAS, MERRF, Kearns-Sayre (ragged red fibres)

Single-gene disorders — mechanisms :

Mechanism	Disease
Enzyme defect	PKU (phenylalanine hydroxylase), Gaucher, Tay-Sachs, galactosemia
Receptor defect	Familial hypercholesterolemia (LDL receptor) — class 1 (no synthesis, most common), 2 (transport), 3, 4, 5
Transport defect	Cystic fibrosis (CFTR — ΔF508, class 2 misfolding)
Structural protein	Marfan (fibrillin-1, FBN1 — 15q21), Ehlers-Danlos (collagen), Duchenne (dystrophin)
Trinucleotide repeat	See below

Marfan syndrome :

- FBN1 (fibrillin-1) → ↑ TGF-β activity; AD.
- Skeletal : tall, long limbs (arm span > height), arachnodactyly, pectus, joint hypermobility, high-arched palate.
- Eye : bilateral ectopia lentis (upward & outward subluxation) — most characteristic.
- CVS : mitral valve prolapse (most common), aortic root dilation → cystic medial degeneration → dissection (most common cause of death).
- Losartan (blocks TGF-β signalling).

Filled : Marfan vs Homocystinuria vs Loeys-Dietz

Feature	Marfan	Homocystinuria	Loeys-Dietz
Gene	FBN1 (AD)	Cystathionine β-synthase (AR)	TGFBR1/2 (AD)
Lens	Upward / outward	Downward / inward	—
Others	Aortic root	Thrombosis, intellectual disability, osteoporosis	Bifid uvula, hypertelorism, arterial tortuosity, early dissection

Lysosomal storage diseases :

Disease	Enzyme deficiency	Accumulates	Key features
Tay-Sachs (GM2)	Hexosaminidase A	GM2 ganglioside	Cherry-red spot, no hepatosplenomegaly, onion-skin lysosomes (whorled), Ashkenazi Jews
Niemann-Pick A/B	Sphingomyelinase	Sphingomyelin	Cherry-red spot + hepatosplenomegaly, foam cells (zebra bodies)
Niemann-Pick C	NPC1/2 (cholesterol transport)	Cholesterol	Ataxia, dystonia, vertical gaze palsy
Gaucher (most common LSD)	Glucocerebrosidase (β-glucosidase)	Glucocerebroside	Gaucher cells — "crumpled tissue paper / wrinkled silk" cytoplasm (PAS+); hepatosplenomegaly, Erlenmeyer flask femur, aseptic necrosis; type 1 (non-neuronopathic) most common; enzyme replacement (imiglucerase)
Fabry (X-linked)	α-galactosidase A	Ceramide trihexoside (Gb3)	Angiokeratomas, painful neuropathy, renal failure, cardiac; zebra bodies in podocytes
Krabbe	Galactocerebrosidase	Galactocerebroside	Globoid cells, optic atrophy
Metachromatic leukodystrophy	Arylsulfatase A	Sulfatides	Metachromatic granules, demyelination
Hurler (MPS I)	α-L-iduronidase	Dermatan, heparan sulfate	Gargoylism, corneal clouding, hepatosplenomegaly, zebra bodies
Hunter (MPS II, X-linked)	Iduronate sulfatase	Dermatan, heparan sulfate	Milder, no corneal clouding, aggressive behaviour
Pompe (GSD II)	Acid maltase (α-1,4-glucosidase)	Glycogen (lysosomal)	Cardiomegaly, hypotonia; only GSD that is lysosomal

Mnemonic : "Hunters see clearly (no corneal clouding) and aim for the X (X-linked)"
Hunter = X-linked, no corneal clouding; Hurler = AR, corneal clouding.

Glycogen storage diseases :

Type	Enzyme	Features
I — von Gierke	Glucose-6-phosphatase	Severe fasting hypoglycemia, hepatomegaly, lactic acidosis, hyperuricemia, hyperlipidemia; doll-like face
II — Pompe	Lysosomal acid maltase	Cardiomegaly, infantile death
III — Cori	Debranching enzyme	Milder type I, normal lactate
V — McArdle	Muscle glycogen phosphorylase (myophosphorylase)	Exercise-induced cramps, myoglobinuria, no rise in lactate on ischemic forearm test, "second wind"

Mnemonic : GSD order — "Very Poor Carbohydrate Metabolism"
Von Gierke (I) · Pompe (II) · Cori (III) · McArdle (V).

Trinucleotide repeat disorders :

Disease	Repeat	Gene / location	Anticipation
Fragile X syndrome	CGG (> 200 full mutation)	FMR1 — 5′ UTR (methylation)	Maternal

Disease	Repeat	Gene / location	Anticipation
Huntington disease	CAG (> 40)	HTT — coding (polyglutamine)	Paternal
Myotonic dystrophy	CTG	DMPK — 3′ UTR	Maternal (congenital)
Friedreich ataxia	GAA (AR)	Frataxin — intron	—
Spinocerebellar ataxias, SBMA (Kennedy)	CAG	Coding	Paternal

- Fragile X : 2nd most common genetic cause of intellectual disability (after Down); macro-orchidism (post-pubertal), long face, large mandible, large everted ears, MVP; premutation carriers (55-200) → FXTAS (males), premature ovarian failure (females).

Cytogenetic Disorders

Disorder	Karyotype	Key features
Down syndrome	Trisomy 21 — 95% meiotic non-disjunction (maternal, ↑ age); 4% Robertsonian translocation t(14;21) (not age-related, recurrence risk ↑); 1% mosaic	Most common chromosomal disorder; flat facies, epicanthic folds, simian crease, endocardial cushion defects (AV canal — most common), duodenal atresia, Hirschsprung, ALL (10-20×), AML-M7 (megakaryoblastic) & transient myeloproliferative disorder, early Alzheimer (APP on chr 21), hypothyroidism, atlantoaxial instability. Screening : ↑ β-hCG, ↑ inhibin A, ↓ AFP, ↓ estriol, ↑ nuchal translucency, ↓ PAPP-A
Edwards syndrome	Trisomy 18	Rocker-bottom feet, clenched fists with overlapping fingers, micrognathia, prominent occiput, VSD; ↓ all screening markers
Patau syndrome	Trisomy 13	Cleft lip/palate, holoprosencephaly, polydactyly, microphthalmia, cutis aplasia, rocker-bottom
22q11.2 deletion	Microdeletion (FISH)	DiGeorge / velocardiofacial — CATCH 22
Klinefelter syndrome	47,XXY (most common cause of male hypogonadism)	Tall, eunuchoid habitus, gynecomastia, small firm testes, azoospermia, ↑ FSH, LH, ↓ testosterone; Barr body +; ↑ breast Ca, SLE
Turner syndrome	45,X (57%), mosaics	Short stature, webbed neck (cystic hygroma), coarctation of aorta, bicuspid aortic valve, streak ovaries (primary amenorrhea), horseshoe kidney, shield chest, lymphedema of hands/feet at birth; no Barr body
Cri-du-chat	5p deletion	Cat-like cry, microcephaly, VSD
Williams	7q11.23 deletion (elastin)	Elfin facies, hypercalcemia, supravalvular aortic stenosis, cocktail-party personality
Prader-Willi	15q11-13 deletion (paternal) or maternal uniparental disomy	Hyperphagia, obesity, hypogonadism, hypotonia
Angelman	15q11-13 deletion (maternal) or paternal UPD (UBE3A)	"Happy puppet" — inappropriate laughter, seizures, ataxia

Filled : Genomic imprinting & special mechanisms

- Imprinting — gene silenced by methylation depending on parental origin : Prader-Willi (paternal loss), Angelman (maternal loss), Beckwith-Wiedemann (11p15 — IGF2).
- Gonadal mosaicism — recurrence in sibs with normal parents (osteogenesis imperfecta).
- Lyonization — random X inactivation → manifesting female carriers (DMD, hemophilia).
- Investigations : karyotyping (G-banding), FISH (microdeletions), chromosomal microarray (CMA — first-tier for intellectual disability), NIPT (cell-free fetal DNA) — screening at ≥ 10 weeks.

Protein–Energy Malnutrition (PEM)

Feature	Marasmus	Kwashiorkor
Deficiency	Calories (total energy)	Protein (relative) with adequate calories
Weight for age	< 60%	60–80%
Edema	Absent	Present (hypoalbuminemia) — mask weight loss
Serum albumin	Normal / mildly ↓	Markedly ↓
Appearance	"Old man" face, extreme muscle wasting, loss of subcutaneous fat, monkey face	Moon face, apathy, irritability
Skin	Loose folds	Flaky paint dermatosis, alternating zones of hyper/hypopigmentation
Hair	Normal / sparse	Flag sign (alternating bands of pale and dark hair)
Liver	Normal	Enlarged fatty liver (↓ apolipoprotein synthesis)
Other	—	Small bowel mucosal atrophy (reversible lactase deficiency), immune deficiency, anemia

- Secondary PEM — chronically ill, elderly, cancer (cachexia — TNF, proteolysis-inducing factor).
- Anorexia nervosa : amenorrhea (↓ GnRH), ↓ thyroid, osteoporosis, arrhythmias (hypokalemia) — most serious.
  Bulimia : hypokalemia, aspiration, Mallory-Weiss tears, Boerhaave, Russell sign.

Vitamin Deficiencies

Vitamin	Deficiency	Key pathology
A (retinol)	Night blindness (earliest), xerophthalmia, Bitot spots, keratomalacia (blindness)	Squamous metaplasia (respiratory, urinary — stones), follicular hyperkeratosis (phrynoderma), ↑ infections (measles mortality); ATRA treats APL
D	Rickets (children), osteomalacia (adults)	Excess unmineralized osteoid; rachitic rosary, Harrison groove, pigeon breast, craniotabes (earliest), frontal bossing, widened wrists, bowlegs; ↓ Ca, ↓ PO_4, ↑ ALP, ↑ PTH
C (ascorbic acid)	Scurvy	Defective collagen hydroxylation → perifollicular hemorrhages, corkscrew hairs, swollen bleeding gums, subperiosteal hemorrhage (children), poor wound healing, scorbutic rosary (sharp); Frankel white line, Wimberger ring
E	Spinocerebellar degeneration, hemolytic anemia (preterm)	Posterior column & spinocerebellar degeneration, retinopathy; fat malabsorption (abetalipoproteinemia, CF)
K	Bleeding diathesis — hemorrhagic disease of newborn	↓ II, VII, IX, X, protein C/S; ↑ PT; newborn (sterile gut, low placental transfer, breast milk) — prophylactic IM vit K at birth
B1 (thiamine)	Beriberi (dry — neuropathy; wet — high-output cardiac failure); Wernicke-Korsakoff (alcoholics)	Wernicke : ophthalmoplegia, ataxia, confusion — mammillary bodies hemorrhage, periaqueductal grey; Korsakoff — confabulation, memory loss
B2 (riboflavin)	Ariboflavinosis	Cheilosis, angular stomatitis, glossitis (magenta tongue), interstitial keratitis, seborrheic dermatitis
B3 (niacin)	Pellagra — "3 Ds" : dermatitis, diarrhea, dementia (+ death)	Casal necklace, Hartnup disease (tryptophan absorption ↓), carcinoid syndrome (tryptophan diverted), isoniazid, maize (jowar) diets
B6 (pyridoxine)	Seborrheic dermatitis, cheilosis, peripheral neuropathy, convulsions (infants), sideroblastic anemia	Isoniazid therapy
B12 / folate	Megaloblastic anemia	B12 also : subacute combined degeneration (posterior + lateral columns) — see Hematology

Vitamin	Deficiency	Key pathology
Biotin	Dermatitis, alopecia	Raw egg white (avidin)

Radiation Injury

- Ionizing radiation → direct DNA damage + free radicals (hydrolysis of water — OH•).
- Units : Gray (Gy) = absorbed dose (1 Gy = 100 rad); Sievert (Sv) = equivalent / effective dose (biological effect).

Radiosensitivity :

Highly radiosensitive	Radioresistant
Lymphocytes (most sensitive), hematopoietic cells, gonads (germ cells), intestinal epithelium, lens (cataract), lymphoid tissue, skin basal cells	Neurons, muscle, bone, cartilage, mature connective tissue
Tumours : seminoma, dysgerminoma, lymphoma, Ewing sarcoma, Wilms, myeloma	Melanoma, osteosarcoma, RCC, pancreas

- Law of Bergonié & Tribondeau — sensitivity ∝ mitotic activity and inversely ∝ differentiation; G2/M phase most sensitive, late S phase most resistant.
- Hypoxic cells are radioresistant (oxygen enhancement effect).

Morphology :

1. Vascular changes — endothelial swelling, fibrinoid necrosis, intimal fibrosis, hyalinization → obliterative ischemia (late)
2. Bizarre fibroblasts (radiation atypia) with enlarged hyperchromatic nuclei
3. Interstitial fibrosis (lung — radiation pneumonitis → fibrosis)
4. Cell death (apoptosis), chromosomal aberrations

Acute radiation syndrome (whole body) :

Dose	Syndrome
0.5-1 Sv	Subclinical / mild — lymphopenia
2-10 Sv	Hematopoietic syndrome — pancytopenia (2-3 weeks), infection, bleeding
10-20 Sv	Gastrointestinal syndrome — diarrhea, sepsis; death in 1-2 weeks
> 50 Sv	Cerebrovascular (CNS) syndrome — death in 1-2 days

- Earliest sign : lymphocyte depletion (absolute lymphocyte count at 48 h = best prognostic indicator).
- Late effects : cancer (leukemia — earliest, papillary thyroid Ca in children — Chernobyl), cataract, sterility, fibrosis, angiosarcoma (breast after RT).

Pigments

Pigment	Source / features	Conditions
Lipofuscin ("wear-and-tear" / age pigment)	Lipid peroxidation of membranes — residual bodies; yellow-brown, perinuclear; autofluorescent	Brown atrophy (heart, liver) of aging, cachexia; not injurious — marker of past free-radical injury
Melanin	Tyrosine → tyrosinase → DOPA → melanin (melanocytes); Masson-Fontana (black), bleached by H_2O_2	Hyperpigmentation : Addison, Peutz-Jeghers, melasma, McCune-Albright (café-au-lait). Hypopigmentation : albinism (tyrosinase deficiency), vitiligo, PKU
Homogentisic acid	Black pigment in connective tissue — ochronosis	Alkaptonuria (homogentisate oxidase deficiency, AR) — urine darkens on standing, arthropathy, ear/sclera pigmentation

Pigment	Source / features	Conditions
Hemosiderin	Hb-derived; golden-yellow to brown granules; ferritin micelles; Perls Prussian blue — blue	Local : bruise, heart failure cells. Systemic (hemosiderosis) : ↑ absorption, hemolytic anemia, repeated transfusions; hereditary hemochromatosis (HFE C282Y) → cirrhosis, DM, skin pigmentation ("bronze diabetes"), cardiomyopathy
Bilirubin	Normal pigment of bile (heme → biliverdin → bilirubin); green-yellow; no iron	Jaundice; kernicterus (unconjugated bilirubin in basal ganglia of neonates); bile lakes, bile infarcts in obstruction
Hematin	Hb degradation (acid hematin)	Malaria pigment (hemozoin), gastric acid (coffee-ground vomitus)
Carbon (anthracotic)	Coal dust (exogenous)	Anthracosis — black lungs, hilar nodes
Tattoo	Exogenous	Dermal macrophages
Argyria / plumbism	Silver / lead	Blue-grey skin; Burton line (lead) at gum margin

Exam pearl

- Hemosiderin (iron) vs lipofuscin (no iron) vs melanin : Perls positive only hemosiderin; melanin bleaches with $KMnO_4/H_2O_2$; lipofuscin stains with Sudan black / PAS and autofluoresces.
- Hemochromatosis : iron in hepatocytes (periportal, zone 1) and pancreatic acini/islets, heart; HFE on chromosome 6p (linked to HLA-A3); ↑ transferrin saturation (> 45%) — best screening; risk of HCC (200×).
- Transfusional (secondary) hemosiderosis — iron first in reticuloendothelial (Kupffer) cells rather than hepatocytes.

Environmental & Nutritional Miscellany

Agent	Key pathology
Lead	Microcytic hypochromic anemia with basophilic stippling, inhibits ALA dehydratase & ferrochelatase (↑ ZPP), ringed sideroblasts, Burton gum line, lead lines in metaphyses, wrist/foot drop, encephalopathy (children); Fanconi-like tubular damage
Mercury	Minamata disease (methylmercury — cerebral palsy, blindness); acrodynia (pink disease); PCT necrosis ($HgCl_2$)
Arsenic	Raindrop pigmentation, palmar/plantar hyperkeratosis, Mees lines (nails), skin SCC, Bowen disease, blackfoot disease; West Bengal groundwater
Cadmium	Itai-itai disease — osteoporosis + renal damage; lung (emphysema)
Carbon monoxide	Cherry-red discoloration; bilateral globus pallidus necrosis
Alcohol (ethanol)	Fatty liver, alcoholic hepatitis (Mallory-Denk bodies), cirrhosis, Wernicke, cardiomyopathy, pancreatitis, fetal alcohol syndrome (smooth philtrum, thin upper lip, microcephaly)
Tobacco	Most common avoidable cause of human death; lung, oral, larynx, esophagus, bladder, pancreas, kidney, cervix cancer; COPD, atherosclerosis
Obesity	Leptin (↓ appetite — from fat), adiponectin (↓ in obesity — insulin sensitizer), ghrelin (↑ appetite — stomach); metabolic syndrome, NAFLD, OSA, cancers (endometrium, breast, colon)

CARDIOVASCULAR SYSTEM

Blood Vessels — Atherosclerosis

- Disease of large and medium-sized elastic and muscular arteries characterized by intimal lesions (atheromas / fibrofatty plaques).
- Most commonly involved vessels (descending order) : abdominal aorta (infrarenal) > coronary arteries > popliteal arteries > internal carotid arteries > circle of Willis. Upper limb, renal, mesenteric arteries usually spared (except at ostia).

Risk factors :

Non-modifiable (constitutional) :

1. Age (40-60 years — increasing incidence)
2. Male gender (premenopausal women protected)
3. Genetics / family history (most important independent risk factor; familial hypercholesterolemia)

Modifiable (major) :

1. Hyperlipidemia (↑ LDL, ↓ HDL) — most important modifiable
2. Hypertension
3. Cigarette smoking
4. Diabetes mellitus (↑ MI 2×, ↑ gangrene of legs 100×)
5. Inflammation — ↑ CRP (hs-CRP)

Additional :

1. Hyperhomocysteinemia
2. Metabolic syndrome, obesity
3. Lipoprotein(a) (altered apo B with apo(a))
4. Factors affecting hemostasis (↑ PAI-1, fibrinogen)
5. Lack of exercise, stress (type A personality)

Pathogenesis — "response to injury" hypothesis :

Chronic endothelial injury / dysfunction (hyperlipidemia, HTN, smoking, turbulence — branch points) → ↑ permeability, leukocyte adhesion (VCAM-1)

↓

LDL accumulates in intima → oxidized LDL

↓

Monocytes adhere, migrate → macrophages ingest oxLDL via scavenger receptors (CD36, SR-A) → foam cells

↓

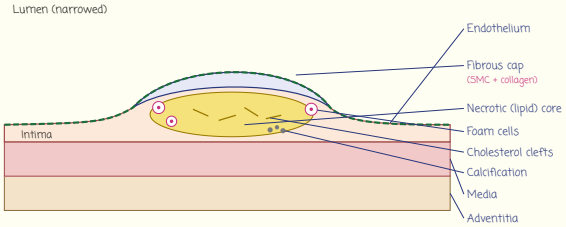
Fatty streak (earliest lesion) ; inflammasome (cholesterol crystals) → IL-1

↓

Platelet adhesion + cytokines (PDGF, FGF, TGF-β) → SMC migration from media to intima & proliferation + ECM (collagen)

↓

Fibrofatty atheroma : fibrous cap + necrotic core



Structure of a fibrofatty atheroma (labelled)

Morphology :

Lesion	Features
Fatty streak	Earliest lesion; flat yellow spots; lipid-filled foam cells; seen in children < 1 year (aorta); not raised, no flow disturbance
Atheroma (fibrofatty plaque)	Raised white-yellow plaque; superficial fibrous cap (SMC, collagen); shoulder (macrophages, T cells, SMC); necrotic core (lipid, cholesterol clefts, foam cells, debris); neovascularization
Complicated plaque	Rupture / ulceration / erosion → thrombosis; hemorrhage into plaque; atheroembolism; aneurysm (media atrophy); calcification

- Vulnerable (unstable) plaque : thin fibrous cap, large lipid core, many inflammatory cells (macrophages — MMPs) → rupture → acute coronary syndrome. Stable plaque : thick cap, small core.

Exam pearl

- Hallmark cell of atherosclerosis : foam cell (macrophage); key cell for plaque growth : smooth muscle cell.
- Statins stabilize plaque (↓ inflammation, ↓ LDL) in addition to lipid lowering.
- Mönckeberg medial calcific sclerosis — calcification of media of muscular arteries, does not narrow lumen, > 50 years; radiologic "pipe-stem".

Arteriolosclerosis & Hypertension

Type	Associated with	Morphology
Hyaline arteriolosclerosis	Benign (essential) hypertension, diabetes, aging	Homogeneous pink hyaline thickening of arteriolar walls (plasma protein leakage + ↑ ECM) → narrowed lumen → benign nephrosclerosis (granular contracted kidney)
Hyperplastic arteriolosclerosis	Malignant hypertension (> 180/120 mmHg)	"Onion-skin" concentric laminated thickening (SMC + reduplicated basement membrane) + fibrinoid necrosis (necrotizing arteriolitis) — "flea-bitten kidney" (petechiae)

Secondary causes of hypertension :

1. Renal — renal artery stenosis (atherosclerosis in elderly, fibromuscular dysplasia in young women — "string of beads"), GN, CKD, PKD, renin-producing tumours
2. Endocrine — primary aldosteronism (Conn — most common endocrine cause), Cushing, pheochromocytoma, acromegaly, thyroid disease, OCPs
3. Cardiovascular — coarctation of aorta (upper limb HTN, rib notching)
4. Neurologic — raised ICP, sleep apnea
5. Pregnancy — preeclampsia

Hypertensive heart disease :

- Systemic (left-sided) : concentric LVH (wall > 1.5–2 cm) without other cause + history/evidence of HTN; myocyte hypertrophy (boxcar nuclei), interstitial fibrosis → diastolic dysfunction, AF, CHF.

- Pulmonary (right-sided) — cor pulmonale : RVH due to pulmonary hypertension from lung disease (COPD — most common, ILD, PE, kyphoscoliosis, OSA).

Aneurysms & Dissection

Type	Cause / features
Abdominal aortic aneurysm (AAA)	Atherosclerosis; infrarenal, above bifurcation; > 50 y, men, smokers; rupture risk ↑ when > 5 cm (surgery at 5.5 cm); inflammatory AAA (IgG4)
Thoracic (ascending) aortic aneurysm	Hypertension, Marfan, Loeys-Dietz, bicuspid valve, tertiary syphilis (tree-bark intima, obliterative endarteritis of vasa vasorum)
Berry (saccular) aneurysm	Circle of Willis (anterior communicating artery most common); ADPKD, Ehlers-Danlos, coarctation; rupture → subarachnoid hemorrhage
Charcot-Bouchard (micro)aneurysm	Small penetrating arteries (lenticulostriate) — hypertension → intracerebral hemorrhage (basal ganglia)
Mycotic aneurysm	Infection of the wall (septic emboli in IE, Salmonella)
Pseudoaneurysm (false)	Breach in wall with extravascular hematoma communicating with lumen (post-MI ventricular rupture, catheterization)
Kawasaki	Coronary artery aneurysm in children

Aortic dissection :

- Blood enters media via intimal tear → separates layers.
- Causes : hypertension (> 90% in men 40-60 y); young — Marfan, Ehlers-Danlos, pregnancy, iatrogenic.
- Underlying lesion : cystic medial degeneration (mucoid degeneration) — fragmentation of elastic tissue + pooling of ground substance (no inflammation).
- Classification : Stanford type A (proximal — ascending ± descending; more common, more dangerous, surgery) ; type B (distal — beyond subclavian; medical management). DeBakey I, II (ascending), III (descending).
- Presentation : sudden excruciating tearing chest pain radiating to back, pulse deficit, widened mediastinum; death from rupture into pericardium (tamponade), pleural or peritoneal cavity.

Vasculitis

Vessel size	Disease	Key features
Large	Giant cell (temporal) arteritis	> 50 y, most common vasculitis in elderly; granulomatous inflammation of media with giant cells, fragmentation of internal elastic lamina; temporal artery — headache, jaw claudication, visual loss (ophthalmic artery); ↑ ESR; polymyalgia rheumatica; skip lesions (biopsy ≥ 1 cm); steroids
	Takayasu arteritis	Young Asian women (< 40 y); "pulseless disease", aortic arch and branches (subclavian); granulomatous; ↓ upper-limb pulses, BP difference; ocular disturbances
Medium	Polyarteritis nodosa (PAN)	Young adults; HBV (30%); renal, mesenteric, skin, nerves — spares lungs; segmental transmural necrotizing inflammation, fibrinoid necrosis; lesions in different stages; microaneurysms (rosary); ANCA negative; mononeuritis multiplex
	Kawasaki disease	< 4 years; mucocutaneous lymph node syndrome — fever > 5 days, conjunctival injection, strawberry tongue, cracked lips, desquamation of palms/soles, cervical lymphadenopathy; coronary artery aneurysm (MI); IVIG + aspirin
	Thromboangiitis obliterans (Buerger)	Heavy smokers (young men, India, bidi); tibial & radial arteries; segmental thrombosing acute & chronic inflammation, microabscesses in thrombus, extends to adjacent veins and nerves; Raynaud, superficial migratory thrombophlebitis, gangrene, claudication of instep

Vessel size	Disease	Key features
Small	Granulomatosis with polyangiitis (Wegener)	c-ANCA (PR3); triad : upper respiratory (nasal ulcers – saddle nose, sinusitis), lungs (nodules, cavities), kidney (crescentic GN – pauci-immune); necrotizing granulomas + vasculitis; cyclophosphamide/rituximab
	Microscopic polyangiitis	p-ANCA (MPO); no granulomas; lung capillaritis (hemoptysis) + necrotizing GN; leukocytoclastic
	EGPA (Churg-Strauss)	p-ANCA (40%); asthma, allergic rhinitis, eosinophilia, granulomas; heart (eosinophilic myocarditis – major cause of death), neuropathy
	IgA vasculitis (Henoch-Schönlein purpura)	Children; palpable purpura (buttocks, legs), arthralgia, abdominal pain (intussusception), GN (IgA mesangial) ; after URTI
	Cryoglobulinemic vasculitis	HCV – mixed cryoglobulins; purpura, arthralgia, MPGN
	Hypersensitivity (leukocytoclastic) vasculitis	Drugs, infection; all lesions same age; nuclear debris (leukocytoclasis)
	Behçet disease	HLA-B51; oral & genital aphthous ulcers, uveitis, pathergy

Mnemonic : ANCA

"C-ANCA = Cytoplasmic = PR3 = granulomatosis (Wegener)" · "P-ANCA = Perinuclear = MPO = MPA, EGPA, UC, PSC, crescentic GN".

Vascular tumours :

Tumour	Key features
Capillary hemangioma (juvenile/strawberry)	Most common tumour of infancy; grows rapidly, regresses by 7 years; GLUT-1 +
Cavernous hemangioma	Large dilated channels; von Hippel-Lindau (cerebellum, retina); does not regress
Pyogenic granuloma (lobular capillary hemangioma)	Rapidly growing, bleeds; pregnancy (granuloma gravidarum)
Glomus tumour	Painful, subungual; modified SMC of glomus body
Kaposi sarcoma	HHV-8; types : classic (elderly Mediterranean men, legs), endemic African, transplant, AIDS (most aggressive); spindle cells + slit-like vascular spaces with extravasated RBCs + hyaline globules; LANA-1 +
Angiosarcoma	Liver – vinyl chloride, arsenic, Thorotrast; breast – post-mastectomy lymphedema (Stewart-Treves), post-RT; CD31, ERG
Bacillary angiomatosis	Bartonella – AIDS; neutrophils
Hemangioblastoma	VHL – cerebellum; erythropoietin (polycythemia)

Ischemic Heart Disease

- > 90% due to coronary atherosclerosis; LAD (40-50%) > RCA (30-40%) > LCX (15-20%).

Territory	Infarct region
LAD	Anterior wall of LV, anterior 2/3 of interventricular septum, apex – "widow maker"
RCA	Posterior (inferior) wall of LV, posterior 1/3 septum, RV free wall; SA node (60%), AV node (90%) – bradyarrhythmias
LCX	Lateral wall of LV

Myocardial infarction — sequence of changes :

Time	Gross	Light microscopy	EM / other
0-30 min	None	None	Reversible — relaxation of myofibrils, glycogen loss, mitochondrial swelling
30 min-4 h	None	Usually none; waviness of fibres at border	Irreversible — sarcolemmal disruption, mitochondrial amorphous densities
4-12 h	Occasionally dark mottling	Early coagulation necrosis, edema, hemorrhage	TTC stain : unstained (pale) infarct
12-24 h	Dark mottling	Ongoing coagulation necrosis; pyknosis, hypereosinophilia; marginal contraction band necrosis; early neutrophilic infiltrate	
1-3 days	Mottling with yellow-tan center	Coagulation necrosis with loss of nuclei and striations; brisk neutrophilic infiltrate	Fibrinous pericarditis (day 2-3)
3-7 days	Hyperemic border, central yellow-tan softening	Beginning disintegration of dead fibres, dying neutrophils, early macrophage phagocytosis at border	Highest risk of free-wall rupture (3-7 days)
7-10 days	Maximally yellow-tan and soft, depressed red-tan margins	Well-developed phagocytosis; early granulation tissue at margins	
10-14 days	Red-grey depressed borders	Well-established granulation tissue, new vessels, collagen	
2-8 weeks	Grey-white scar, progressive from border to core	↑ collagen, ↓ cellularity	Dressler syndrome (2-10 weeks — autoimmune pericarditis)
> 2 months	Scarring complete	Dense collagenous scar	Ventricular aneurysm

Cardiac biomarkers :

Marker	Rises	Peaks	Returns to normal	Notes
Troponin I / T (cTnI, cTnT)	2-4 h (hs-Tn 1 h)	24-48 h	7-10 days (I), 10-14 days (T)	Most sensitive & specific; diagnosis of late MI
CK-MB	2-4 h	24-48 h	48-72 h	Best for re-infarction
Myoglobin	1-2 h (earliest)	6-12 h	24 h	Not specific
LDH-1 (flipped LDH 1>2)	24 h	3-6 days	8-14 days	Obsolete

Complications of MI :

1. Arrhythmias — most common cause of death in first hours (VF); most common complication
2. Contractile dysfunction / LV failure, cardiogenic shock (infarct > 40% of LV)
3. Myocardial rupture (3-7 days) :
   - Free wall rupture → hemopericardium, cardiac tamponade (most common rupture; elderly women, first MI, no LVH)
   - Interventricular septal rupture → left-to-right shunt
   - Papillary muscle rupture → acute mitral regurgitation (posteromedial — RCA)
4. Pericarditis — fibrinous (2-3 days); Dressler (weeks)

5. Right ventricular infarction (with inferior MI) — hypotension, clear lungs, ↑ JVP
6. Infarct expansion, mural thrombus → systemic embolism
7. Ventricular aneurysm (late) — persistent ST elevation
8. Papillary muscle dysfunction, progressive late heart failure

Filled : Rank boosters — MI & reperfusion

- Subendocardial infarct — inner 1/3-1/2 of wall; NSTEMI; hypotension, diffuse coronary disease. Transmural — STEMI.
- Reperfusion injury : contraction band necrosis (Ca^{2+} influx), hemorrhagic infarct, myocardial stunning (reversible dysfunction for days), reperfusion arrhythmias, microvascular obstruction (no-reflow).
- Hibernating myocardium — chronic ischemia, reduced function, recovers after revascularization.
- Sudden cardiac death : most common cause — coronary atherosclerosis (lethal arrhythmia); young — HOCM, myocarditis, long QT, Brugada, ARVC, anomalous coronary.
- Earliest histological change of MI on light microscopy : wavy fibres (within hours); earliest EM change : glycogen depletion / mitochondrial swelling.
- Takotsubo (stress) cardiomyopathy — apical ballooning, catecholamines, women.

Rheumatic Fever & Rheumatic Heart Disease

- Acute immune-mediated multisystem disease following group A β-hemolytic streptococcal pharyngitis (not skin infections) after 2-3 weeks; 5-15 years.
- Mechanism : molecular mimicry — antibodies & CD4+ T cells against streptococcal M protein cross-react with cardiac myosin, laminin, valve glycoproteins (type II + type IV hypersensitivity).

Jones criteria (revised 2015) :

Major (J♥NES)	Minor
Joints — migratory polyarthritis (large joints) — most common	Fever (≥ 38.5 °C)
♥ Carditis (pancarditis) — clinical / subclinical (echo)	Arthralgia
Nodules (subcutaneous) — painless, extensor surfaces	↑ ESR (≥ 60) / CRP (≥ 3)
Erythema marginatum — trunk, evanescent	Prolonged PR interval
Sydenham chorea — late, may be isolated	

- Diagnosis : evidence of preceding GAS infection (↑ ASO, anti-DNase B, throat culture) + 2 major or 1 major + 2 minor.

Morphology :

- Aschoff body (pathognomonic) : foci of fibrinoid necrosis surrounded by T lymphocytes, plasma cells, and plump macrophages — Anitschkow cells (caterpillar cells / owl-eye nuclei — chromatin condensed into central wavy ribbon); multinucleate = Aschoff giant cells. Found in any layer of heart (pancarditis).
- Pericardium : fibrinous / serofibrinous "bread-and-butter" pericarditis.
- Myocardium : myocarditis — interstitial Aschoff bodies → cardiac dilation, most common cause of death in acute RF.
- Endocardium & valves : verrucous vegetations — small (1-2 mm), warty, along lines of closure; MacCallum plaques — subendocardial lesions in left atrium (jet lesions).
- Chronic RHD : organization → fibrous thickening, commissural fusion, shortening & thickening of chordae → fish-mouth / buttonhole stenosis; neovascularization; mitral (65-70%) alone > mitral + aortic (25%) ; tricuspid less, pulmonary rarely.

Exam pearl

- Mitral stenosis — most common valvular lesion in RHD; virtually always rheumatic.
- MS → LA dilation → AF, mural thrombi (left atrial appendage) → embolism, pulmonary venous HTN → brown induration, RVH, hemoptysis, hoarseness (Ortner — recurrent laryngeal compression).
- Anitschkow cells = activated macrophages; Aschoff bodies are replaced by scar in chronic RHD.

Infective Endocarditis (IE)

Feature	Acute IE	Subacute IE
Organism	Staphylococcus aureus (highly virulent)	Viridans streptococci (S. sanguinis, mutans) (low virulence)
Valve	Previously normal valves	Previously damaged valves (RHD, congenital, MVP)
Course	Destructive, rapid (days), high mortality	Insidious (weeks-months); recovery after antibiotics
Vegetation	Large, destructive (ring abscess, perforation)	Smaller, less destructive

Other organisms :

1. IV drug users — S. aureus, tricuspid valve, also Pseudomonas, Candida
2. Prosthetic valve — Staph. epidermidis (CoNS) (early), streptococci (late)
3. Streptococcus gallolyticus (bovis) → colon carcinoma — colonoscopy mandatory
4. HACEK group (Haemophilus, Aggregatibacter, Cardiobacterium, Eikenella, Kingella) — culture-negative IE; also Coxiella, Bartonella, Brucella, fungi
5. Enterococci — elderly men after GU procedures

Morphology & complications :

- Friable, bulky, destructive vegetations (fibrin, inflammatory cells, bacteria) on atrial surface of AV valves / ventricular surface of semilunar valves; mitral > aortic.
- Cardiac : valvular insufficiency (perforation), ring abscess, suppurative pericarditis, conduction defects.
- Embolic : septic infarcts (brain, spleen, kidney), mycotic aneurysms.
- Immune complex : focal segmental (embolic) GN — "flea-bitten kidney"; Roth spots (retina), Osler nodes (painful, finger pads), Janeway lesions (painless, palms/soles — microemboli), splinter hemorrhages; clubbing, splenomegaly.
- Modified Duke criteria : major — positive blood cultures (typical organism, 2 separate), echo evidence of endocardial involvement; minor — predisposition, fever, vascular & immunologic phenomena.

Non-infective vegetations :

Type	Features	Associated with
Rheumatic (verrucous)	Small, warty, along lines of closure, non-destructive	Acute RF
Infective	Large, irregular, friable, destructive; bacteria	IE
NBTE (marantic)	Small, sterile, bland, loosely attached along lines of closure; no inflammation	Hypercoagulable states, mucin-secreting adenocarcinoma (pancreas), chronic debilitating illness; embolize
Libman-Sacks (SLE)	Small-medium, sterile, on both surfaces (undersurface) of valves, chordae; fibrinoid necrosis	SLE, APS
Carcinoid heart disease	Fibrous plaques on endocardium — right heart (tricuspid, pulmonary)	Serotonin from liver metastases of carcinoid (inactivated in lung)

Valvular Heart Disease — Other

Lesion	Cause	Key features
Calcific aortic stenosis	Age-related ("wear and tear", 70-80 y); bicuspid aortic valve (50-60 y)	Heaped-up calcified masses within cusps (sinuses of Valsalva), no commissural fusion (vs rheumatic); angina, syncope, CHF ; concentric LVH
Bicuspid aortic valve	Most common congenital heart anomaly (1-2%); NOTCH1	Premature calcific AS, aortic dilatation / dissection, coarctation, Turner
Mitral valve prolapse (myxomatous degeneration)	Marfan, idiopathic (2-3%), young women	Ballooning (hooding) of mitral leaflets into LA ; mid-systolic click; intercalated myxoid degeneration of spongiosa
Mitral annular calcification	Elderly women	Calcium in annulus; arrhythmias, embolism

Cardiomyopathies

Feature	Dilated (DCM)	Hypertrophic (HCM)	Restrictive (RCM)
Function	Systolic dysfunction (↓ EF < 40%)	Diastolic dysfunction, outflow obstruction (asymmetric septal)	Diastolic dysfunction (stiff, non-dilated)
Causes	Idiopathic, genetic (TTN — titin; most common), alcohol, myocarditis (Coxsackie B), doxorubicin, trastuzumab, peripartum, Chagas, hemochromatosis, thyrotoxicosis, beriberi, cocaine, DMD	Genetic (AD) — mutations in sarcomere proteins : β-myosin heavy chain (MYH7 — most common), myosin-binding protein C (MYBPC3) ; Friedreich ataxia, Fabry, Pompe	Amyloidosis (most common), sarcoidosis, radiation, endomyocardial fibrosis (tropical — Kerala, children), Loeffler endocarditis (eosinophils), hemochromatosis, endocardial fibroelastosis (infants)
Morphology	Four-chamber dilatation, mural thrombi, flabby	Massive myocardial hypertrophy without dilation, asymmetric septal hypertrophy (septum : free wall > 1.3), banana-shaped LV cavity ; myocyte disarray (haphazard), interstitial fibrosis	Normal size, firm; biatrial dilation
Clinical	CHF, arrhythmias, embolism	Sudden death in young athletes, systolic murmur ↑ with Valsalva / standing (↓ preload); SAM of mitral valve	

- Arrhythmogenic RV cardiomyopathy (ARVC) : desmosomal protein mutations (plakoglobin, desmoplakin, plakophilin-2); fibrofatty replacement of RV myocardium; Naxos syndrome (palmoplantar keratoderma + woolly hair); sudden death.
- Left ventricular non-compaction — spongy trabeculated myocardium.

Myocarditis :

Cause	Features
Viral (Coxsackie A & B — most common), enteroviruses, CMV, HIV, influenza, SARS-CoV-2	Lymphocytic myocarditis with myocyte necrosis (Dallas criteria)
Hypersensitivity (drug)	Eosinophils, perivascular
Giant cell myocarditis	Multinucleate giant cells, eosinophils, necrosis — poor prognosis

Cause	Features
Chagas disease (T. cruzi)	Parasites in myofibres, Latin America
Diphtheria	Toxin — myocyte necrosis
Lyme disease	Conduction block

Pericardial Disease

Type	Causes / features
Serous	Non-infectious — RF, SLE, scleroderma, tumours, uremia
Fibrinous / serofibrinous	Most common type — acute MI, Dressler, uremia, radiation, RF, SLE, trauma ; friction rub
Purulent (suppurative)	Bacterial, fungal, parasitic invasion
Hemorrhagic	Malignancy (most common), TB, bleeding diathesis, post-cardiac surgery
Caseous	Tuberculosis — leads to constrictive pericarditis
Chronic constrictive pericarditis	TB (India), post-surgical, radiation; thick dense fibrous / calcified "concrete" pericardium (up to 1 cm) → impaired diastolic filling, Kussmaul sign, pericardial knock, pulsus paradoxus (less)
Cardiac tamponade	Beck triad (hypotension, ↑ JVP, muffled heart sounds), pulsus paradoxus, electrical alternans

Cardiac Tumours

Tumour	Features
Metastasis	Most common cardiac tumours (lung, breast, melanoma, lymphoma)
Myxoma	Most common primary cardiac tumour in adults; left atrium (90%), fossa ovalis; pedunculated, gelatinous; "ball-valve" obstruction, tumour plop, syncope, embolism, constitutional symptoms (IL-6); Carney complex (PRKAR1A)
Rhabdomyoma	Most common primary cardiac tumour in children/infants; tuberous sclerosis; spider cells; ventricles; regress
Papillary fibroelastoma	Most common tumour of valves; "sea anemone"
Lipoma, angiosarcoma (most common primary malignant — right atrium)	—

Congenital Heart Disease

Category	Lesions
Left-to-right shunts (acyanotic, late cyanosis)	VSD (most common CHD overall), ASD (most common in adults; secundum type), PDA, AV septal defect (Down)
Right-to-left shunts (cyanotic — early)	"5 Ts" : Tetralogy of Fallot (most common cyanotic), Transposition of great arteries (most common cyanotic at birth), Truncus arteriosus, Tricuspid atresia, TAPVC
Obstructive	Coarctation of aorta, pulmonary stenosis, aortic stenosis/atresia

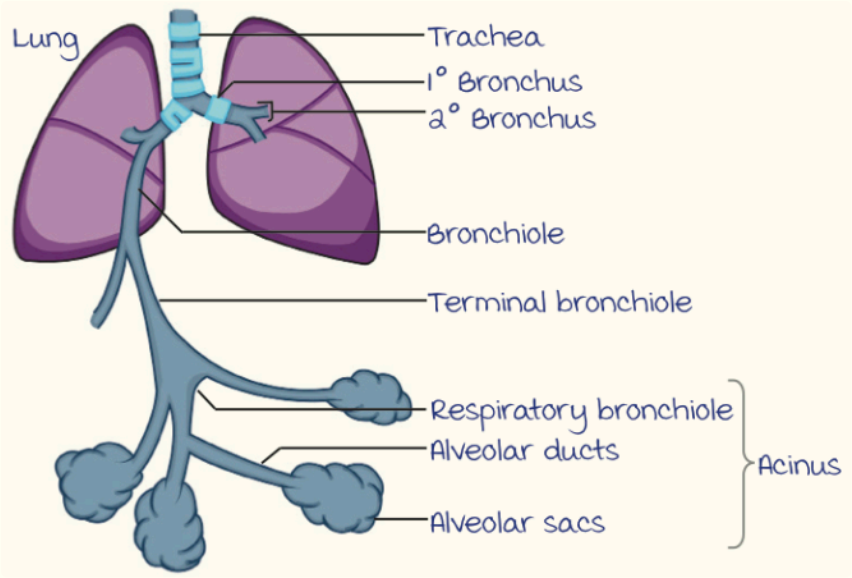
- Eisenmenger syndrome : long-standing L→R shunt → pulmonary HTN → reversal (R→L) → late cyanosis, clubbing, polycythemia.
- Tetralogy of Fallot : VSD + overriding aorta + RV outflow (pulmonary) obstruction (determines severity) + RVH → boot-shaped heart (coeur en sabot), tet spells (squatting ↑ SVR).
- PDA : machinery murmur, congenital rubella; closure with indomethacin (NSAIDs); keep open with PGE1.
- Coarctation : infantile (preductal) — Turner, with PDA ; adult (postductal) — upper-limb HTN, weak femoral pulses, rib notching (intercostal collaterals), bicuspid AV.

Filled : Embryology & associations

- TGA – maternal diabetes; TOF, truncus, interrupted arch – DiGeorge (22q11); ASD – Holt-Oram (TBX5); supravalvular AS – Williams; PDA – rubella; AV canal – Down; Ebstein anomaly – lithium; coarctation – Turner; Marfan – MVP, aortic root.
- VSD – membranous septum most common (90%).
- ASD types : secundum (most common, fossa ovalis), primum (Down, AV canal), sinus venosus (PAPVR).

RESPIRATORY SYSTEM

Structure of the Respiratory Tract



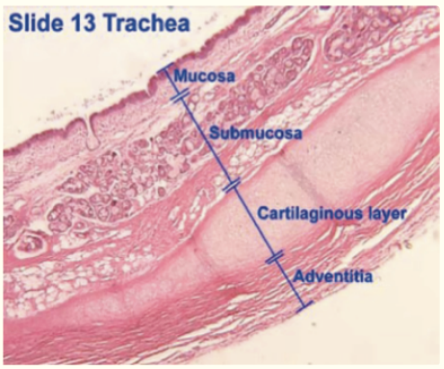
Bronchial tree : trachea → bronchi → bronchioles → acinus

- Acinus : part of lung distal to the terminal bronchiole — respiratory bronchiole + alveolar ducts + alveolar sacs (terminal respiratory unit). 3-5 acini = lobule.

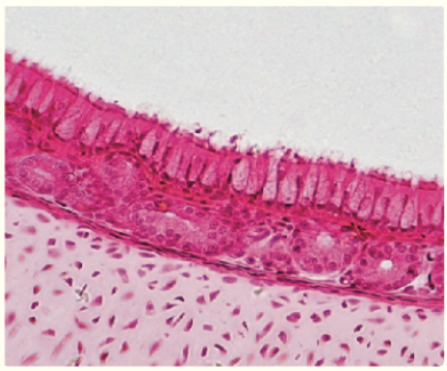
Layers of respiratory tract :

Layers of respiratory tract

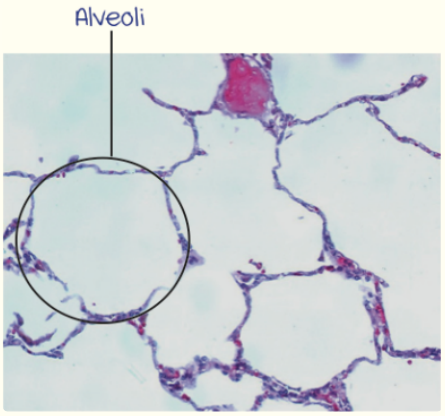
Mucosa	Submucosa	Muscularis	Cartilage
• Epithelium (pseudostratified ciliated columnar with goblet cells) • Lamina propria • Muscularis mucosa	• Submucosal glands (seromucous)		• C-shaped in trachea; plates in bronchi; absent in bronchioles



Respiratory tract (trachea) histology — mucosa, submucosa, cartilaginous layer, adventitia



Pseudostratified ciliated columnar epithelium



Alveoli — alveolar biopsy (circle marks one alveolus)

Alveoli — lined by :

1. Type I pneumocytes – 95% of alveolar surface, flat, gas exchange; cannot divide.
2. Type II pneumocytes – cuboidal; produce surfactant (dipalmitoyl phosphatidylcholine / lecithin; lamellar bodies); stem cells for type I (repair).

- Alveolar macrophages (dust cells); pores of Kohn (collateral ventilation, spread of bacteria in lobar pneumonia).
- Clara (club) cells – non-ciliated bronchiolar cells : CC10 protein, CYP450 detoxification, progenitors.

Lung Diseases — Classification

Lung diseases

Obstructive (↓ FEV1/FVC < 0.7)

- COPD : Emphysema, Chronic bronchitis
- Asthma
- Bronchiectasis

Restrictive (↓ TLC, FEV1/FVC normal or ↑)

- Pneumoconiosis : Coal worker's pneumoconiosis, Silicosis, Asbestosis
- Cryptogenic organising pneumonia
- Sarcoidosis
- IPF, chest wall disorders

Feature	Obstructive	Restrictive
FEV1	↓↓	↓
FVC	Normal / ↓	↓↓
FEV1/FVC	↓ (< 0.7)	Normal or ↑
TLC	↑ (air trapping)	↓
DLCO	↓ emphysema, normal bronchitis, ↑/normal asthma	↓ in ILD; normal in extrapulmonary (chest wall, neuromuscular)

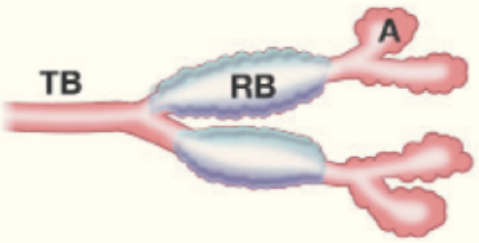
Emphysema

- Irreversible enlargement of air spaces distal to terminal bronchioles (acinus) with destruction of walls without obvious fibrosis.

Types :

1. Centrilobular / centriacinar :

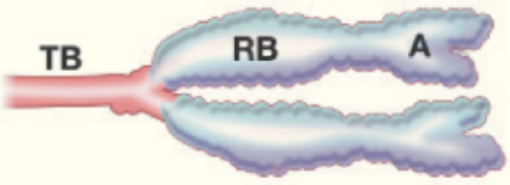
- M/c type (95%).
- Central acini affected, distal acini spared (respiratory bronchioles).
- Associated with smoking.
- Usually affects upper lobes of lung (apical segments).



Centriacinar – RB dilated, alveoli spared

3. Panacinar :

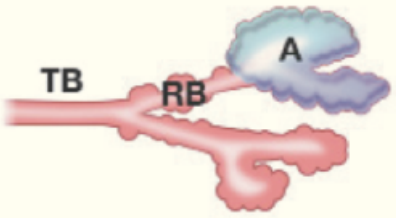
- Dilatation of the entire acinus (RB → alveoli).
- Associated with α1-antitrypsin deficiency.
- Usually affects lower lobes of lung (bases).



Panacinar – whole acinus uniformly dilated

4. Paraseptal / distal acinar :

- Distal acini affected, central acini spared (alveolar ducts & sacs, along pleura and septa).
- Associated with spontaneous pneumothorax in young adults (bullae).

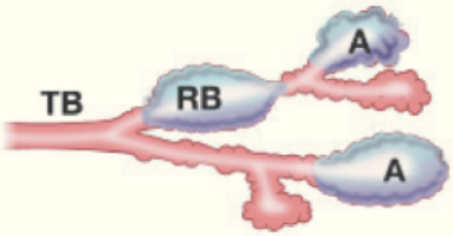


Paraseptal – distal alveoli dilated

5. Irregular :

- Common histologically – associated with scarring (healed inflammation); usually asymptomatic.

TB : Terminal bronchiole · RB : Respiratory bronchiole · A : Alveoli.



Irregular – acinus irregularly involved

Pathogenesis – protease-antiprotease & oxidant hypothesis :

Smoke / irritants → neutrophils & macrophages recruited (IL-8, LTB4, TNF)

↓

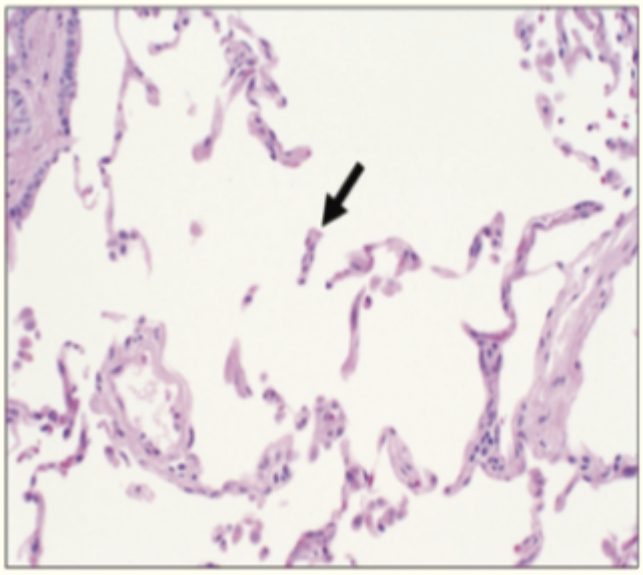
Neutrophil elastase, MMPs (macrophage elastase – MMP-12) | Oxidants in smoke + ROS → inactivate α1-antitrypsin

↓

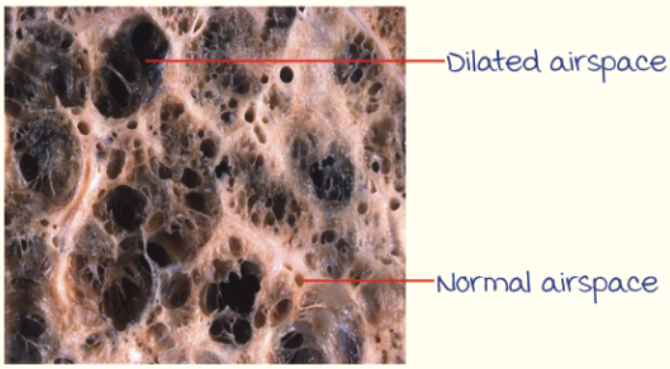
Protease > antiprotease → destruction of elastic tissue in alveolar walls

↓

Loss of radial traction on small airways → collapse during expiration → air trapping



Histology : broken alveoli / floating septae (arrow)



Gross : centriacinar emphysema – dilated vs normal airspaces

Note : α-1-antitrypsin deficiency :

- Genetics :
  - Chromosome 14 : gene for α-1-antitrypsin (α-1-AT; SERPINA1); autosomal codominant.
  - Genotype → Normal : PiMM. Diseased : PiZZ (Glu342Lys – misfolding, retained in ER).

- Organs affected :
  - Lungs : panacinar emphysema (loss of antiprotease; smoking accelerates).
  - Liver : cirrhosis (biopsy → PAS +ve, diastase-resistant globules in periportal hepatocytes — toxic gain of function); neonatal hepatitis, HCC.

> Filled : Emphysema — rank boosters
> - Pink puffers : emphysema — dyspnea, pursed-lip breathing, barrel chest, ↓ DLCO, hyperinflated lungs, flat diaphragm.
> - Blue bloaters : chronic bronchitis — cyanosis, hypercapnia, cor pulmonale, edema.
> - Bullous emphysema : bullae > 1 cm — subpleural, apex; rupture → pneumothorax.
> - Compensatory hyperinflation (after lobectomy) and obstructive overinflation (foreign body — ball-valve) are not true emphysema (no destruction).
> - Interstitial emphysema — air in connective tissue (mediastinum, subcutaneous) — ventilator, trauma, whooping cough.

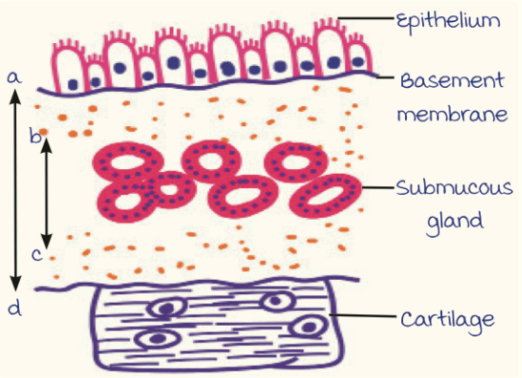
Chronic Bronchitis

- Persistent productive cough for at least 3 consecutive months in each of 2 consecutive years, in the absence of other identifiable causes (D/d : tuberculosis).
- 90% patients → smokers; also air pollution (SO_2, NO_2), cold damp climate.

Reid index (RI) :

$$ RI = Thickness of submucous gland layer / Thickness of wall between epithelium & cartilage = bc / ad

- Normal : 0.4.
- ↑ed in chronic bronchitis (> 0.5) : D/t submucosal gland hyperplasia (mucus hypersecretion — earliest feature).



Reid index — a-d : epithelium to cartilage; b-c : submucous gland layer

Morphology :

1. Hypertrophy of submucosal mucous glands (trachea, large bronchi) — ↑ Reid index
2. Goblet cell hyperplasia (small airways — chronic bronchiolitis, fibrosis → obstruction)
3. Chronic inflammation (lymphocytes), squamous metaplasia, dysplasia (→ carcinoma)

Bronchial Asthma

- Chronic inflammatory disorder of airways → episodic reversible bronchoconstriction, wheeze, ↑ airway responsiveness.

Feature	Atopic (extrinsic)	Non-atopic (intrinsic)
Trigger	Environmental allergens (dust mite, pollen)	Viral infections, pollutants, aspirin, cold, exercise
Mechanism	Type I hypersensitivity (Th2 — IL-4, IL-5, IL-13), IgE	Not IgE-mediated; airway hyperreactivity
Family history	Positive (atopy)	Negative
Serum IgE	↑	Normal

Morphology (airway remodelling) :

1. Thickened basement membrane (subepithelial collagen IV deposition) — most characteristic
2. Hypertrophy / hyperplasia of bronchial smooth muscle
3. ↑ submucosal glands, goblet cells; eosinophil infiltration; ↑ vascularity
4. Mucus plugs — occlude bronchi (status asthmaticus) ; overdistended lungs

Sputum findings :

Finding	Description
Curschmann spirals	Mucus plugs with whorled shed epithelium
Charcot-Leyden crystals	Crystalloids of eosinophil galectin-10 (lysophospholipase)
Creola bodies	Clusters of shed epithelial cells

> Exam pearl
> - Aspirin-exacerbated respiratory disease : asthma + nasal polyps + aspirin sensitivity (COX block → ↑ leukotrienes).
> - ADAM33 (airway remodelling), IL-13, TSLP genes; hygiene hypothesis.
> - Status asthmaticus — severe, unremitting; mucus plugs at autopsy.

Bronchiectasis

- Permanent dilation of bronchi and bronchioles caused by destruction of muscle and elastic tissue, from chronic necrotizing infection.

Causes :

1. Congenital / hereditary :
   - Cystic fibrosis (most common in children / West)
   - Primary ciliary dyskinesia — Kartagener syndrome (situs inversus + sinusitis + bronchiectasis + infertility) — dynein arm defect (DNAH5, DNAI1)
   - Immunodeficiency (hypogammaglobulinemia)
2. Post-infectious — necrotizing pneumonia (TB — most common in India), measles, pertussis, adenovirus
3. Bronchial obstruction — tumour, foreign body, mucus impaction
4. Allergic bronchopulmonary aspergillosis (ABPA) — central / proximal bronchiectasis
5. Others : RA, SLE, IBD, post-transplant (bronchiolitis obliterans)

Morphology :

- Lower lobes, bilateral, vertical air passages; airways dilated up to 4× (cylindroid, fusiform, saccular); can be followed to pleural surface.
- Clinical : copious foul-smelling purulent sputum, hemoptysis, clubbing; HRCT — signet ring sign, tram-track.
- Complications : lung abscess, brain abscess, amyloidosis (AA), cor pulmonale.

Pneumonias

Type	Organism	Features
Community-acquired typical	S. pneumoniae (most common), H. influenzae (COPD), Moraxella, S. aureus (post-influenza), Klebsiella (alcoholics), Pseudomonas (CF)	Lobar / bronchopneumonia
Atypical	Mycoplasma (most common), Chlamydophila, Legionella, viruses (influenza, RSV, SARS-CoV-2)	Interstitial pneumonia, patchy; cold agglutinins (Mycoplasma)

Type	Organism	Features
Hospital-acquired	Gram-negative (Klebsiella, Pseudomonas, E. coli), MRSA	Ventilator-associated
Aspiration	Anaerobes (oral flora) + aerobes	Right lower lobe (posterior segment of upper lobe / superior lower lobe when supine) — abscess

Lobar pneumonia — stages (Laennec) :

Stage	Time	Gross	Microscopy
Congestion	Day 1-2	Heavy, boggy, red	Vascular engorgement, intra-alveolar fluid with few neutrophils, many bacteria
Red hepatization	Day 2-4	Red, firm, airless, liver-like	Massive exudation of neutrophils, RBCs, fibrin filling alveoli
Grey hepatization	Day 4-8	Greyish-brown, dry	Disintegration of RBCs, persistence of fibrinosuppurative exudate
Resolution	Day 8+	—	Enzymatic digestion of exudate; macrophages; restoration

- Complications : fibrinous pleuritis → organization (pleural adhesions), empyema, lung abscess, bacteremia (meningitis, endocarditis, arthritis), organization → carnification.

Feature	Lobar pneumonia	Bronchopneumonia
Distribution	Entire lobe, uniform	Patchy, centred on bronchioles, bilateral, basal
Age	Adults (20-50 y)	Extremes of age, debilitated
Organism	S. pneumoniae (90%), Klebsiella	Staph, Strep, H. influenzae, Pseudomonas

Interstitial (atypical) pneumonia :

- Inflammation confined to alveolar septa (widened, edematous; mononuclear infiltrate); alveolar spaces free of exudate; hyaline membranes in severe cases (DAD).
- Mild symptoms disproportionate to X-ray changes ("walking pneumonia").

Pulmonary Suppuration — Lung Abscess

Causes :

1. Aspiration of infective material (most common) — alcoholics, unconscious, anesthesia, seizures, dental sepsis → anaerobes (Bacteroides, Fusobacterium, Peptostreptococcus)
2. Antecedent primary bacterial infection — S. aureus, Klebsiella, Pseudomonas (post-influenza)
3. Septic embolism — IE of right heart (IV drug users), thrombophlebitis — multiple abscesses
4. Neoplasia — post-obstructive (bronchogenic carcinoma — 10-15% of abscesses)
5. Others — penetrating trauma, spread from subphrenic abscess

- Site (aspiration) : right lung (more vertical main bronchus) — posterior segment of right upper lobe, apical segment of right lower lobe.
- Morphology : cavity with air-fluid level, necrotic debris surrounded by fibrous wall.
- Complications : empyema, brain abscess / meningitis, secondary AA amyloidosis, hemorrhage.

Pulmonary Tuberculosis — Lung Specifics

- Primary : Ghon focus (subpleural — lower upper lobe / upper lower lobe) + hilar node = Ghon complex.
- Secondary : apical (Simon focus) — cavitation; fibrocaseous TB; tuberculous bronchopneumonia; miliary TB.

- Pleural involvement : serous effusion (lymphocytic, ADA ↑), tuberculous empyema, obliterative fibrous pleuritis, pleural thickening.
- Complications : hemoptysis (Rasmussen aneurysm), aspergilloma, bronchiectasis, broncho-pleural fistula, laryngeal & intestinal TB (swallowed sputum), amyloidosis, cor pulmonale.

Atelectasis & Neonatal Respiratory Distress

Type	Mechanism	Example
Resorption (obstruction)	Complete airway obstruction → air resorbed	Mucus plug (post-operative), foreign body, tumour; mediastinum shifts toward atelectatic lung
Compression	Pleural cavity filled with fluid / air	Pleural effusion, pneumothorax; mediastinum shifts away
Contraction (cicatrization)	Local / generalized fibrosis	Irreversible
Patchy (microatelectasis)	Loss of surfactant	Neonatal RDS, ARDS

Hyaline membrane disease (Neonatal RDS) :

- Deficiency of surfactant (immaturity of type II pneumocytes) — prematurity (< 28 weeks: 60%); maternal diabetes (insulin suppresses surfactant), caesarean section before labour (no glucocorticoid surge), male sex, twins.
- Morphology : lungs solid, airless, reddish-purple (liver-like), sink in water; poorly developed atelectatic alveoli; eosinophilic hyaline membranes (fibrin + necrotic epithelium) lining respiratory bronchioles, alveolar ducts; little neutrophil inflammation.
- Prevention : antenatal corticosteroids (betamethasone) ; L/S ratio ≥ 2 (amniotic fluid), phosphatidylglycerol presence = lung maturity.
- Complications (O_2 / ventilation — ROS) : retinopathy of prematurity (VEGF), bronchopulmonary dysplasia (↓ alveolar septation — "alveolar hypoplasia"), PDA, IVH, NEC.

ARDS (diffuse alveolar damage) :

- Causes : sepsis (most common), diffuse pulmonary infections (viral — COVID-19), gastric aspiration, trauma, pancreatitis, transfusion (TRALI), fat embolism.
- Stages : exudative (hyaline membranes — day 3-7) → proliferative (type II pneumocyte hyperplasia, organization) → fibrotic.
- Berlin definition : acute (< 1 week), bilateral opacities, $PaO_2/FiO_2 \leq 300$, not cardiac.

Restrictive & Interstitial Lung Diseases

Disease	Key pathology
Idiopathic pulmonary fibrosis (UIP pattern)	Patchy interstitial fibrosis, temporal heterogeneity, fibroblastic foci, honeycombing — subpleural, lower lobes; > 50 y men, smokers; TERT, MUC5B; nintedanib, pirfenidone
Nonspecific interstitial pneumonia (NSIP)	Uniform (temporally homogeneous) inflammation / fibrosis; CTD-associated; better prognosis
Cryptogenic organizing pneumonia (BOOP)	Polypoid plugs of loose organizing connective tissue (Masson bodies) in alveolar ducts, alveoli, bronchioles; responds to steroids
Sarcoidosis	Non-caseating granulomas (lymphangitic distribution), bilateral hilar lymphadenopathy, asteroid & Schaumann bodies, ↑ ACE, hypercalcemia; Löfgren syndrome (erythema nodosum + BHL + arthritis); Heerfordt (uveoparotid fever); CD4:CD8 > 3.5 in BAL ; African-American women, non-smokers

Disease	Key pathology
Hypersensitivity pneumonitis	Type III + IV; farmer's lung (Saccharopolyspora / thermophilic actinomycetes), bird fancier's (pigeon), bagassosis (sugarcane), byssinosis (cotton — "Monday chest tightness"); poorly formed non-necrotizing granulomas, bronchiolocentric
Pulmonary eosinophilia	Loeffler (transient, parasites), tropical (microfilariae), chronic eosinophilic pneumonia, drug-induced
Pulmonary alveolar proteinosis	Anti-GM-CSF antibodies → defective macrophage clearance of surfactant; PAS + granular intra-alveolar material; "crazy paving" CT; whole-lung lavage
Langerhans cell histiocytosis (pulmonary)	Smokers, upper-lobe nodules & cysts; Birbeck granules, CD1a+
Goodpasture syndrome	Anti-GBM (α3 type IV collagen) → pulmonary hemorrhage + RPGN; linear IgG; young men, smokers

Occupational Lung Diseases (Pneumoconioses)

Disease	Agent	Key pathology
Coal worker's pneumoconiosis	Coal dust (carbon)	Anthracosis (asymptomatic, black pigment in macrophages along lymphatics) → simple CWP (coal macules 1-2 mm, upper lobes; coal nodules) → complicated CWP / progressive massive fibrosis (> 2 cm black scars); Caplan syndrome (RA + pneumoconiosis — rheumatoid nodules)
Silicosis	Crystalline silica (quartz) — most common occupational disease worldwide; mining, sand-blasting, stone cutting, glass, pottery, foundry	Upper lobes; silicotic nodules — concentric "onion-skin" hyalinized collagen with birefringent particles (weakly); eggshell calcification of hilar nodes; ↑ susceptibility to TB (silica kills macrophages — inflammasome); lung cancer (IARC group 1)
Asbestosis	Asbestos (amphiboles — crocidolite most pathogenic; serpentine — chrysotile) — shipyard, insulation, roofing, brake linings	Lower lobes, subpleural diffuse interstitial fibrosis (UIP-like); asbestos (ferruginous) bodies — golden-brown beaded dumbbell rods, Prussian blue +; pleural plaques (most common lesion — parietal pleura, dome of diaphragm, calcified, no asbestos bodies); bronchogenic carcinoma (most common malignancy; synergistic with smoking ×55), malignant mesothelioma
Berylliosis	Beryllium (aerospace, electronics)	Non-caseating granulomas (sarcoid-like), type IV hypersensitivity; lung cancer
Byssinosis	Cotton, flax, hemp	Monday fever; not granulomatous
Siderosis	Iron (welders)	Benign
Bagassosis	Sugarcane bagasse (thermophilic actinomycetes)	Hypersensitivity pneumonitis
Farmer's lung	Mouldy hay (Saccharopolyspora rectivirgula)	HP

Mnemonic : Upper vs lower lobe pneumoconioses

"Upper — SCB" : Silicosis, Coal, Berylliosis (and TB, sarcoid upper/mid) · "Lower — Asbestosis" (and IPF, scleroderma).

Lung Tumours

Classification (WHO) :

Type	Frequency	Key features
Adenocarcinoma	Most common overall (~40%); most common in women, non-smokers, young	Peripheral; glandular, mucin; TTF-1, Napsin A +; mutations : EGFR (Asian non-smoking women), ALK (EML4-ALK), ROS1, KRAS (smokers), BRAF, MET, RET; scar carcinoma; adenocarcinoma in situ (lepidic, < 3 cm, formerly BAC)
Squamous cell carcinoma	~25-30%; strongest smoking association (with SCLC); men	Central (hilar); keratin pearls, intercellular bridges; p40, p63, CK5/6 +; cavitation, PTHrP → hypercalcemia; SOX2, FGFR1 amplification, TP53, CDKN2A; arises from squamous metaplasia → dysplasia → CIS
Small cell carcinoma (SCLC)	~15%; almost exclusively smokers	Central; small round blue cells, scant cytoplasm, nuclear molding, salt-and-pepper chromatin, no nucleoli, crush artifact (Azzopardi effect), high mitotic rate; neuroendocrine (chromogranin, synaptophysin, CD56, INSM1); TP53 + RB1 loss (nearly 100%), L-MYC; most aggressive, early metastasis; ectopic ACTH, ADH, Lambert-Eaton; chemo-radiosensitive, not surgical
Large cell carcinoma	~10%	Undifferentiated, diagnosis of exclusion; large cells, prominent nucleoli; gynecomastia (hCG)
Carcinoid tumour	1-5%	Low-grade neuroendocrine; young (< 40 y), not smoking-related; central bronchus — polypoid "collar-button / mushroom" lesion; organoid / trabecular nests, salt-and-pepper chromatin; carcinoid syndrome rare; typical (< 2 mitoses/2 mm²) vs atypical

Risk factors :

1. Tobacco smoking — most important (85-90%) — dose (pack-years) related; benzo[a]pyrene, nitrosamines (NNK)
2. Radon — most common cause in non-smokers; uranium miners
3. Asbestos (synergistic with smoking)
4. Arsenic, chromium, nickel, vinyl chloride, bis-chloromethyl ether, mustard gas
5. Air pollution (PM2.5), indoor biomass fuel (India — women)
6. Genetic — CYP1A1 polymorphisms, CHRNA (nicotinic receptor)

Local effects & syndromes :

Syndrome	Mechanism
Pancoast (superior sulcus) tumour	Apical tumour → Horner syndrome (ptosis, miosis, anhidrosis — cervical sympathetic chain), C8-T1 brachial plexus (ulnar pain), rib destruction
Superior vena cava syndrome	SCLC (most common cause) — facial swelling, dilated neck veins
Recurrent laryngeal nerve	Hoarseness (left)
Phrenic nerve	Diaphragmatic paralysis
Pericardial / pleural	Effusion (malignant, hemorrhagic)

Filled : Rank boosters — lung tumours

- Most common lung tumour overall : metastasis (breast, colon, kidney — cannonball).
- Most common benign lung tumour : hamartoma — "popcorn" calcification, coin lesion, mature cartilage + fat + respiratory epithelium.
- Lung adenocarcinoma in never-smokers : EGFR exon 19 deletion / L858R → gefitinib, osimertinib.
- PD-L1 ≥ 50% → pembrolizumab first line.
- Mesothelioma : asbestos (latency 25-45 y; smoking NOT synergistic); encases lung like a rind; epithelioid (most common), sarcomatoid, biphasic; calretinin, WT1, D2-40, CK5/6 +; CEA, TTF-1 – (vs adenocarcinoma); long microvilli (EM); BAP1 loss, CDKN2A deletion.
- Solitary fibrous tumour of pleura — STAT6 + (NAB2-STAT6), hypoglycemia (IGF-2 — Doege-Potter).